

# Exams Preparation

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2026-09-21

```
library(tidyverse)
```

```
## Warning: le package 'tidyverse' a été compilé avec la version R 4.4.3
```

```
## Warning: le package 'ggplot2' a été compilé avec la version R 4.4.3
```

```
## Warning: le package 'tibble' a été compilé avec la version R 4.4.3
```

```
## Warning: le package 'tidyr' a été compilé avec la version R 4.4.3
```

```
## Warning: le package 'readr' a été compilé avec la version R 4.4.3
```

```
## Warning: le package 'purrr' a été compilé avec la version R 4.4.3
```

```
## Warning: le package 'dplyr' a été compilé avec la version R 4.4.3
```

```
## Warning: le package 'stringr' a été compilé avec la version R 4.4.3
```

```
## Warning: le package 'forcats' a été compilé avec la version R 4.4.3
```

```
## Warning: le package 'lubridate' a été compilé avec la version R 4.4.3
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
```

```
## v dplyr      1.2.1      v readr      2.2.0
```

```
## v forcats    1.0.1      v stringr    1.6.0
```

```
## v ggplot2    4.0.3      v tibble     3.3.1
```

```
## v lubridate  1.9.5      v tidyr      1.3.2
```

```
## v purrr      1.2.2
```

```
## -- Conflicts ----- tidyverse_conflicts() --
```

```
## x dplyr::filter() masks stats::filter()
```

```
## x dplyr::lag()     masks stats::lag()
```

```
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
table1
```

```
## # A tibble: 6 x 4
##   country      year cases population
##   <chr>      <dbl> <dbl>      <dbl>
## 1 Afghanistan 1999     745   19987071
## 2 Afghanistan 2000    2666   20595360
## 3 Brazil      1999   37737   172006362
## 4 Brazil      2000   80488   174504898
## 5 China       1999  212258  1272915272
## 6 China       2000  213766  1280428583
```

table2

```
## # A tibble: 12 x 4
##   country      year type      count
##   <chr>      <dbl> <chr>      <dbl>
## 1 Afghanistan 1999 cases         745
## 2 Afghanistan 1999 population 19987071
## 3 Afghanistan 2000 cases         2666
## 4 Afghanistan 2000 population 20595360
## 5 Brazil      1999 cases         37737
## 6 Brazil      1999 population 172006362
## 7 Brazil      2000 cases         80488
## 8 Brazil      2000 population 174504898
## 9 China       1999 cases        212258
## 10 China      1999 population 1272915272
## 11 China      2000 cases        213766
## 12 China      2000 population 1280428583
```

table3

```
## # A tibble: 6 x 3
##   country      year rate
##   <chr>      <dbl> <chr>
## 1 Afghanistan 1999 745/19987071
## 2 Afghanistan 2000 2666/20595360
## 3 Brazil      1999 37737/172006362
## 4 Brazil      2000 80488/174504898
## 5 China       1999 212258/1272915272
## 6 China       2000 213766/1280428583
```

```
table2 |>
  spread(
    type, count
  )
```

```
## # A tibble: 6 x 4
##   country      year cases population
##   <chr>      <dbl> <dbl>      <dbl>
## 1 Afghanistan 1999     745   19987071
## 2 Afghanistan 2000    2666   20595360
## 3 Brazil      1999   37737   172006362
## 4 Brazil      2000   80488   174504898
## 5 China       1999  212258  1272915272
## 6 China       2000  213766  1280428583
```

```
table1 |>
  gather(
    type, count, -country, -year
  )
```

```
## # A tibble: 12 x 4
##   country      year type      count
##   <chr>      <dbl> <chr>    <dbl>
## 1 Afghanistan 1999 cases      745
## 2 Afghanistan 2000 cases     2666
## 3 Brazil      1999 cases    37737
## 4 Brazil      2000 cases   80488
## 5 China       1999 cases  212258
## 6 China       2000 cases  213766
## 7 Afghanistan 1999 population 19987071
## 8 Afghanistan 2000 population 20595360
## 9 Brazil      1999 population 172006362
## 10 Brazil     2000 population 174504898
## 11 China       1999 population 1272915272
## 12 China       2000 population 1280428583
```

```
# Question
# function to generate 1D latin hypercube sample
lhs1 <- function(n) {
  x <- sample(1:n)
  x <- x-runif(n)
  x <- x/n
  return(x)
}

int.approx <- function(u) {
  mean(sin(u))
}

## The simulation code
set.seed(10)
n <- 10
nrep <- 1000
lhs.mean <- replicate(nrep, int.approx(lhs1(n)))
iid.mean <- apply(matrix(runif(n*nrep), nrep, n), 1,
  int.approx)

cbind(lhs.mean, iid.mean)
```

```
##           lhs.mean iid.mean
## [1,] 0.4492382 0.4111729
## [2,] 0.4740878 0.5336689
## [3,] 0.4686392 0.2831740
## [4,] 0.4630862 0.5202691
## [5,] 0.4486171 0.4977472
```

```

##      [6,] 0.4678334 0.5558633
##      [7,] 0.4506937 0.5211396
##      [8,] 0.4666926 0.4505082
##      [9,] 0.4481356 0.4342126
##     [10,] 0.4586429 0.4886705
##     [11,] 0.4575018 0.5354966
##     [12,] 0.4457905 0.5112394
##     [13,] 0.4429602 0.5077432
##     [14,] 0.4649693 0.4039388
##     [15,] 0.4536420 0.4726236
##     [16,] 0.4574113 0.4989835
##     [17,] 0.4719839 0.4802305
##     [18,] 0.4686634 0.3352840
##     [19,] 0.4625846 0.4379407
##     [20,] 0.4675316 0.4159490
##     [21,] 0.4538213 0.4843395
##     [22,] 0.4610735 0.4861898
##     [23,] 0.4451018 0.1534040
##     [24,] 0.4593846 0.4822176
##     [25,] 0.4590963 0.5652945
##     [26,] 0.4539921 0.4370029
##     [27,] 0.4600417 0.5169427
##     [28,] 0.4391114 0.4038098
##     [29,] 0.4578232 0.3648320
##     [30,] 0.4603820 0.5833314
##     [31,] 0.4456751 0.3343359
##     [32,] 0.4559072 0.3912344
##     [33,] 0.4591181 0.3866992
##     [34,] 0.4675378 0.4471281
##     [35,] 0.4539707 0.3777160
##     [36,] 0.4649583 0.3917972
##     [37,] 0.4594288 0.4301338
##     [38,] 0.4537041 0.5656042
##     [39,] 0.4556006 0.5033595
##     [40,] 0.4630064 0.3859980
##     [41,] 0.4677663 0.4689194
##     [42,] 0.4621455 0.4610742
##     [43,] 0.4596831 0.3496947
##     [44,] 0.4460224 0.4142556
##     [45,] 0.4691450 0.6551731
##     [46,] 0.4561753 0.4063886
##     [47,] 0.4773234 0.4446645
##     [48,] 0.4678150 0.4846266
##     [49,] 0.4515931 0.4702511
##     [50,] 0.4667391 0.6127111
##     [51,] 0.4620470 0.3879099
##     [52,] 0.4482420 0.5027142
##     [53,] 0.4672723 0.5330597
##     [54,] 0.4666497 0.5059314
##     [55,] 0.4544189 0.4222346
##     [56,] 0.4689939 0.5386421
##     [57,] 0.4581096 0.4417185
##     [58,] 0.4435505 0.5456838
##     [59,] 0.4610206 0.4252292

```

```

## [60,] 0.4543042 0.5763939
## [61,] 0.4580870 0.4207081
## [62,] 0.4552394 0.4444731
## [63,] 0.4739727 0.5796451
## [64,] 0.4702112 0.3459953
## [65,] 0.4472579 0.3881540
## [66,] 0.4543019 0.4785282
## [67,] 0.4699996 0.3781036
## [68,] 0.4493528 0.4762305
## [69,] 0.4676894 0.4227060
## [70,] 0.4723298 0.6096162
## [71,] 0.4456663 0.4879608
## [72,] 0.4614731 0.5910294
## [73,] 0.4538210 0.3908505
## [74,] 0.4664193 0.5786256
## [75,] 0.4582512 0.5952466
## [76,] 0.4561929 0.3956808
## [77,] 0.4503692 0.5322499
## [78,] 0.4622341 0.4384585
## [79,] 0.4506678 0.4964759
## [80,] 0.4625496 0.4974113
## [81,] 0.4469759 0.4504464
## [82,] 0.4582674 0.4627628
## [83,] 0.4585003 0.4585315
## [84,] 0.4738734 0.4513831
## [85,] 0.4536932 0.2814120
## [86,] 0.4655069 0.3523355
## [87,] 0.4572818 0.4982128
## [88,] 0.4606436 0.3263331
## [89,] 0.4681678 0.4151744
## [90,] 0.4715944 0.4020116
## [91,] 0.4588905 0.5231772
## [92,] 0.4654687 0.4672324
## [93,] 0.4530080 0.5037612
## [94,] 0.4605101 0.5515832
## [95,] 0.4808530 0.4314672
## [96,] 0.4600141 0.5266933
## [97,] 0.4496866 0.4979257
## [98,] 0.4640013 0.3534647
## [99,] 0.4591138 0.6767510
## [100,] 0.4639082 0.4681774
## [101,] 0.4638486 0.3993466
## [102,] 0.4582132 0.3829161
## [103,] 0.4578682 0.5561038
## [104,] 0.4649639 0.5245502
## [105,] 0.4743307 0.4648904
## [106,] 0.4492904 0.3657232
## [107,] 0.4615177 0.3815915
## [108,] 0.4490056 0.5674841
## [109,] 0.4543945 0.4423475
## [110,] 0.4570306 0.3658813
## [111,] 0.4567321 0.5100508
## [112,] 0.4482287 0.4212014
## [113,] 0.4679550 0.3736440

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```

## [114,] 0.4515908 0.5018932
## [115,] 0.4574960 0.4041629
## [116,] 0.4644010 0.5445603
## [117,] 0.4626922 0.5800585
## [118,] 0.4542139 0.4485383
## [119,] 0.4618540 0.5477516
## [120,] 0.4588785 0.4193020
## [121,] 0.4545817 0.5002335
## [122,] 0.4491281 0.4219507
## [123,] 0.4448482 0.5010653
## [124,] 0.4647998 0.4238089
## [125,] 0.4688008 0.4935096
## [126,] 0.4464512 0.4307960
## [127,] 0.4531582 0.3550956
## [128,] 0.4571022 0.4464631
## [129,] 0.4613707 0.2027528
## [130,] 0.4577271 0.4810019
## [131,] 0.4615398 0.6546957
## [132,] 0.4626330 0.4652083
## [133,] 0.4536754 0.3026372
## [134,] 0.4597356 0.6594623
## [135,] 0.4761758 0.5187737
## [136,] 0.4643026 0.5603863
## [137,] 0.4558668 0.4676432
## [138,] 0.4639657 0.5379229
## [139,] 0.4710845 0.3913728
## [140,] 0.4618139 0.5170564
## [141,] 0.4512346 0.5071433
## [142,] 0.4588654 0.3980388
## [143,] 0.4502851 0.3931177
## [144,] 0.4662249 0.5505165
## [145,] 0.4631958 0.3583543
## [146,] 0.4560222 0.4683178
## [147,] 0.4540164 0.2832091
## [148,] 0.4530651 0.5649008
## [149,] 0.4674971 0.5189278
## [150,] 0.4663743 0.4568832
## [151,] 0.4688420 0.4726493
## [152,] 0.4587446 0.3062036
## [153,] 0.4792534 0.3586989
## [154,] 0.4554535 0.4506619
## [155,] 0.4551169 0.4166211
## [156,] 0.4448539 0.4536018
## [157,] 0.4584618 0.4410436
## [158,] 0.4584290 0.4471117
## [159,] 0.4648204 0.5946776
## [160,] 0.4586429 0.3540083
## [161,] 0.4533682 0.3776030
## [162,] 0.4506396 0.5271735
## [163,] 0.4747083 0.4447113
## [164,] 0.4500028 0.4468244
## [165,] 0.4691475 0.5149788
## [166,] 0.4652896 0.3604354
## [167,] 0.4583647 0.3782486

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## [168,] 0.4696257 0.4316286
## [169,] 0.4545706 0.3559827
## [170,] 0.4688290 0.4921979
## [171,] 0.4461224 0.5815313
## [172,] 0.4706878 0.5789285
## [173,] 0.4616292 0.5391881
## [174,] 0.4575274 0.3488571
## [175,] 0.4586779 0.4516578
## [176,] 0.4633216 0.4442726
## [177,] 0.4436629 0.4730222
## [178,] 0.4654907 0.4441287
## [179,] 0.4585364 0.4070221
## [180,] 0.4586909 0.3879733
## [181,] 0.4585255 0.4623218
## [182,] 0.4584501 0.3117514
## [183,] 0.4561375 0.5400680
## [184,] 0.4601609 0.4938297
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## [186,] 0.4511500 0.5106704
## [187,] 0.4545064 0.6090147
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## [190,] 0.4706318 0.3628268
## [191,] 0.4619415 0.3207736
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## [194,] 0.4631263 0.4143972
## [195,] 0.4585754 0.4786421
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## [200,] 0.4562940 0.4681311
## [201,] 0.4713694 0.4867821
## [202,] 0.4762160 0.5812935
## [203,] 0.4654375 0.3220716
## [204,] 0.4470964 0.5667732
## [205,] 0.4695678 0.4739801
## [206,] 0.4465591 0.4669783
## [207,] 0.4577460 0.4317542
## [208,] 0.4597049 0.3060745
## [209,] 0.4613979 0.5143420
## [210,] 0.4488883 0.5242501
## [211,] 0.4506691 0.4434165
## [212,] 0.4670144 0.4537021
## [213,] 0.4699683 0.5876266
## [214,] 0.4718423 0.4141106
## [215,] 0.4688029 0.5401150
## [216,] 0.4642160 0.4335682
## [217,] 0.4553449 0.4828636
## [218,] 0.4547795 0.3998899
## [219,] 0.4578234 0.4020541
## [220,] 0.4572155 0.4546080
## [221,] 0.4641532 0.3128476

```

```

## [222,] 0.4553084 0.3122389
## [223,] 0.4665640 0.3573905
## [224,] 0.4564801 0.5026090
## [225,] 0.4505630 0.5234178
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## [227,] 0.4537212 0.4971472
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## [232,] 0.4497134 0.5175543
## [233,] 0.4654710 0.6026702
## [234,] 0.4431213 0.4459138
## [235,] 0.4568650 0.4131727
## [236,] 0.4476652 0.3644487
## [237,] 0.4544627 0.5998041
## [238,] 0.4533883 0.3229539
## [239,] 0.4697455 0.4776343
## [240,] 0.4526264 0.4962587
## [241,] 0.4490630 0.4409474
## [242,] 0.4557836 0.4531983
## [243,] 0.4593536 0.5116104
## [244,] 0.4742937 0.4955683
## [245,] 0.4586304 0.3193380
## [246,] 0.4563249 0.5956672
## [247,] 0.4678714 0.3830796
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## [249,] 0.4631204 0.3867485
## [250,] 0.4572106 0.2864953
## [251,] 0.4756832 0.5332008
## [252,] 0.4454407 0.4803737
## [253,] 0.4617781 0.4090945
## [254,] 0.4627058 0.4376972
## [255,] 0.4463827 0.5371841
## [256,] 0.4676613 0.4560047
## [257,] 0.4499309 0.3681738
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## [259,] 0.4595207 0.5663247
## [260,] 0.4713499 0.4547619
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## [269,] 0.4491298 0.5732195
## [270,] 0.4472508 0.4038756
## [271,] 0.4657711 0.3885980
## [272,] 0.4585619 0.4478136
## [273,] 0.4654152 0.4331766
## [274,] 0.4698743 0.3840115
## [275,] 0.4547786 0.4483827

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## [276,] 0.4631698 0.4855680
## [277,] 0.4553744 0.4890770
## [278,] 0.4549000 0.4477689
## [279,] 0.4592473 0.3895351
## [280,] 0.4615471 0.4403896
## [281,] 0.4660042 0.4447546
## [282,] 0.4501626 0.4632167
## [283,] 0.4593811 0.3928862
## [284,] 0.4548281 0.4928483
## [285,] 0.4542310 0.6711797
## [286,] 0.4638118 0.3796829
## [287,] 0.4551207 0.4811426
## [288,] 0.4475077 0.4602362
## [289,] 0.4680395 0.5855279
## [290,] 0.4529439 0.4430806
## [291,] 0.4543235 0.5253098
## [292,] 0.4715275 0.4932123
## [293,] 0.4645351 0.3490064
## [294,] 0.4587206 0.4990311
## [295,] 0.4616445 0.4934867
## [296,] 0.4690273 0.5563377
## [297,] 0.4595019 0.5534281
## [298,] 0.4617268 0.5278319
## [299,] 0.4659741 0.3090726
## [300,] 0.4673600 0.4765552
## [301,] 0.4605743 0.5672143
## [302,] 0.4487208 0.5298982
## [303,] 0.4680612 0.4066626
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## [305,] 0.4601116 0.3465467
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## [309,] 0.4682173 0.4103723
## [310,] 0.4484181 0.3345682
## [311,] 0.4631438 0.3342025
## [312,] 0.4587852 0.5058187
## [313,] 0.4600615 0.4502322
## [314,] 0.4534857 0.3396024
## [315,] 0.4727158 0.5478868
## [316,] 0.4592509 0.4510790
## [317,] 0.4605016 0.3749071
## [318,] 0.4546772 0.5178800
## [319,] 0.4559267 0.4691224
## [320,] 0.4644385 0.4501781
## [321,] 0.4528168 0.5166388
## [322,] 0.4599350 0.5953166
## [323,] 0.4551601 0.3275772
## [324,] 0.4663535 0.4459031
## [325,] 0.4515761 0.4801127
## [326,] 0.4612998 0.4233086
## [327,] 0.4452991 0.5545393
## [328,] 0.4647428 0.3894086
## [329,] 0.4514000 0.5531938

```

## [330,] 0.4506152 0.5092876  
## [331,] 0.4589990 0.6416360  
## [332,] 0.4522443 0.4538399  
## [333,] 0.4691872 0.5021117  
## [334,] 0.4584851 0.5333875  
## [335,] 0.4677185 0.5385529  
## [336,] 0.4730250 0.3896840  
## [337,] 0.4515219 0.4336391  
## [338,] 0.4628874 0.3773857  
## [339,] 0.4621164 0.3980614  
## [340,] 0.4725553 0.3647571  
## [341,] 0.4483723 0.4106129  
## [342,] 0.4615115 0.4013547  
## [343,] 0.4556812 0.5120904  
## [344,] 0.4597259 0.4232413  
## [345,] 0.4468780 0.5188508  
## [346,] 0.4716168 0.4866391  
## [347,] 0.4710539 0.4811492  
## [348,] 0.4606681 0.4302806  
## [349,] 0.4654280 0.6192882  
## [350,] 0.4652305 0.5252306  
## [351,] 0.4733094 0.5380557  
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## [353,] 0.4519001 0.5174011  
## [354,] 0.4581475 0.3995318  
## [355,] 0.4587782 0.4211945  
## [356,] 0.4625962 0.6060794  
## [357,] 0.4653480 0.5017707  
## [358,] 0.4406603 0.5237123  
## [359,] 0.4546443 0.3987125  
## [360,] 0.4620235 0.4152447  
## [361,] 0.4715076 0.4563781  
## [362,] 0.4427992 0.4505268  
## [363,] 0.4621490 0.3475100  
## [364,] 0.4493473 0.4269282  
## [365,] 0.4738135 0.4252399  
## [366,] 0.4571672 0.4386075  
## [367,] 0.4593042 0.3489821  
## [368,] 0.4553167 0.4774092  
## [369,] 0.4658650 0.5483851  
## [370,] 0.4566997 0.6108461  
## [371,] 0.4658452 0.2961180  
## [372,] 0.4750305 0.3272538  
## [373,] 0.4550193 0.5355752  
## [374,] 0.4672345 0.6077746  
## [375,] 0.4553368 0.2965291  
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## [840,] 0.4507495 0.4995895
## [841,] 0.4537139 0.6420090
## [842,] 0.4625441 0.3402894
## [843,] 0.4619676 0.5317339
## [844,] 0.4580340 0.4548745
## [845,] 0.4497635 0.6275281
## [846,] 0.4615597 0.4996808
## [847,] 0.4731771 0.4899561
## [848,] 0.4742731 0.4392701
## [849,] 0.4760780 0.4852373
## [850,] 0.4644193 0.3710826
## [851,] 0.4688363 0.5006645
## [852,] 0.4616018 0.4067812
## [853,] 0.4603875 0.5394467
## [854,] 0.4635732 0.4800967
## [855,] 0.4740941 0.3952649
## [856,] 0.4488024 0.4751931
## [857,] 0.4588301 0.4781337
## [858,] 0.4512984 0.5155058
## [859,] 0.4632326 0.4207150
## [860,] 0.4519550 0.3607609
## [861,] 0.4674932 0.3336236
## [862,] 0.4661686 0.5477842
## [863,] 0.4764511 0.3836351
## [864,] 0.4562016 0.5147837
## [865,] 0.4661287 0.5204791
## [866,] 0.4536946 0.3947150
## [867,] 0.4615182 0.6171343
## [868,] 0.4634440 0.5093673
## [869,] 0.4495813 0.4372707

```

```

## [870,] 0.4618782 0.2926626
## [871,] 0.4514552 0.5610606
## [872,] 0.4709737 0.4299599
## [873,] 0.4524648 0.4942835
## [874,] 0.4532247 0.3924351
## [875,] 0.4639425 0.2948680
## [876,] 0.4651289 0.5487018
## [877,] 0.4599360 0.4312837
## [878,] 0.4682137 0.4035657
## [879,] 0.4563097 0.4146019
## [880,] 0.4676667 0.4849093
## [881,] 0.4593781 0.4997039
## [882,] 0.4601253 0.4992561
## [883,] 0.4455808 0.3167608
## [884,] 0.4614112 0.4264070
## [885,] 0.4564431 0.4852389
## [886,] 0.4522971 0.5689331
## [887,] 0.4557058 0.4449808
## [888,] 0.4623011 0.4395294
## [889,] 0.4376291 0.5721331
## [890,] 0.4716505 0.4384841
## [891,] 0.4569987 0.5521757
## [892,] 0.4721406 0.5055316
## [893,] 0.4709536 0.4477240
## [894,] 0.4536195 0.3017418
## [895,] 0.4486795 0.3869085
## [896,] 0.4557115 0.3758464
## [897,] 0.4622530 0.3630011
## [898,] 0.4637300 0.2993232
## [899,] 0.4522074 0.4019102
## [900,] 0.4532164 0.3915915
## [901,] 0.4619910 0.4009837
## [902,] 0.4715919 0.4801348
## [903,] 0.4498604 0.3010425
## [904,] 0.4585722 0.4578353
## [905,] 0.4621788 0.4680945
## [906,] 0.4534922 0.4705566
## [907,] 0.4564327 0.5678970
## [908,] 0.4572842 0.4500379
## [909,] 0.4462610 0.3583967
## [910,] 0.4589054 0.3941128
## [911,] 0.4523719 0.4130318
## [912,] 0.4756053 0.4363022
## [913,] 0.4550077 0.4545298
## [914,] 0.4628744 0.4763874
## [915,] 0.4560041 0.3852730
## [916,] 0.4515660 0.4218062
## [917,] 0.4551391 0.4384610
## [918,] 0.4627599 0.4733850
## [919,] 0.4709025 0.3782532
## [920,] 0.4518945 0.4078650
## [921,] 0.4562898 0.3678956
## [922,] 0.4607882 0.4447556
## [923,] 0.4598189 0.3700671

```

```

## [924,] 0.4536797 0.3857430
## [925,] 0.4612091 0.2827032
## [926,] 0.4532057 0.5655371
## [927,] 0.4645697 0.2642776
## [928,] 0.4617083 0.4958864
## [929,] 0.4633347 0.3810818
## [930,] 0.4547514 0.3963084
## [931,] 0.4561873 0.5214483
## [932,] 0.4738726 0.4007466
## [933,] 0.4601256 0.3383873
## [934,] 0.4594580 0.5597324
## [935,] 0.4605126 0.3645350
## [936,] 0.4438417 0.4637500
## [937,] 0.4711276 0.4887476
## [938,] 0.4588765 0.4877849
## [939,] 0.4631990 0.5546875
## [940,] 0.4554981 0.4760159
## [941,] 0.4654662 0.4756591
## [942,] 0.4632057 0.4075802
## [943,] 0.4615862 0.4895455
## [944,] 0.4601428 0.4881391
## [945,] 0.4675794 0.4480013
## [946,] 0.4644804 0.5421879
## [947,] 0.4484765 0.5183477
## [948,] 0.4573075 0.4039070
## [949,] 0.4643943 0.5561833
## [950,] 0.4477768 0.4601516
## [951,] 0.4599654 0.5159700
## [952,] 0.4553065 0.4665280
## [953,] 0.4662195 0.6232392
## [954,] 0.4678549 0.4584632
## [955,] 0.4517304 0.5326216
## [956,] 0.4686818 0.4692121
## [957,] 0.4644938 0.4293321
## [958,] 0.4488518 0.5633885
## [959,] 0.4621075 0.4832890
## [960,] 0.4714268 0.4473254
## [961,] 0.4703682 0.4834074
## [962,] 0.4614769 0.4210589
## [963,] 0.4566545 0.5512563
## [964,] 0.4604279 0.5519345
## [965,] 0.4511093 0.5442469
## [966,] 0.4611357 0.4154795
## [967,] 0.4478714 0.4719814
## [968,] 0.4616582 0.4631108
## [969,] 0.4761040 0.3750211
## [970,] 0.4729715 0.2644836
## [971,] 0.4654790 0.4254625
## [972,] 0.4621911 0.4391791
## [973,] 0.4768523 0.4758939
## [974,] 0.4555833 0.3605688
## [975,] 0.4560469 0.4342380
## [976,] 0.4504118 0.5434615
## [977,] 0.4747051 0.5242554

```

```
## [978,] 0.4473871 0.5006905
## [979,] 0.4577505 0.4676838
## [980,] 0.4679769 0.4607645
## [981,] 0.4579977 0.3892182
## [982,] 0.4654443 0.4196048
## [983,] 0.4562725 0.4252728
## [984,] 0.4582884 0.4153635
## [985,] 0.4536808 0.3299570
## [986,] 0.4443648 0.4944398
## [987,] 0.4655313 0.5555375
## [988,] 0.4489014 0.3092714
## [989,] 0.4475195 0.6035789
## [990,] 0.4612706 0.4216266
## [991,] 0.4605150 0.5982261
## [992,] 0.4570782 0.5054368
## [993,] 0.4476485 0.4419917
## [994,] 0.4635524 0.3963362
## [995,] 0.4605728 0.4384398
## [996,] 0.4503636 0.4964623
## [997,] 0.4571508 0.4821531
## [998,] 0.4624286 0.4303004
## [999,] 0.4553331 0.4833466
## [1000,] 0.4603769 0.5692220
```

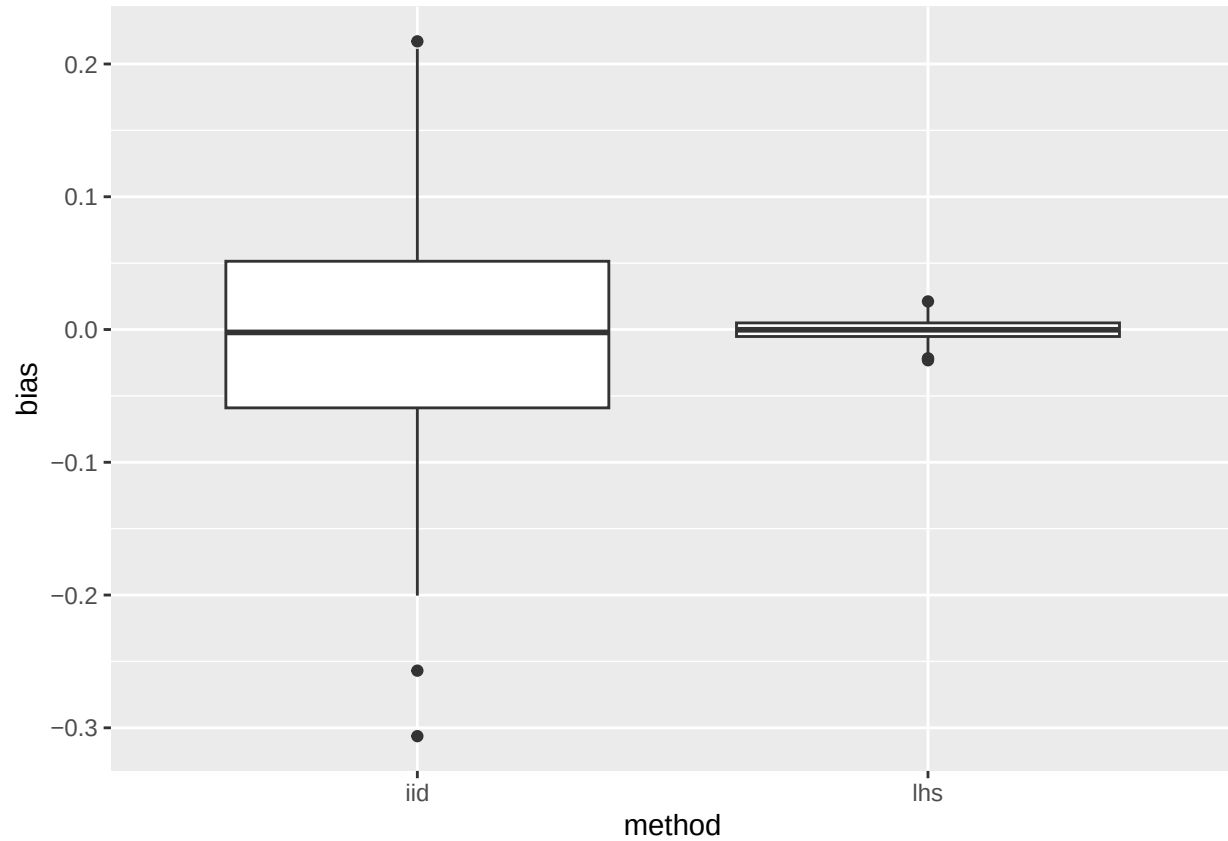
```
results1 <- tibble(
  rep.id = 1:nrep,
  lhs = lhs.mean,
  iid = iid.mean
)

results2 <- results1 |>
  gather(
    method, est, -rep.id
  )

true_mean <- 1-cos(1)
results2 |>
  group_by(method) |>
  summarize(
    bias = mean(est)-true_mean,
    var = var(est),
    mse = var(est)+bias^2
  )
```

```
## # A tibble: 2 x 4
##   method      bias      var      mse
##   <chr>      <dbl>    <dbl>    <dbl>
## 1 iid      -0.00389  0.00645  0.00647
## 2 lhs      -0.000119 0.0000573 0.0000573
```

```
results2 |>
  ggplot(aes(x=method, y=est-true_mean)) +
  geom_boxplot()+
  ylab("bias")
```



## Simulation

```
## Lab Quiz 2
```

```
# What is the mean of a normal random with mean 0, sd=1
```

```
x <- rnorm(100)
```

```
mean(x)
```

```
## [1] 0.0001524749
```

```
# what is the probability of  $X < 3$ 
```

```
pnorm(3)
```

```
## [1] 0.9986501
```

```
mean(x<3)
```

```
## [1] 1
```

```
# What is mean of  $Z=X1*X2$ , both of them are standard normal random  
# variables, they are independent  
nrep <- 100  
x <- matrix(rnorm(2*nrep), nrep, 2)  
z <- x[,1]*x[,2]  
mean(z)
```

```
## [1] 0.04496375
```

```
quantile(z, prob=0.9)
```

```
##      90%  
## 1.1548
```

```
d <- 10  
nrep <- 1000  
prods <- replicate(nrep,  
                    prod(runif(d))  
                    )  
quantile(prods, prob=0.9)
```

```
##      90%  
## 0.001890821
```

## Data Analysis

```
library(tidyverse)  
bschool <- read.csv("C://Users//FRT//Downloads//Fall-2026//MATH-8830//Dataset//Bschoo1.csv")  
# some useful function  
filter(bschool, student == "MBA") # filter the student who does the MBA class
```

```
##           title gender student year semester size  score  
## 1 Associate Professor      f   MBA 2002     fall  114 12.700  
## 2 Associate Professor      f   MBA 2002     fall   89 14.092  
## 3 Associate Professor      f   MBA 2003     fall   84 17.832  
## 4 Associate Professor      f   MBA 2002     fall   67 13.880  
## 5 Associate Professor      f   MBA 2003     fall   46 18.520  
## 6 Associate Professor      f   MBA 2004     fall   45 18.044  
## 7 Associate Professor      f   MBA 2002     fall   38 12.736  
## 8 Associate Professor      f   MBA 2002     fall   32 14.192  
## 9 Associate Professor      f   MBA 2003     fall   31 17.732  
## 10 Associate Professor     f   MBA 2004     fall   30 17.200  
## 11 Associate Professor     f   MBA 2003     fall   29 18.144  
## 12 Associate Professor     f   MBA 2004     fall   29 18.144
```

|       |                     |   |          |      |     |        |
|-------|---------------------|---|----------|------|-----|--------|
| ## 13 | Associate Professor | f | MBA 2003 | fall | 23  | 13.740 |
| ## 14 | Associate Professor | f | MBA 2004 | fall | 20  | 19.000 |
| ## 15 | Associate Professor | f | MBA 2004 | fall | 20  | 18.104 |
| ## 16 | Associate Professor | f | MBA 2002 | fall | 6   | 19.332 |
| ## 17 | Associate Professor | m | MBA 2003 | fall | 135 | 17.196 |
| ## 18 | Associate Professor | m | MBA 2004 | fall | 121 | 18.512 |
| ## 19 | Associate Professor | m | MBA 2003 | fall | 110 | 13.176 |
| ## 20 | Associate Professor | m | MBA 2003 | fall | 107 | 19.020 |
| ## 21 | Associate Professor | m | MBA 2004 | fall | 92  | 13.956 |
| ## 22 | Associate Professor | m | MBA 2003 | fall | 92  | 16.912 |
| ## 23 | Associate Professor | m | MBA 2002 | fall | 88  | 18.668 |
| ## 24 | Associate Professor | m | MBA 2003 | fall | 86  | 16.000 |
| ## 25 | Associate Professor | m | MBA 2002 | fall | 76  | 16.852 |
| ## 26 | Associate Professor | m | MBA 2003 | fall | 67  | 12.544 |
| ## 27 | Associate Professor | m | MBA 2002 | fall | 66  | 11.692 |
| ## 28 | Associate Professor | m | MBA 2003 | fall | 66  | 16.308 |
| ## 29 | Associate Professor | m | MBA 2002 | fall | 62  | 17.444 |
| ## 30 | Associate Professor | m | MBA 2002 | fall | 62  | 16.852 |
| ## 31 | Associate Professor | m | MBA 2002 | fall | 61  | 19.068 |
| ## 32 | Associate Professor | m | MBA 2004 | fall | 54  | 17.208 |
| ## 33 | Associate Professor | m | MBA 2002 | fall | 54  | 13.508 |
| ## 34 | Associate Professor | m | MBA 2002 | fall | 52  | 18.152 |
| ## 35 | Associate Professor | m | MBA 2002 | fall | 52  | 17.492 |
| ## 36 | Associate Professor | m | MBA 2003 | fall | 48  | 17.916 |
| ## 37 | Associate Professor | m | MBA 2004 | fall | 45  | 18.488 |
| ## 38 | Associate Professor | m | MBA 2002 | fall | 42  | 13.072 |
| ## 39 | Associate Professor | m | MBA 2003 | fall | 41  | 16.000 |
| ## 40 | Associate Professor | m | MBA 2004 | fall | 40  | 17.300 |
| ## 41 | Associate Professor | m | MBA 2003 | fall | 41  | 16.500 |
| ## 42 | Associate Professor | m | MBA 2002 | fall | 37  | 10.668 |
| ## 43 | Associate Professor | m | MBA 2004 | fall | 35  | 16.916 |
| ## 44 | Associate Professor | m | MBA 2002 | fall | 33  | 10.544 |
| ## 45 | Associate Professor | m | MBA 2002 | fall | 33  | 14.624 |
| ## 46 | Associate Professor | m | MBA 2002 | fall | 33  | 15.500 |
| ## 47 | Associate Professor | m | MBA 2004 | fall | 32  | 18.708 |
| ## 48 | Associate Professor | m | MBA 2004 | fall | 27  | 16.296 |
| ## 49 | Associate Professor | m | MBA 2003 | fall | 27  | 15.076 |
| ## 50 | Associate Professor | m | MBA 2004 | fall | 24  | 18.668 |
| ## 51 | Associate Professor | m | MBA 2004 | fall | 22  | 16.000 |
| ## 52 | Associate Professor | m | MBA 2003 | fall | 20  | 15.400 |
| ## 53 | Associate Professor | m | MBA 2004 | fall | 15  | 14.132 |
| ## 54 | Associate Professor | m | MBA 2002 | fall | 16  | 14.132 |
| ## 55 | Associate Professor | m | MBA 2002 | fall | 15  | 14.000 |
| ## 56 | Associate Professor | m | MBA 2002 | fall | 15  | 18.572 |
| ## 57 | Associate Professor | m | MBA 2002 | fall | 15  | 17.144 |
| ## 58 | AssistantProfessor  | f | MBA 2002 | fall | 74  | 16.540 |
| ## 59 | AssistantProfessor  | f | MBA 2004 | fall | 55  | 10.224 |
| ## 60 | AssistantProfessor  | f | MBA 2003 | fall | 53  | 16.692 |
| ## 61 | AssistantProfessor  | f | MBA 2003 | fall | 33  | 13.576 |
| ## 62 | AssistantProfessor  | f | MBA 2002 | fall | 30  | 15.312 |
| ## 63 | AssistantProfessor  | m | MBA 2002 | fall | 81  | 17.300 |
| ## 64 | AssistantProfessor  | m | MBA 2002 | fall | 79  | 18.076 |
| ## 65 | AssistantProfessor  | m | MBA 2003 | fall | 63  | 13.904 |
| ## 66 | AssistantProfessor  | m | MBA 2002 | fall | 59  | 17.084 |

|        |                    |   |          |        |     |        |
|--------|--------------------|---|----------|--------|-----|--------|
| ## 67  | AssistantProfessor | m | MBA 2002 | fall   | 51  | 18.432 |
| ## 68  | AssistantProfessor | m | MBA 2003 | fall   | 50  | 14.204 |
| ## 69  | AssistantProfessor | m | MBA 2004 | fall   | 46  | 14.784 |
| ## 70  | AssistantProfessor | m | MBA 2003 | fall   | 37  | 17.728 |
| ## 71  | AssistantProfessor | m | MBA 2002 | fall   | 37  | 12.332 |
| ## 72  | AssistantProfessor | m | MBA 2004 | fall   | 31  | 14.268 |
| ## 73  | AssistantProfessor | m | MBA 2004 | fall   | 28  | 18.428 |
| ## 74  | AssistantProfessor | m | MBA 2002 | fall   | 22  | 14.856 |
| ## 75  | AssistantProfessor | m | MBA 2003 | fall   | 15  | 14.400 |
| ## 76  | AssistantProfessor | m | MBA 2002 | fall   | 12  | 17.820 |
| ## 77  | AssistantProfessor | m | MBA 2004 | fall   | 12  | 19.272 |
| ## 78  | Professor          | f | MBA 2002 | fall   | 102 | 17.568 |
| ## 79  | Professor          | f | MBA 2002 | fall   | 70  | 12.800 |
| ## 80  | Professor          | f | MBA 2003 | fall   | 54  | 15.556 |
| ## 81  | Professor          | f | MBA 2004 | fall   | 49  | 17.632 |
| ## 82  | Professor          | f | MBA 2002 | fall   | 45  | 16.180 |
| ## 83  | Professor          | f | MBA 2002 | fall   | 40  | 16.800 |
| ## 84  | Professor          | m | MBA 2002 | fall   | 136 | 17.028 |
| ## 85  | Professor          | m | MBA 2003 | fall   | 135 | 18.104 |
| ## 86  | Professor          | m | MBA 2003 | fall   | 131 | 18.460 |
| ## 87  | Professor          | m | MBA 2003 | fall   | 106 | 19.696 |
| ## 88  | Professor          | m | MBA 2004 | fall   | 101 | 16.440 |
| ## 89  | Professor          | m | MBA 2004 | fall   | 86  | 19.304 |
| ## 90  | Professor          | m | MBA 2002 | fall   | 84  | 16.904 |
| ## 91  | Professor          | m | MBA 2004 | fall   | 82  | 18.096 |
| ## 92  | Professor          | m | MBA 2003 | fall   | 78  | 17.076 |
| ## 93  | Professor          | m | MBA 2002 | fall   | 76  | 17.172 |
| ## 94  | Professor          | m | MBA 2004 | fall   | 71  | 15.492 |
| ## 95  | Professor          | m | MBA 2002 | fall   | 68  | 19.176 |
| ## 96  | Professor          | m | MBA 2002 | fall   | 67  | 18.028 |
| ## 97  | Professor          | m | MBA 2002 | fall   | 67  | 16.120 |
| ## 98  | Professor          | m | MBA 2003 | fall   | 67  | 18.788 |
| ## 99  | Professor          | m | MBA 2003 | fall   | 65  | 13.624 |
| ## 100 | Professor          | m | MBA 2002 | fall   | 58  | 13.864 |
| ## 101 | Professor          | m | MBA 2002 | fall   | 54  | 15.184 |
| ## 102 | Professor          | m | MBA 2002 | fall   | 53  | 14.568 |
| ## 103 | Professor          | m | MBA 2004 | fall   | 53  | 15.924 |
| ## 104 | Professor          | m | MBA 2002 | fall   | 50  | 17.920 |
| ## 105 | Professor          | m | MBA 2002 | fall   | 51  | 16.880 |
| ## 106 | Professor          | m | MBA 2004 | fall   | 51  | 17.680 |
| ## 107 | Professor          | m | MBA 2002 | fall   | 47  | 18.980 |
| ## 108 | Professor          | m | MBA 2002 | fall   | 48  | 18.128 |
| ## 109 | Professor          | m | MBA 2004 | fall   | 46  | 19.044 |
| ## 110 | Professor          | m | MBA 2002 | fall   | 45  | 18.756 |
| ## 111 | Professor          | m | MBA 2002 | fall   | 41  | 16.196 |
| ## 112 | Professor          | m | MBA 2002 | spring | 42  | 15.708 |
| ## 113 | Professor          | m | MBA 2004 | spring | 40  | 18.300 |
| ## 114 | Professor          | m | MBA 2004 | spring | 38  | 18.596 |
| ## 115 | Professor          | m | MBA 2002 | spring | 36  | 18.516 |
| ## 116 | Professor          | m | MBA 2002 | spring | 34  | 18.352 |
| ## 117 | Professor          | m | MBA 2004 | spring | 32  | 18.624 |
| ## 118 | Professor          | m | MBA 2002 | spring | 31  | 15.468 |
| ## 119 | Professor          | m | MBA 2003 | spring | 28  | 14.572 |
| ## 120 | Professor          | m | MBA 2002 | spring | 28  | 15.260 |

|        |           |   |          |        |    |        |
|--------|-----------|---|----------|--------|----|--------|
| ## 121 | Professor | m | MBA 2002 | spring | 26 | 16.768 |
| ## 122 | Professor | m | MBA 2003 | spring | 25 | 16.332 |
| ## 123 | Professor | m | MBA 2003 | spring | 23 | 18.608 |
| ## 124 | Professor | m | MBA 2003 | spring | 22 | 18.364 |
| ## 125 | Professor | m | MBA 2004 | spring | 23 | 19.272 |
| ## 126 | Professor | m | MBA 2002 | spring | 21 | 18.400 |
| ## 127 | Professor | m | MBA 2004 | spring | 17 | 15.764 |
| ## 128 | Professor | m | MBA 2003 | spring | 18 | 18.824 |
| ## 129 | Professor | m | MBA 2002 | spring | 15 | 15.428 |
| ## 130 | Professor | m | MBA 2002 | spring | 14 | 16.616 |
| ## 131 | Professor | m | MBA 2002 | spring | 10 | 18.000 |
| ## 132 | Professor | m | MBA 2004 | spring | 10 | 20.000 |

```
arrange(bschool, size, score) # orders the rows of the data
```

| ##    |           | title     | gender | student       | year | semester | size | score  |
|-------|-----------|-----------|--------|---------------|------|----------|------|--------|
| ## 1  |           | Professor | m      | Graduate      | 2002 | spring   | 4    | 20.000 |
| ## 2  |           | Professor | m      | Graduate      | 2002 | spring   | 5    | 19.000 |
| ## 3  |           | Professor | m      | Graduate      | 2004 | spring   | 5    | 20.000 |
| ## 4  | Associate | Professor | m      | Graduate      | 2002 | fall     | 6    | 19.200 |
| ## 5  | Associate | Professor | f      | MBA           | 2002 | fall     | 6    | 19.332 |
| ## 6  | Assistant | Professor | m      | Graduate      | 2004 | fall     | 7    | 15.428 |
| ## 7  | Associate | Professor | m      | Graduate      | 2002 | fall     | 7    | 17.144 |
| ## 8  |           | Professor | m      | Graduate      | 2002 | spring   | 7    | 17.332 |
| ## 9  |           | Professor | f      | Graduate      | 2003 | fall     | 7    | 18.856 |
| ## 10 |           | Professor | m      | Graduate      | 2003 | spring   | 7    | 19.428 |
| ## 11 |           | Professor | m      | Graduate      | 2002 | spring   | 7    | 19.428 |
| ## 12 | Associate | Professor | m      | Graduate      | 2003 | fall     | 8    | 18.000 |
| ## 13 | Assistant | Professor | m      | Graduate      | 2003 | fall     | 8    | 18.500 |
| ## 14 |           | Professor | m      | Graduate      | 2002 | spring   | 9    | 16.888 |
| ## 15 | Associate | Professor | m      | Graduate      | 2002 | fall     | 9    | 17.332 |
| ## 16 |           | Professor | f      | Graduate      | 2002 | fall     | 9    | 18.668 |
| ## 17 | Associate | Professor | m      | Graduate      | 2004 | fall     | 9    | 19.112 |
| ## 18 |           | Professor | m      | Graduate      | 2002 | spring   | 9    | 19.556 |
| ## 19 |           | Professor | m      | MBA           | 2002 | spring   | 10   | 18.000 |
| ## 20 |           | Professor | m      | Graduate      | 2002 | spring   | 10   | 18.400 |
| ## 21 | Assistant | Professor | m      | Undergraduate | 2003 | fall     | 10   | 19.112 |
| ## 22 |           | Professor | m      | Graduate      | 2004 | spring   | 10   | 20.000 |
| ## 23 |           | Professor | m      | MBA           | 2004 | spring   | 10   | 20.000 |
| ## 24 |           | Professor | m      | Graduate      | 2002 | spring   | 11   | 16.400 |
| ## 25 | Assistant | Professor | m      | Graduate      | 2002 | fall     | 11   | 16.728 |
| ## 26 | Assistant | Professor | m      | Undergraduate | 2004 | fall     | 11   | 17.820 |
| ## 27 | Assistant | Professor | m      | Graduate      | 2003 | fall     | 11   | 18.000 |
| ## 28 |           | Professor | m      | Graduate      | 2004 | spring   | 11   | 18.800 |
| ## 29 | Associate | Professor | m      | Graduate      | 2002 | fall     | 11   | 19.200 |
| ## 30 |           | Professor | m      | Graduate      | 2002 | spring   | 11   | 19.200 |
| ## 31 |           | Professor | m      | Graduate      | 2002 | spring   | 11   | 20.000 |
| ## 32 |           | Professor | m      | Graduate      | 2002 | spring   | 12   | 15.668 |
| ## 33 |           | Professor | m      | Graduate      | 2002 | spring   | 12   | 15.668 |
| ## 34 | Associate | Professor | m      | Graduate      | 2002 | fall     | 12   | 16.364 |
| ## 35 | Assistant | Professor | m      | MBA           | 2002 | fall     | 12   | 17.820 |
| ## 36 | Associate | Professor | m      | Graduate      | 2003 | fall     | 12   | 18.908 |
| ## 37 |           | Professor | m      | Graduate      | 2003 | spring   | 12   | 18.908 |
| ## 38 | Assistant | Professor | m      | MBA           | 2004 | fall     | 12   | 19.272 |

|       |                     |   |               |      |        |    |        |
|-------|---------------------|---|---------------|------|--------|----|--------|
| ## 39 | Associate Professor | m | Graduate      | 2002 | fall   | 12 | 20.000 |
| ## 40 | AssistantProfessor  | m | Graduate      | 2002 | fall   | 13 | 13.848 |
| ## 41 | Associate Professor | m | Graduate      | 2002 | fall   | 13 | 16.308 |
| ## 42 | Associate Professor | m | Graduate      | 2004 | fall   | 13 | 16.668 |
| ## 43 | Professor           | m | Graduate      | 2002 | spring | 13 | 17.848 |
| ## 44 | Professor           | m | Graduate      | 2002 | spring | 13 | 18.000 |
| ## 45 | Professor           | m | Graduate      | 2004 | spring | 13 | 18.332 |
| ## 46 | Professor           | m | Graduate      | 2004 | spring | 13 | 19.384 |
| ## 47 | Professor           | f | Graduate      | 2004 | fall   | 13 | 19.668 |
| ## 48 | Professor           | m | MBA           | 2002 | spring | 14 | 16.616 |
| ## 49 | AssistantProfessor  | m | Graduate      | 2003 | fall   | 14 | 17.540 |
| ## 50 | AssistantProfessor  | m | Graduate      | 2003 | fall   | 14 | 19.692 |
| ## 51 | Associate Professor | m | Graduate      | 2002 | fall   | 15 | 13.600 |
| ## 52 | Associate Professor | m | MBA           | 2002 | fall   | 15 | 14.000 |
| ## 53 | Associate Professor | m | MBA           | 2004 | fall   | 15 | 14.132 |
| ## 54 | AssistantProfessor  | m | MBA           | 2003 | fall   | 15 | 14.400 |
| ## 55 | Professor           | m | MBA           | 2002 | spring | 15 | 15.428 |
| ## 56 | Associate Professor | m | Graduate      | 2002 | fall   | 15 | 16.000 |
| ## 57 | Professor           | m | Graduate      | 2004 | spring | 15 | 16.800 |
| ## 58 | Associate Professor | m | MBA           | 2002 | fall   | 15 | 17.144 |
| ## 59 | Associate Professor | m | MBA           | 2002 | fall   | 15 | 18.572 |
| ## 60 | Associate Professor | f | Graduate      | 2004 | fall   | 15 | 18.856 |
| ## 61 | AssistantProfessor  | m | Graduate      | 2004 | fall   | 15 | 18.856 |
| ## 62 | Associate Professor | m | Graduate      | 2002 | fall   | 15 | 18.932 |
| ## 63 | Professor           | m | Graduate      | 2002 | spring | 15 | 18.932 |
| ## 64 | Professor           | m | Graduate      | 2004 | spring | 15 | 19.428 |
| ## 65 | Associate Professor | m | Graduate      | 2004 | fall   | 15 | 19.468 |
| ## 66 | Associate Professor | m | MBA           | 2002 | fall   | 16 | 14.132 |
| ## 67 | Associate Professor | m | Graduate      | 2002 | fall   | 16 | 16.500 |
| ## 68 | Professor           | m | Graduate      | 2002 | spring | 16 | 16.532 |
| ## 69 | Professor           | m | Graduate      | 2002 | spring | 16 | 18.668 |
| ## 70 | Professor           | m | Graduate      | 2003 | spring | 16 | 18.752 |
| ## 71 | Professor           | m | Graduate      | 2003 | spring | 16 | 19.200 |
| ## 72 | Professor           | m | Graduate      | 2003 | spring | 17 | 14.352 |
| ## 73 | AssistantProfessor  | m | Graduate      | 2002 | fall   | 17 | 15.500 |
| ## 74 | Professor           | m | MBA           | 2004 | spring | 17 | 15.764 |
| ## 75 | AssistantProfessor  | m | Graduate      | 2002 | fall   | 17 | 16.252 |
| ## 76 | Associate Professor | m | Graduate      | 2002 | fall   | 17 | 16.752 |
| ## 77 | Professor           | m | Graduate      | 2004 | spring | 17 | 17.176 |
| ## 78 | Associate Professor | m | Graduate      | 2002 | fall   | 17 | 17.500 |
| ## 79 | Associate Professor | m | Graduate      | 2002 | fall   | 17 | 17.752 |
| ## 80 | Professor           | m | Graduate      | 2003 | spring | 17 | 19.500 |
| ## 81 | Professor           | m | Graduate      | 2002 | spring | 18 | 12.224 |
| ## 82 | Professor           | m | MBA           | 2003 | spring | 18 | 18.824 |
| ## 83 | Associate Professor | m | Graduate      | 2004 | fall   | 18 | 19.296 |
| ## 84 | Associate Professor | m | Undergraduate | 2002 | fall   | 19 | 13.332 |
| ## 85 | Associate Professor | m | Graduate      | 2002 | fall   | 19 | 13.896 |
| ## 86 | Associate Professor | f | Graduate      | 2003 | fall   | 19 | 17.556 |
| ## 87 | Professor           | m | Graduate      | 2002 | spring | 19 | 18.104 |
| ## 88 | Professor           | m | Graduate      | 2002 | spring | 19 | 18.668 |
| ## 89 | Professor           | m | Graduate      | 2002 | spring | 19 | 19.112 |
| ## 90 | Associate Professor | m | MBA           | 2003 | fall   | 20 | 15.400 |
| ## 91 | Professor           | m | Graduate      | 2002 | spring | 20 | 15.600 |
| ## 92 | Professor           | m | Graduate      | 2002 | spring | 20 | 16.800 |

|        |                     |   |               |      |        |    |        |
|--------|---------------------|---|---------------|------|--------|----|--------|
| ## 93  | Associate Professor | m | Graduate      | 2004 | fall   | 20 | 17.800 |
| ## 94  | Associate Professor | f | MBA           | 2004 | fall   | 20 | 18.104 |
| ## 95  | Associate Professor | m | Graduate      | 2004 | fall   | 20 | 18.736 |
| ## 96  | Associate Professor | f | MBA           | 2004 | fall   | 20 | 19.000 |
| ## 97  | Associate Professor | m | Undergraduate | 2004 | fall   | 20 | 19.580 |
| ## 98  | Associate Professor | m | Undergraduate | 2003 | fall   | 21 | 17.144 |
| ## 99  | AssistantProfessor  | m | Graduate      | 2003 | fall   | 21 | 17.716 |
| ## 100 | Professor           | m | MBA           | 2002 | spring | 21 | 18.400 |
| ## 101 | Professor           | m | Graduate      | 2003 | spring | 21 | 18.668 |
| ## 102 | Associate Professor | m | Undergraduate | 2003 | fall   | 22 | 13.456 |
| ## 103 | AssistantProfessor  | m | MBA           | 2002 | fall   | 22 | 14.856 |
| ## 104 | Associate Professor | m | MBA           | 2004 | fall   | 22 | 16.000 |
| ## 105 | Associate Professor | f | Undergraduate | 2003 | fall   | 22 | 16.544 |
| ## 106 | Professor           | m | MBA           | 2003 | spring | 22 | 18.364 |
| ## 107 | Professor           | m | Graduate      | 2002 | spring | 22 | 18.908 |
| ## 108 | Associate Professor | f | MBA           | 2003 | fall   | 23 | 13.740 |
| ## 109 | Professor           | m | Graduate      | 2003 | spring | 23 | 16.544 |
| ## 110 | Professor           | m | MBA           | 2003 | spring | 23 | 18.608 |
| ## 111 | Professor           | m | MBA           | 2004 | spring | 23 | 19.272 |
| ## 112 | Professor           | m | Graduate      | 2003 | spring | 24 | 17.332 |
| ## 113 | Associate Professor | m | Graduate      | 2002 | fall   | 24 | 17.740 |
| ## 114 | Associate Professor | m | MBA           | 2004 | fall   | 24 | 18.668 |
| ## 115 | Professor           | m | Graduate      | 2002 | spring | 24 | 19.132 |
| ## 116 | Associate Professor | m | Undergraduate | 2004 | fall   | 25 | 16.332 |
| ## 117 | Professor           | m | MBA           | 2003 | spring | 25 | 16.332 |
| ## 118 | Associate Professor | m | Graduate      | 2004 | fall   | 25 | 18.500 |
| ## 119 | Associate Professor | m | Undergraduate | 2002 | fall   | 26 | 13.280 |
| ## 120 | Associate Professor | m | Undergraduate | 2002 | fall   | 26 | 16.152 |
| ## 121 | AssistantProfessor  | f | Undergraduate | 2004 | fall   | 26 | 16.616 |
| ## 122 | Professor           | m | MBA           | 2002 | spring | 26 | 16.768 |
| ## 123 | Professor           | m | Graduate      | 2002 | spring | 26 | 17.920 |
| ## 124 | Associate Professor | m | MBA           | 2003 | fall   | 27 | 15.076 |
| ## 125 | AssistantProfessor  | f | Undergraduate | 2003 | fall   | 27 | 16.000 |
| ## 126 | Professor           | m | Graduate      | 2004 | spring | 27 | 16.152 |
| ## 127 | Associate Professor | m | MBA           | 2004 | fall   | 27 | 16.296 |
| ## 128 | Professor           | m | Graduate      | 2004 | spring | 27 | 16.888 |
| ## 129 | Professor           | m | Graduate      | 2002 | spring | 27 | 16.924 |
| ## 130 | AssistantProfessor  | m | Graduate      | 2002 | fall   | 27 | 18.308 |
| ## 131 | Professor           | m | MBA           | 2003 | spring | 28 | 14.572 |
| ## 132 | Professor           | m | MBA           | 2002 | spring | 28 | 15.260 |
| ## 133 | Professor           | m | Graduate      | 2002 | spring | 28 | 15.260 |
| ## 134 | Associate Professor | f | Undergraduate | 2004 | fall   | 28 | 15.852 |
| ## 135 | Professor           | m | Graduate      | 2002 | spring | 28 | 16.144 |
| ## 136 | Associate Professor | m | Graduate      | 2002 | fall   | 28 | 16.856 |
| ## 137 | Professor           | f | Undergraduate | 2003 | fall   | 28 | 17.716 |
| ## 138 | Associate Professor | m | Undergraduate | 2004 | fall   | 28 | 18.144 |
| ## 139 | AssistantProfessor  | m | MBA           | 2004 | fall   | 28 | 18.428 |
| ## 140 | AssistantProfessor  | m | Undergraduate | 2004 | fall   | 29 | 14.208 |
| ## 141 | Associate Professor | f | Graduate      | 2002 | fall   | 29 | 16.412 |
| ## 142 | Professor           | m | Graduate      | 2002 | spring | 29 | 17.144 |
| ## 143 | AssistantProfessor  | m | Graduate      | 2004 | fall   | 29 | 17.716 |
| ## 144 | Associate Professor | f | MBA           | 2003 | fall   | 29 | 18.144 |
| ## 145 | Associate Professor | f | MBA           | 2004 | fall   | 29 | 18.144 |
| ## 146 | AssistantProfessor  | m | Undergraduate | 2003 | fall   | 29 | 18.572 |

|        |                     |   |               |      |        |    |        |
|--------|---------------------|---|---------------|------|--------|----|--------|
| ## 147 | AssistantProfessor  | m | Graduate      | 2004 | fall   | 29 | 19.284 |
| ## 148 | Professor           | m | Graduate      | 2002 | spring | 29 | 19.572 |
| ## 149 | AssistantProfessor  | f | MBA           | 2002 | fall   | 30 | 15.312 |
| ## 150 | Associate Professor | f | MBA           | 2004 | fall   | 30 | 17.200 |
| ## 151 | AssistantProfessor  | f | Undergraduate | 2003 | fall   | 30 | 17.516 |
| ## 152 | AssistantProfessor  | m | MBA           | 2004 | fall   | 31 | 14.268 |
| ## 153 | Professor           | m | MBA           | 2002 | spring | 31 | 15.468 |
| ## 154 | AssistantProfessor  | m | Undergraduate | 2004 | fall   | 31 | 16.388 |
| ## 155 | Associate Professor | f | MBA           | 2003 | fall   | 31 | 17.732 |
| ## 156 | Professor           | m | Graduate      | 2004 | spring | 31 | 19.468 |
| ## 157 | Associate Professor | f | MBA           | 2002 | fall   | 32 | 14.192 |
| ## 158 | Professor           | m | Graduate      | 2002 | spring | 32 | 17.252 |
| ## 159 | Associate Professor | m | Undergraduate | 2002 | fall   | 32 | 17.500 |
| ## 160 | Professor           | m | Graduate      | 2002 | spring | 32 | 18.064 |
| ## 161 | Professor           | m | MBA           | 2004 | spring | 32 | 18.624 |
| ## 162 | Associate Professor | m | MBA           | 2004 | fall   | 32 | 18.708 |
| ## 163 | Associate Professor | m | MBA           | 2002 | fall   | 33 | 10.544 |
| ## 164 | AssistantProfessor  | f | MBA           | 2003 | fall   | 33 | 13.576 |
| ## 165 | Associate Professor | m | MBA           | 2002 | fall   | 33 | 14.624 |
| ## 166 | Associate Professor | m | MBA           | 2002 | fall   | 33 | 15.500 |
| ## 167 | Associate Professor | m | Undergraduate | 2003 | fall   | 33 | 16.364 |
| ## 168 | Associate Professor | m | Graduate      | 2002 | fall   | 33 | 17.752 |
| ## 169 | Professor           | m | Graduate      | 2002 | spring | 33 | 19.376 |
| ## 170 | AssistantProfessor  | f | Undergraduate | 2004 | fall   | 34 | 17.764 |
| ## 171 | Professor           | m | MBA           | 2002 | spring | 34 | 18.352 |
| ## 172 | Associate Professor | m | Graduate      | 2003 | fall   | 34 | 18.824 |
| ## 173 | Professor           | m | Graduate      | 2004 | spring | 34 | 19.032 |
| ## 174 | Professor           | f | Graduate      | 2003 | fall   | 34 | 19.528 |
| ## 175 | AssistantProfessor  | f | Undergraduate | 2004 | fall   | 35 | 16.456 |
| ## 176 | Associate Professor | m | MBA           | 2004 | fall   | 35 | 16.916 |
| ## 177 | Associate Professor | m | Graduate      | 2004 | fall   | 35 | 17.764 |
| ## 178 | Professor           | m | Graduate      | 2004 | spring | 35 | 18.116 |
| ## 179 | AssistantProfessor  | m | Undergraduate | 2002 | fall   | 35 | 18.236 |
| ## 180 | Professor           | f | Graduate      | 2002 | fall   | 35 | 18.284 |
| ## 181 | Professor           | f | Graduate      | 2002 | fall   | 35 | 18.352 |
| ## 182 | Professor           | m | Undergraduate | 2004 | spring | 35 | 18.588 |
| ## 183 | Professor           | m | Undergraduate | 2002 | spring | 36 | 17.600 |
| ## 184 | Associate Professor | m | Undergraduate | 2002 | fall   | 36 | 18.056 |
| ## 185 | Professor           | m | MBA           | 2002 | spring | 36 | 18.516 |
| ## 186 | Professor           | f | Graduate      | 2004 | fall   | 36 | 18.668 |
| ## 187 | Associate Professor | m | Graduate      | 2003 | fall   | 36 | 18.856 |
| ## 188 | Associate Professor | m | MBA           | 2002 | fall   | 37 | 10.668 |
| ## 189 | AssistantProfessor  | m | MBA           | 2002 | fall   | 37 | 12.332 |
| ## 190 | AssistantProfessor  | m | MBA           | 2003 | fall   | 37 | 17.728 |
| ## 191 | Associate Professor | f | MBA           | 2002 | fall   | 38 | 12.736 |
| ## 192 | Associate Professor | m | Undergraduate | 2002 | fall   | 38 | 16.864 |
| ## 193 | Professor           | m | Undergraduate | 2003 | spring | 38 | 17.836 |
| ## 194 | Professor           | m | MBA           | 2004 | spring | 38 | 18.596 |
| ## 195 | Associate Professor | m | Graduate      | 2003 | fall   | 39 | 14.564 |
| ## 196 | AssistantProfessor  | f | Undergraduate | 2004 | fall   | 39 | 15.156 |
| ## 197 | Associate Professor | m | Undergraduate | 2004 | fall   | 39 | 16.844 |
| ## 198 | Associate Professor | m | Graduate      | 2003 | fall   | 39 | 16.924 |
| ## 199 | Professor           | m | Undergraduate | 2004 | spring | 39 | 19.488 |
| ## 200 | Professor           | m | Undergraduate | 2004 | spring | 40 | 15.600 |

|        |                     |   |               |      |        |    |        |
|--------|---------------------|---|---------------|------|--------|----|--------|
| ## 201 | Associate Professor | m | Undergraduate | 2002 | fall   | 40 | 16.000 |
| ## 202 | Professor           | f | MBA           | 2002 | fall   | 40 | 16.800 |
| ## 203 | Associate Professor | m | MBA           | 2004 | fall   | 40 | 17.300 |
| ## 204 | Associate Professor | m | Graduate      | 2002 | fall   | 40 | 17.948 |
| ## 205 | Professor           | m | Graduate      | 2002 | spring | 40 | 18.200 |
| ## 206 | Professor           | m | MBA           | 2004 | spring | 40 | 18.300 |
| ## 207 | Associate Professor | m | Undergraduate | 2004 | fall   | 41 | 12.900 |
| ## 208 | Associate Professor | m | MBA           | 2003 | fall   | 41 | 16.000 |
| ## 209 | Professor           | m | MBA           | 2002 | fall   | 41 | 16.196 |
| ## 210 | Professor           | m | Graduate      | 2003 | spring | 41 | 16.200 |
| ## 211 | Associate Professor | m | MBA           | 2003 | fall   | 41 | 16.500 |
| ## 212 | Professor           | f | Undergraduate | 2002 | fall   | 41 | 16.976 |
| ## 213 | Professor           | f | Undergraduate | 2004 | fall   | 41 | 18.000 |
| ## 214 | Associate Professor | m | MBA           | 2002 | fall   | 42 | 13.072 |
| ## 215 | Professor           | m | Graduate      | 2002 | fall   | 42 | 15.620 |
| ## 216 | Professor           | m | MBA           | 2002 | spring | 42 | 15.708 |
| ## 217 | Professor           | m | Graduate      | 2002 | spring | 42 | 18.244 |
| ## 218 | AssistantProfessor  | f | Graduate      | 2004 | fall   | 42 | 18.476 |
| ## 219 | Professor           | m | Graduate      | 2002 | fall   | 44 | 15.440 |
| ## 220 | Professor           | m | Graduate      | 2002 | fall   | 44 | 18.512 |
| ## 221 | Professor           | f | MBA           | 2002 | fall   | 45 | 16.180 |
| ## 222 | Professor           | m | Graduate      | 2002 | fall   | 45 | 16.888 |
| ## 223 | AssistantProfessor  | f | Graduate      | 2002 | fall   | 45 | 16.976 |
| ## 224 | Associate Professor | f | MBA           | 2004 | fall   | 45 | 18.044 |
| ## 225 | Professor           | m | Graduate      | 2002 | fall   | 45 | 18.456 |
| ## 226 | Associate Professor | m | MBA           | 2004 | fall   | 45 | 18.488 |
| ## 227 | Professor           | m | MBA           | 2002 | fall   | 45 | 18.756 |
| ## 228 | AssistantProfessor  | m | MBA           | 2004 | fall   | 46 | 14.784 |
| ## 229 | Associate Professor | m | Undergraduate | 2002 | fall   | 46 | 15.392 |
| ## 230 | Associate Professor | m | Undergraduate | 2003 | fall   | 46 | 17.156 |
| ## 231 | Associate Professor | f | MBA           | 2003 | fall   | 46 | 18.520 |
| ## 232 | Professor           | m | MBA           | 2004 | fall   | 46 | 19.044 |
| ## 233 | AssistantProfessor  | f | Undergraduate | 2002 | fall   | 47 | 17.044 |
| ## 234 | Professor           | m | MBA           | 2002 | fall   | 47 | 18.980 |
| ## 235 | Associate Professor | m | Graduate      | 2002 | fall   | 48 | 17.360 |
| ## 236 | Associate Professor | m | Graduate      | 2003 | fall   | 48 | 17.532 |
| ## 237 | Associate Professor | m | MBA           | 2003 | fall   | 48 | 17.916 |
| ## 238 | Professor           | m | MBA           | 2002 | fall   | 48 | 18.128 |
| ## 239 | Professor           | m | Undergraduate | 2004 | fall   | 48 | 18.892 |
| ## 240 | Professor           | f | MBA           | 2004 | fall   | 49 | 17.632 |
| ## 241 | Professor           | f | Undergraduate | 2003 | fall   | 49 | 18.776 |
| ## 242 | AssistantProfessor  | m | MBA           | 2003 | fall   | 50 | 14.204 |
| ## 243 | Professor           | m | MBA           | 2002 | fall   | 50 | 17.920 |
| ## 244 | AssistantProfessor  | f | Undergraduate | 2002 | fall   | 51 | 16.240 |
| ## 245 | Professor           | m | MBA           | 2002 | fall   | 51 | 16.880 |
| ## 246 | Professor           | m | MBA           | 2004 | fall   | 51 | 17.680 |
| ## 247 | AssistantProfessor  | m | MBA           | 2002 | fall   | 51 | 18.432 |
| ## 248 | Associate Professor | m | MBA           | 2002 | fall   | 52 | 17.492 |
| ## 249 | Associate Professor | m | MBA           | 2002 | fall   | 52 | 18.152 |
| ## 250 | Professor           | m | MBA           | 2002 | fall   | 53 | 14.568 |
| ## 251 | Professor           | m | MBA           | 2004 | fall   | 53 | 15.924 |
| ## 252 | AssistantProfessor  | f | MBA           | 2003 | fall   | 53 | 16.692 |
| ## 253 | Associate Professor | m | MBA           | 2002 | fall   | 54 | 13.508 |
| ## 254 | Professor           | m | MBA           | 2002 | fall   | 54 | 15.184 |

|        |                     |   |                    |      |    |        |
|--------|---------------------|---|--------------------|------|----|--------|
| ## 255 | Professor           | f | MBA 2003           | fall | 54 | 15.556 |
| ## 256 | Associate Professor | m | Undergraduate 2003 | fall | 54 | 16.964 |
| ## 257 | Associate Professor | m | MBA 2004           | fall | 54 | 17.208 |
| ## 258 | AssistantProfessor  | f | MBA 2004           | fall | 55 | 10.224 |
| ## 259 | Professor           | m | Undergraduate 2002 | fall | 56 | 16.944 |
| ## 260 | Professor           | m | Undergraduate 2002 | fall | 56 | 17.572 |
| ## 261 | Associate Professor | m | Graduate 2004      | fall | 57 | 18.456 |
| ## 262 | AssistantProfessor  | m | Undergraduate 2004 | fall | 58 | 13.864 |
| ## 263 | Professor           | m | MBA 2002           | fall | 58 | 13.864 |
| ## 264 | Associate Professor | m | Graduate 2002      | fall | 58 | 16.280 |
| ## 265 | Professor           | m | Undergraduate 2002 | fall | 58 | 18.276 |
| ## 266 | AssistantProfessor  | m | MBA 2002           | fall | 59 | 17.084 |
| ## 267 | Associate Professor | m | Graduate 2002      | fall | 59 | 18.068 |
| ## 268 | Professor           | m | Undergraduate 2002 | fall | 60 | 17.732 |
| ## 269 | Associate Professor | m | MBA 2002           | fall | 61 | 19.068 |
| ## 270 | Associate Professor | m | MBA 2002           | fall | 62 | 16.852 |
| ## 271 | Associate Professor | m | MBA 2002           | fall | 62 | 17.444 |
| ## 272 | Associate Professor | m | Graduate 2004      | fall | 62 | 19.096 |
| ## 273 | AssistantProfessor  | m | MBA 2003           | fall | 63 | 13.904 |
| ## 274 | AssistantProfessor  | f | Undergraduate 2004 | fall | 63 | 16.888 |
| ## 275 | Associate Professor | m | Graduate 2004      | fall | 63 | 18.452 |
| ## 276 | Professor           | f | Undergraduate 2002 | fall | 64 | 17.000 |
| ## 277 | Professor           | m | MBA 2003           | fall | 65 | 13.624 |
| ## 278 | AssistantProfessor  | f | Undergraduate 2003 | fall | 65 | 17.048 |
| ## 279 | Associate Professor | m | MBA 2002           | fall | 66 | 11.692 |
| ## 280 | Associate Professor | m | MBA 2003           | fall | 66 | 16.308 |
| ## 281 | Associate Professor | m | MBA 2003           | fall | 67 | 12.544 |
| ## 282 | Associate Professor | f | MBA 2002           | fall | 67 | 13.880 |
| ## 283 | AssistantProfessor  | m | Undergraduate 2002 | fall | 67 | 14.608 |
| ## 284 | Professor           | m | MBA 2002           | fall | 67 | 16.120 |
| ## 285 | Professor           | m | MBA 2002           | fall | 67 | 18.028 |
| ## 286 | Professor           | m | MBA 2003           | fall | 67 | 18.788 |
| ## 287 | Professor           | m | Graduate 2003      | fall | 68 | 17.296 |
| ## 288 | Associate Professor | m | Undergraduate 2004 | fall | 68 | 17.972 |
| ## 289 | Professor           | m | MBA 2002           | fall | 68 | 19.176 |
| ## 290 | AssistantProfessor  | f | Graduate 2003      | fall | 69 | 18.940 |
| ## 291 | Professor           | f | MBA 2002           | fall | 70 | 12.800 |
| ## 292 | Associate Professor | m | Graduate 2004      | fall | 70 | 14.628 |
| ## 293 | Professor           | m | MBA 2004           | fall | 71 | 15.492 |
| ## 294 | Professor           | m | Undergraduate 2002 | fall | 72 | 15.000 |
| ## 295 | Associate Professor | m | Graduate 2002      | fall | 72 | 18.424 |
| ## 296 | Professor           | f | Graduate 2002      | fall | 73 | 19.500 |
| ## 297 | AssistantProfessor  | m | Undergraduate 2003 | fall | 74 | 13.040 |
| ## 298 | AssistantProfessor  | f | MBA 2002           | fall | 74 | 16.540 |
| ## 299 | Associate Professor | m | Graduate 2003      | fall | 75 | 18.488 |
| ## 300 | Associate Professor | m | MBA 2002           | fall | 76 | 16.852 |
| ## 301 | Professor           | m | Graduate 2003      | fall | 76 | 16.908 |
| ## 302 | Professor           | m | MBA 2002           | fall | 76 | 17.172 |
| ## 303 | AssistantProfessor  | f | Undergraduate 2002 | fall | 77 | 17.196 |
| ## 304 | Professor           | m | MBA 2003           | fall | 78 | 17.076 |
| ## 305 | AssistantProfessor  | m | MBA 2002           | fall | 79 | 18.076 |
| ## 306 | Professor           | m | Graduate 2002      | fall | 80 | 17.112 |
| ## 307 | AssistantProfessor  | m | Undergraduate 2003 | fall | 81 | 14.800 |
| ## 308 | AssistantProfessor  | m | MBA 2002           | fall | 81 | 17.300 |

|        |                     |   |               |      |      |     |        |
|--------|---------------------|---|---------------|------|------|-----|--------|
| ## 309 | Professor           | m | Undergraduate | 2002 | fall | 81  | 17.400 |
| ## 310 | Professor           | m | Graduate      | 2002 | fall | 82  | 10.880 |
| ## 311 | Professor           | m | Graduate      | 2004 | fall | 82  | 16.636 |
| ## 312 | Professor           | m | MBA           | 2004 | fall | 82  | 18.096 |
| ## 313 | Associate Professor | m | Graduate      | 2002 | fall | 82  | 18.780 |
| ## 314 | Professor           | m | Graduate      | 2004 | fall | 83  | 17.492 |
| ## 315 | Professor           | m | MBA           | 2002 | fall | 84  | 16.904 |
| ## 316 | Associate Professor | f | MBA           | 2003 | fall | 84  | 17.832 |
| ## 317 | Associate Professor | m | MBA           | 2003 | fall | 86  | 16.000 |
| ## 318 | Professor           | m | MBA           | 2004 | fall | 86  | 19.304 |
| ## 319 | Associate Professor | m | Graduate      | 2003 | fall | 88  | 18.436 |
| ## 320 | Associate Professor | m | MBA           | 2002 | fall | 88  | 18.668 |
| ## 321 | Associate Professor | f | MBA           | 2002 | fall | 89  | 14.092 |
| ## 322 | Associate Professor | m | Undergraduate | 2003 | fall | 91  | 16.312 |
| ## 323 | Associate Professor | m | MBA           | 2004 | fall | 92  | 13.956 |
| ## 324 | Associate Professor | m | MBA           | 2003 | fall | 92  | 16.912 |
| ## 325 | Associate Professor | m | Undergraduate | 2002 | fall | 98  | 16.452 |
| ## 326 | Professor           | m | MBA           | 2004 | fall | 101 | 16.440 |
| ## 327 | Professor           | f | MBA           | 2002 | fall | 102 | 17.568 |
| ## 328 | Professor           | m | MBA           | 2003 | fall | 106 | 19.696 |
| ## 329 | Associate Professor | m | MBA           | 2003 | fall | 107 | 19.020 |
| ## 330 | Associate Professor | m | MBA           | 2003 | fall | 110 | 13.176 |
| ## 331 | Associate Professor | m | Undergraduate | 2003 | fall | 113 | 18.964 |
| ## 332 | Associate Professor | f | MBA           | 2002 | fall | 114 | 12.700 |
| ## 333 | Associate Professor | m | Undergraduate | 2002 | fall | 119 | 16.780 |
| ## 334 | Associate Professor | m | MBA           | 2004 | fall | 121 | 18.512 |
| ## 335 | Professor           | m | Graduate      | 2004 | fall | 125 | 18.208 |
| ## 336 | Associate Professor | m | Undergraduate | 2002 | fall | 130 | 17.924 |
| ## 337 | Professor           | m | MBA           | 2003 | fall | 131 | 18.460 |
| ## 338 | Associate Professor | m | MBA           | 2003 | fall | 135 | 17.196 |
| ## 339 | Professor           | m | MBA           | 2003 | fall | 135 | 18.104 |
| ## 340 | Professor           | m | MBA           | 2002 | fall | 136 | 17.028 |

```
select(bschool, semester, size, score) # select a specific column
```

| ##    | semester | size | score  |
|-------|----------|------|--------|
| ## 1  | fall     | 114  | 12.700 |
| ## 2  | fall     | 89   | 14.092 |
| ## 3  | fall     | 84   | 17.832 |
| ## 4  | fall     | 67   | 13.880 |
| ## 5  | fall     | 46   | 18.520 |
| ## 6  | fall     | 45   | 18.044 |
| ## 7  | fall     | 38   | 12.736 |
| ## 8  | fall     | 32   | 14.192 |
| ## 9  | fall     | 31   | 17.732 |
| ## 10 | fall     | 30   | 17.200 |
| ## 11 | fall     | 29   | 16.412 |
| ## 12 | fall     | 29   | 18.144 |
| ## 13 | fall     | 29   | 18.144 |
| ## 14 | fall     | 28   | 15.852 |
| ## 15 | fall     | 23   | 13.740 |
| ## 16 | fall     | 22   | 16.544 |
| ## 17 | fall     | 20   | 19.000 |
| ## 18 | fall     | 20   | 18.104 |

|       |      |     |        |
|-------|------|-----|--------|
| ## 19 | fall | 19  | 17.556 |
| ## 20 | fall | 15  | 18.856 |
| ## 21 | fall | 6   | 19.332 |
| ## 22 | fall | 135 | 17.196 |
| ## 23 | fall | 130 | 17.924 |
| ## 24 | fall | 121 | 18.512 |
| ## 25 | fall | 119 | 16.780 |
| ## 26 | fall | 113 | 18.964 |
| ## 27 | fall | 110 | 13.176 |
| ## 28 | fall | 107 | 19.020 |
| ## 29 | fall | 98  | 16.452 |
| ## 30 | fall | 92  | 13.956 |
| ## 31 | fall | 92  | 16.912 |
| ## 32 | fall | 91  | 16.312 |
| ## 33 | fall | 88  | 18.668 |
| ## 34 | fall | 88  | 18.436 |
| ## 35 | fall | 86  | 16.000 |
| ## 36 | fall | 82  | 18.780 |
| ## 37 | fall | 76  | 16.852 |
| ## 38 | fall | 75  | 18.488 |
| ## 39 | fall | 72  | 18.424 |
| ## 40 | fall | 70  | 14.628 |
| ## 41 | fall | 68  | 17.972 |
| ## 42 | fall | 67  | 12.544 |
| ## 43 | fall | 66  | 11.692 |
| ## 44 | fall | 66  | 16.308 |
| ## 45 | fall | 63  | 18.452 |
| ## 46 | fall | 62  | 19.096 |
| ## 47 | fall | 62  | 17.444 |
| ## 48 | fall | 62  | 16.852 |
| ## 49 | fall | 61  | 19.068 |
| ## 50 | fall | 59  | 18.068 |
| ## 51 | fall | 58  | 16.280 |
| ## 52 | fall | 57  | 18.456 |
| ## 53 | fall | 54  | 16.964 |
| ## 54 | fall | 54  | 17.208 |
| ## 55 | fall | 54  | 13.508 |
| ## 56 | fall | 52  | 18.152 |
| ## 57 | fall | 52  | 17.492 |
| ## 58 | fall | 48  | 17.916 |
| ## 59 | fall | 48  | 17.360 |
| ## 60 | fall | 48  | 17.532 |
| ## 61 | fall | 46  | 15.392 |
| ## 62 | fall | 45  | 18.488 |
| ## 63 | fall | 46  | 17.156 |
| ## 64 | fall | 42  | 13.072 |
| ## 65 | fall | 41  | 16.000 |
| ## 66 | fall | 40  | 17.300 |
| ## 67 | fall | 41  | 16.500 |
| ## 68 | fall | 41  | 12.900 |
| ## 69 | fall | 40  | 16.000 |
| ## 70 | fall | 39  | 14.564 |
| ## 71 | fall | 40  | 17.948 |
| ## 72 | fall | 39  | 16.924 |

|        |      |           |
|--------|------|-----------|
| ## 73  | fall | 39 16.844 |
| ## 74  | fall | 38 16.864 |
| ## 75  | fall | 37 10.668 |
| ## 76  | fall | 35 16.916 |
| ## 77  | fall | 36 18.056 |
| ## 78  | fall | 36 18.856 |
| ## 79  | fall | 35 17.764 |
| ## 80  | fall | 34 18.824 |
| ## 81  | fall | 33 10.544 |
| ## 82  | fall | 33 16.364 |
| ## 83  | fall | 33 14.624 |
| ## 84  | fall | 33 15.500 |
| ## 85  | fall | 32 17.500 |
| ## 86  | fall | 33 17.752 |
| ## 87  | fall | 32 18.708 |
| ## 88  | fall | 28 18.144 |
| ## 89  | fall | 28 16.856 |
| ## 90  | fall | 27 16.296 |
| ## 91  | fall | 27 15.076 |
| ## 92  | fall | 26 16.152 |
| ## 93  | fall | 26 13.280 |
| ## 94  | fall | 24 18.668 |
| ## 95  | fall | 25 16.332 |
| ## 96  | fall | 25 18.500 |
| ## 97  | fall | 24 17.740 |
| ## 98  | fall | 22 16.000 |
| ## 99  | fall | 22 13.456 |
| ## 100 | fall | 21 17.144 |
| ## 101 | fall | 20 15.400 |
| ## 102 | fall | 20 17.800 |
| ## 103 | fall | 20 19.580 |
| ## 104 | fall | 20 18.736 |
| ## 105 | fall | 19 13.896 |
| ## 106 | fall | 19 13.332 |
| ## 107 | fall | 18 19.296 |
| ## 108 | fall | 17 17.752 |
| ## 109 | fall | 16 16.500 |
| ## 110 | fall | 17 17.500 |
| ## 111 | fall | 17 16.752 |
| ## 112 | fall | 15 14.132 |
| ## 113 | fall | 16 14.132 |
| ## 114 | fall | 15 18.932 |
| ## 115 | fall | 15 13.600 |
| ## 116 | fall | 15 19.468 |
| ## 117 | fall | 15 14.000 |
| ## 118 | fall | 15 18.572 |
| ## 119 | fall | 15 17.144 |
| ## 120 | fall | 15 16.000 |
| ## 121 | fall | 13 16.308 |
| ## 122 | fall | 13 16.668 |
| ## 123 | fall | 12 20.000 |
| ## 124 | fall | 12 18.908 |
| ## 125 | fall | 12 16.364 |
| ## 126 | fall | 11 19.200 |

|        |      |           |
|--------|------|-----------|
| ## 127 | fall | 9 17.332  |
| ## 128 | fall | 9 19.112  |
| ## 129 | fall | 8 18.000  |
| ## 130 | fall | 7 17.144  |
| ## 131 | fall | 6 19.200  |
| ## 132 | fall | 77 17.196 |
| ## 133 | fall | 74 16.540 |
| ## 134 | fall | 69 18.940 |
| ## 135 | fall | 65 17.048 |
| ## 136 | fall | 63 16.888 |
| ## 137 | fall | 55 10.224 |
| ## 138 | fall | 53 16.692 |
| ## 139 | fall | 51 16.240 |
| ## 140 | fall | 47 17.044 |
| ## 141 | fall | 45 16.976 |
| ## 142 | fall | 42 18.476 |
| ## 143 | fall | 39 15.156 |
| ## 144 | fall | 35 16.456 |
| ## 145 | fall | 34 17.764 |
| ## 146 | fall | 33 13.576 |
| ## 147 | fall | 30 15.312 |
| ## 148 | fall | 30 17.516 |
| ## 149 | fall | 26 16.616 |
| ## 150 | fall | 27 16.000 |
| ## 151 | fall | 81 17.300 |
| ## 152 | fall | 81 14.800 |
| ## 153 | fall | 79 18.076 |
| ## 154 | fall | 74 13.040 |
| ## 155 | fall | 67 14.608 |
| ## 156 | fall | 63 13.904 |
| ## 157 | fall | 59 17.084 |
| ## 158 | fall | 58 13.864 |
| ## 159 | fall | 51 18.432 |
| ## 160 | fall | 50 14.204 |
| ## 161 | fall | 46 14.784 |
| ## 162 | fall | 37 17.728 |
| ## 163 | fall | 37 12.332 |
| ## 164 | fall | 35 18.236 |
| ## 165 | fall | 31 16.388 |
| ## 166 | fall | 31 14.268 |
| ## 167 | fall | 29 14.208 |
| ## 168 | fall | 28 18.428 |
| ## 169 | fall | 29 18.572 |
| ## 170 | fall | 29 19.284 |
| ## 171 | fall | 29 17.716 |
| ## 172 | fall | 27 18.308 |
| ## 173 | fall | 22 14.856 |
| ## 174 | fall | 21 17.716 |
| ## 175 | fall | 17 16.252 |
| ## 176 | fall | 17 15.500 |
| ## 177 | fall | 15 14.400 |
| ## 178 | fall | 15 18.856 |
| ## 179 | fall | 14 17.540 |
| ## 180 | fall | 13 13.848 |

|        |      |     |        |
|--------|------|-----|--------|
| ## 181 | fall | 14  | 19.692 |
| ## 182 | fall | 12  | 17.820 |
| ## 183 | fall | 12  | 19.272 |
| ## 184 | fall | 11  | 17.820 |
| ## 185 | fall | 11  | 16.728 |
| ## 186 | fall | 11  | 18.000 |
| ## 187 | fall | 10  | 19.112 |
| ## 188 | fall | 8   | 18.500 |
| ## 189 | fall | 7   | 15.428 |
| ## 190 | fall | 102 | 17.568 |
| ## 191 | fall | 73  | 19.500 |
| ## 192 | fall | 70  | 12.800 |
| ## 193 | fall | 64  | 17.000 |
| ## 194 | fall | 54  | 15.556 |
| ## 195 | fall | 49  | 17.632 |
| ## 196 | fall | 49  | 18.776 |
| ## 197 | fall | 45  | 16.180 |
| ## 198 | fall | 41  | 16.976 |
| ## 199 | fall | 40  | 16.800 |
| ## 200 | fall | 41  | 18.000 |
| ## 201 | fall | 36  | 18.668 |
| ## 202 | fall | 35  | 18.284 |
| ## 203 | fall | 35  | 18.352 |
| ## 204 | fall | 34  | 19.528 |
| ## 205 | fall | 28  | 17.716 |
| ## 206 | fall | 13  | 19.668 |
| ## 207 | fall | 9   | 18.668 |
| ## 208 | fall | 7   | 18.856 |
| ## 209 | fall | 136 | 17.028 |
| ## 210 | fall | 135 | 18.104 |
| ## 211 | fall | 131 | 18.460 |
| ## 212 | fall | 125 | 18.208 |
| ## 213 | fall | 106 | 19.696 |
| ## 214 | fall | 101 | 16.440 |
| ## 215 | fall | 86  | 19.304 |
| ## 216 | fall | 84  | 16.904 |
| ## 217 | fall | 83  | 17.492 |
| ## 218 | fall | 82  | 18.096 |
| ## 219 | fall | 82  | 10.880 |
| ## 220 | fall | 82  | 16.636 |
| ## 221 | fall | 81  | 17.400 |
| ## 222 | fall | 80  | 17.112 |
| ## 223 | fall | 78  | 17.076 |
| ## 224 | fall | 76  | 17.172 |
| ## 225 | fall | 76  | 16.908 |
| ## 226 | fall | 72  | 15.000 |
| ## 227 | fall | 71  | 15.492 |
| ## 228 | fall | 68  | 19.176 |
| ## 229 | fall | 68  | 17.296 |
| ## 230 | fall | 67  | 18.028 |
| ## 231 | fall | 67  | 16.120 |
| ## 232 | fall | 67  | 18.788 |
| ## 233 | fall | 65  | 13.624 |
| ## 234 | fall | 60  | 17.732 |

|        |        |    |        |
|--------|--------|----|--------|
| ## 235 | fall   | 58 | 13.864 |
| ## 236 | fall   | 58 | 18.276 |
| ## 237 | fall   | 56 | 17.572 |
| ## 238 | fall   | 56 | 16.944 |
| ## 239 | fall   | 54 | 15.184 |
| ## 240 | fall   | 53 | 14.568 |
| ## 241 | fall   | 53 | 15.924 |
| ## 242 | fall   | 50 | 17.920 |
| ## 243 | fall   | 51 | 16.880 |
| ## 244 | fall   | 51 | 17.680 |
| ## 245 | fall   | 47 | 18.980 |
| ## 246 | fall   | 48 | 18.128 |
| ## 247 | fall   | 48 | 18.892 |
| ## 248 | fall   | 46 | 19.044 |
| ## 249 | fall   | 45 | 18.756 |
| ## 250 | fall   | 45 | 16.888 |
| ## 251 | fall   | 45 | 18.456 |
| ## 252 | fall   | 44 | 15.440 |
| ## 253 | fall   | 44 | 18.512 |
| ## 254 | fall   | 42 | 15.620 |
| ## 255 | fall   | 41 | 16.196 |
| ## 256 | spring | 42 | 15.708 |
| ## 257 | spring | 42 | 18.244 |
| ## 258 | spring | 40 | 18.300 |
| ## 259 | spring | 40 | 15.600 |
| ## 260 | spring | 40 | 18.200 |
| ## 261 | spring | 41 | 16.200 |
| ## 262 | spring | 39 | 19.488 |
| ## 263 | spring | 38 | 18.596 |
| ## 264 | spring | 38 | 17.836 |
| ## 265 | spring | 36 | 18.516 |
| ## 266 | spring | 36 | 17.600 |
| ## 267 | spring | 34 | 18.352 |
| ## 268 | spring | 35 | 18.588 |
| ## 269 | spring | 35 | 18.116 |
| ## 270 | spring | 34 | 19.032 |
| ## 271 | spring | 32 | 18.624 |
| ## 272 | spring | 32 | 17.252 |
| ## 273 | spring | 33 | 19.376 |
| ## 274 | spring | 32 | 18.064 |
| ## 275 | spring | 31 | 15.468 |
| ## 276 | spring | 31 | 19.468 |
| ## 277 | spring | 28 | 14.572 |
| ## 278 | spring | 28 | 16.144 |
| ## 279 | spring | 29 | 17.144 |
| ## 280 | spring | 29 | 19.572 |
| ## 281 | spring | 28 | 15.260 |
| ## 282 | spring | 28 | 15.260 |
| ## 283 | spring | 27 | 16.888 |
| ## 284 | spring | 26 | 16.768 |
| ## 285 | spring | 27 | 16.924 |
| ## 286 | spring | 27 | 16.152 |
| ## 287 | spring | 26 | 17.920 |
| ## 288 | spring | 25 | 16.332 |

|        |        |    |        |
|--------|--------|----|--------|
| ## 289 | spring | 24 | 17.332 |
| ## 290 | spring | 23 | 18.608 |
| ## 291 | spring | 24 | 19.132 |
| ## 292 | spring | 22 | 18.364 |
| ## 293 | spring | 23 | 19.272 |
| ## 294 | spring | 22 | 18.908 |
| ## 295 | spring | 23 | 16.544 |
| ## 296 | spring | 21 | 18.668 |
| ## 297 | spring | 21 | 18.400 |
| ## 298 | spring | 20 | 16.800 |
| ## 299 | spring | 20 | 15.600 |
| ## 300 | spring | 19 | 18.104 |
| ## 301 | spring | 18 | 12.224 |
| ## 302 | spring | 19 | 18.668 |
| ## 303 | spring | 19 | 19.112 |
| ## 304 | spring | 17 | 15.764 |
| ## 305 | spring | 18 | 18.824 |
| ## 306 | spring | 17 | 14.352 |
| ## 307 | spring | 17 | 17.176 |
| ## 308 | spring | 16 | 18.752 |
| ## 309 | spring | 17 | 19.500 |
| ## 310 | spring | 16 | 16.532 |
| ## 311 | spring | 15 | 16.800 |
| ## 312 | spring | 16 | 19.200 |
| ## 313 | spring | 15 | 18.932 |
| ## 314 | spring | 16 | 18.668 |
| ## 315 | spring | 15 | 15.428 |
| ## 316 | spring | 15 | 19.428 |
| ## 317 | spring | 14 | 16.616 |
| ## 318 | spring | 13 | 19.384 |
| ## 319 | spring | 13 | 17.848 |
| ## 320 | spring | 12 | 15.668 |
| ## 321 | spring | 13 | 18.332 |
| ## 322 | spring | 13 | 18.000 |
| ## 323 | spring | 12 | 15.668 |
| ## 324 | spring | 12 | 18.908 |
| ## 325 | spring | 10 | 18.000 |
| ## 326 | spring | 10 | 20.000 |
| ## 327 | spring | 11 | 16.400 |
| ## 328 | spring | 11 | 19.200 |
| ## 329 | spring | 10 | 18.400 |
| ## 330 | spring | 11 | 20.000 |
| ## 331 | spring | 11 | 18.800 |
| ## 332 | spring | 10 | 20.000 |
| ## 333 | spring | 9  | 16.888 |
| ## 334 | spring | 9  | 19.556 |
| ## 335 | spring | 7  | 19.428 |
| ## 336 | spring | 7  | 19.428 |
| ## 337 | spring | 7  | 17.332 |
| ## 338 | spring | 5  | 20.000 |
| ## 339 | spring | 5  | 19.000 |
| ## 340 | spring | 4  | 20.000 |

```
mutate(bschool, smallSize = 1*(size<30)) # create a new column that have a small size lesser
```

| ##    |                     | title | gender        | student | year | semester | size   | score |
|-------|---------------------|-------|---------------|---------|------|----------|--------|-------|
| ## 1  | Associate Professor | f     | MBA           | 2002    | fall | 114      | 12.700 |       |
| ## 2  | Associate Professor | f     | MBA           | 2002    | fall | 89       | 14.092 |       |
| ## 3  | Associate Professor | f     | MBA           | 2003    | fall | 84       | 17.832 |       |
| ## 4  | Associate Professor | f     | MBA           | 2002    | fall | 67       | 13.880 |       |
| ## 5  | Associate Professor | f     | MBA           | 2003    | fall | 46       | 18.520 |       |
| ## 6  | Associate Professor | f     | MBA           | 2004    | fall | 45       | 18.044 |       |
| ## 7  | Associate Professor | f     | MBA           | 2002    | fall | 38       | 12.736 |       |
| ## 8  | Associate Professor | f     | MBA           | 2002    | fall | 32       | 14.192 |       |
| ## 9  | Associate Professor | f     | MBA           | 2003    | fall | 31       | 17.732 |       |
| ## 10 | Associate Professor | f     | MBA           | 2004    | fall | 30       | 17.200 |       |
| ## 11 | Associate Professor | f     | Graduate      | 2002    | fall | 29       | 16.412 |       |
| ## 12 | Associate Professor | f     | MBA           | 2003    | fall | 29       | 18.144 |       |
| ## 13 | Associate Professor | f     | MBA           | 2004    | fall | 29       | 18.144 |       |
| ## 14 | Associate Professor | f     | Undergraduate | 2004    | fall | 28       | 15.852 |       |
| ## 15 | Associate Professor | f     | MBA           | 2003    | fall | 23       | 13.740 |       |
| ## 16 | Associate Professor | f     | Undergraduate | 2003    | fall | 22       | 16.544 |       |
| ## 17 | Associate Professor | f     | MBA           | 2004    | fall | 20       | 19.000 |       |
| ## 18 | Associate Professor | f     | MBA           | 2004    | fall | 20       | 18.104 |       |
| ## 19 | Associate Professor | f     | Graduate      | 2003    | fall | 19       | 17.556 |       |
| ## 20 | Associate Professor | f     | Graduate      | 2004    | fall | 15       | 18.856 |       |
| ## 21 | Associate Professor | f     | MBA           | 2002    | fall | 6        | 19.332 |       |
| ## 22 | Associate Professor | m     | MBA           | 2003    | fall | 135      | 17.196 |       |
| ## 23 | Associate Professor | m     | Undergraduate | 2002    | fall | 130      | 17.924 |       |
| ## 24 | Associate Professor | m     | MBA           | 2004    | fall | 121      | 18.512 |       |
| ## 25 | Associate Professor | m     | Undergraduate | 2002    | fall | 119      | 16.780 |       |
| ## 26 | Associate Professor | m     | Undergraduate | 2003    | fall | 113      | 18.964 |       |
| ## 27 | Associate Professor | m     | MBA           | 2003    | fall | 110      | 13.176 |       |
| ## 28 | Associate Professor | m     | MBA           | 2003    | fall | 107      | 19.020 |       |
| ## 29 | Associate Professor | m     | Undergraduate | 2002    | fall | 98       | 16.452 |       |
| ## 30 | Associate Professor | m     | MBA           | 2004    | fall | 92       | 13.956 |       |
| ## 31 | Associate Professor | m     | MBA           | 2003    | fall | 92       | 16.912 |       |
| ## 32 | Associate Professor | m     | Undergraduate | 2003    | fall | 91       | 16.312 |       |
| ## 33 | Associate Professor | m     | MBA           | 2002    | fall | 88       | 18.668 |       |
| ## 34 | Associate Professor | m     | Graduate      | 2003    | fall | 88       | 18.436 |       |
| ## 35 | Associate Professor | m     | MBA           | 2003    | fall | 86       | 16.000 |       |
| ## 36 | Associate Professor | m     | Graduate      | 2002    | fall | 82       | 18.780 |       |
| ## 37 | Associate Professor | m     | MBA           | 2002    | fall | 76       | 16.852 |       |
| ## 38 | Associate Professor | m     | Graduate      | 2003    | fall | 75       | 18.488 |       |
| ## 39 | Associate Professor | m     | Graduate      | 2002    | fall | 72       | 18.424 |       |
| ## 40 | Associate Professor | m     | Graduate      | 2004    | fall | 70       | 14.628 |       |
| ## 41 | Associate Professor | m     | Undergraduate | 2004    | fall | 68       | 17.972 |       |
| ## 42 | Associate Professor | m     | MBA           | 2003    | fall | 67       | 12.544 |       |
| ## 43 | Associate Professor | m     | MBA           | 2002    | fall | 66       | 11.692 |       |
| ## 44 | Associate Professor | m     | MBA           | 2003    | fall | 66       | 16.308 |       |
| ## 45 | Associate Professor | m     | Graduate      | 2004    | fall | 63       | 18.452 |       |
| ## 46 | Associate Professor | m     | Graduate      | 2004    | fall | 62       | 19.096 |       |
| ## 47 | Associate Professor | m     | MBA           | 2002    | fall | 62       | 17.444 |       |
| ## 48 | Associate Professor | m     | MBA           | 2002    | fall | 62       | 16.852 |       |
| ## 49 | Associate Professor | m     | MBA           | 2002    | fall | 61       | 19.068 |       |
| ## 50 | Associate Professor | m     | Graduate      | 2002    | fall | 59       | 18.068 |       |

|        |                     |   |               |      |      |    |        |
|--------|---------------------|---|---------------|------|------|----|--------|
| ## 51  | Associate Professor | m | Graduate      | 2002 | fall | 58 | 16.280 |
| ## 52  | Associate Professor | m | Graduate      | 2004 | fall | 57 | 18.456 |
| ## 53  | Associate Professor | m | Undergraduate | 2003 | fall | 54 | 16.964 |
| ## 54  | Associate Professor | m | MBA           | 2004 | fall | 54 | 17.208 |
| ## 55  | Associate Professor | m | MBA           | 2002 | fall | 54 | 13.508 |
| ## 56  | Associate Professor | m | MBA           | 2002 | fall | 52 | 18.152 |
| ## 57  | Associate Professor | m | MBA           | 2002 | fall | 52 | 17.492 |
| ## 58  | Associate Professor | m | MBA           | 2003 | fall | 48 | 17.916 |
| ## 59  | Associate Professor | m | Graduate      | 2002 | fall | 48 | 17.360 |
| ## 60  | Associate Professor | m | Graduate      | 2003 | fall | 48 | 17.532 |
| ## 61  | Associate Professor | m | Undergraduate | 2002 | fall | 46 | 15.392 |
| ## 62  | Associate Professor | m | MBA           | 2004 | fall | 45 | 18.488 |
| ## 63  | Associate Professor | m | Undergraduate | 2003 | fall | 46 | 17.156 |
| ## 64  | Associate Professor | m | MBA           | 2002 | fall | 42 | 13.072 |
| ## 65  | Associate Professor | m | MBA           | 2003 | fall | 41 | 16.000 |
| ## 66  | Associate Professor | m | MBA           | 2004 | fall | 40 | 17.300 |
| ## 67  | Associate Professor | m | MBA           | 2003 | fall | 41 | 16.500 |
| ## 68  | Associate Professor | m | Undergraduate | 2004 | fall | 41 | 12.900 |
| ## 69  | Associate Professor | m | Undergraduate | 2002 | fall | 40 | 16.000 |
| ## 70  | Associate Professor | m | Graduate      | 2003 | fall | 39 | 14.564 |
| ## 71  | Associate Professor | m | Graduate      | 2002 | fall | 40 | 17.948 |
| ## 72  | Associate Professor | m | Graduate      | 2003 | fall | 39 | 16.924 |
| ## 73  | Associate Professor | m | Undergraduate | 2004 | fall | 39 | 16.844 |
| ## 74  | Associate Professor | m | Undergraduate | 2002 | fall | 38 | 16.864 |
| ## 75  | Associate Professor | m | MBA           | 2002 | fall | 37 | 10.668 |
| ## 76  | Associate Professor | m | MBA           | 2004 | fall | 35 | 16.916 |
| ## 77  | Associate Professor | m | Undergraduate | 2002 | fall | 36 | 18.056 |
| ## 78  | Associate Professor | m | Graduate      | 2003 | fall | 36 | 18.856 |
| ## 79  | Associate Professor | m | Graduate      | 2004 | fall | 35 | 17.764 |
| ## 80  | Associate Professor | m | Graduate      | 2003 | fall | 34 | 18.824 |
| ## 81  | Associate Professor | m | MBA           | 2002 | fall | 33 | 10.544 |
| ## 82  | Associate Professor | m | Undergraduate | 2003 | fall | 33 | 16.364 |
| ## 83  | Associate Professor | m | MBA           | 2002 | fall | 33 | 14.624 |
| ## 84  | Associate Professor | m | MBA           | 2002 | fall | 33 | 15.500 |
| ## 85  | Associate Professor | m | Undergraduate | 2002 | fall | 32 | 17.500 |
| ## 86  | Associate Professor | m | Graduate      | 2002 | fall | 33 | 17.752 |
| ## 87  | Associate Professor | m | MBA           | 2004 | fall | 32 | 18.708 |
| ## 88  | Associate Professor | m | Undergraduate | 2004 | fall | 28 | 18.144 |
| ## 89  | Associate Professor | m | Graduate      | 2002 | fall | 28 | 16.856 |
| ## 90  | Associate Professor | m | MBA           | 2004 | fall | 27 | 16.296 |
| ## 91  | Associate Professor | m | MBA           | 2003 | fall | 27 | 15.076 |
| ## 92  | Associate Professor | m | Undergraduate | 2002 | fall | 26 | 16.152 |
| ## 93  | Associate Professor | m | Undergraduate | 2002 | fall | 26 | 13.280 |
| ## 94  | Associate Professor | m | MBA           | 2004 | fall | 24 | 18.668 |
| ## 95  | Associate Professor | m | Undergraduate | 2004 | fall | 25 | 16.332 |
| ## 96  | Associate Professor | m | Graduate      | 2004 | fall | 25 | 18.500 |
| ## 97  | Associate Professor | m | Graduate      | 2002 | fall | 24 | 17.740 |
| ## 98  | Associate Professor | m | MBA           | 2004 | fall | 22 | 16.000 |
| ## 99  | Associate Professor | m | Undergraduate | 2003 | fall | 22 | 13.456 |
| ## 100 | Associate Professor | m | Undergraduate | 2003 | fall | 21 | 17.144 |
| ## 101 | Associate Professor | m | MBA           | 2003 | fall | 20 | 15.400 |
| ## 102 | Associate Professor | m | Graduate      | 2004 | fall | 20 | 17.800 |
| ## 103 | Associate Professor | m | Undergraduate | 2004 | fall | 20 | 19.580 |
| ## 104 | Associate Professor | m | Graduate      | 2004 | fall | 20 | 18.736 |

|        |                     |   |               |      |      |    |        |
|--------|---------------------|---|---------------|------|------|----|--------|
| ## 105 | Associate Professor | m | Graduate      | 2002 | fall | 19 | 13.896 |
| ## 106 | Associate Professor | m | Undergraduate | 2002 | fall | 19 | 13.332 |
| ## 107 | Associate Professor | m | Graduate      | 2004 | fall | 18 | 19.296 |
| ## 108 | Associate Professor | m | Graduate      | 2002 | fall | 17 | 17.752 |
| ## 109 | Associate Professor | m | Graduate      | 2002 | fall | 16 | 16.500 |
| ## 110 | Associate Professor | m | Graduate      | 2002 | fall | 17 | 17.500 |
| ## 111 | Associate Professor | m | Graduate      | 2002 | fall | 17 | 16.752 |
| ## 112 | Associate Professor | m | MBA           | 2004 | fall | 15 | 14.132 |
| ## 113 | Associate Professor | m | MBA           | 2002 | fall | 16 | 14.132 |
| ## 114 | Associate Professor | m | Graduate      | 2002 | fall | 15 | 18.932 |
| ## 115 | Associate Professor | m | Graduate      | 2002 | fall | 15 | 13.600 |
| ## 116 | Associate Professor | m | Graduate      | 2004 | fall | 15 | 19.468 |
| ## 117 | Associate Professor | m | MBA           | 2002 | fall | 15 | 14.000 |
| ## 118 | Associate Professor | m | MBA           | 2002 | fall | 15 | 18.572 |
| ## 119 | Associate Professor | m | MBA           | 2002 | fall | 15 | 17.144 |
| ## 120 | Associate Professor | m | Graduate      | 2002 | fall | 15 | 16.000 |
| ## 121 | Associate Professor | m | Graduate      | 2002 | fall | 13 | 16.308 |
| ## 122 | Associate Professor | m | Graduate      | 2004 | fall | 13 | 16.668 |
| ## 123 | Associate Professor | m | Graduate      | 2002 | fall | 12 | 20.000 |
| ## 124 | Associate Professor | m | Graduate      | 2003 | fall | 12 | 18.908 |
| ## 125 | Associate Professor | m | Graduate      | 2002 | fall | 12 | 16.364 |
| ## 126 | Associate Professor | m | Graduate      | 2002 | fall | 11 | 19.200 |
| ## 127 | Associate Professor | m | Graduate      | 2002 | fall | 9  | 17.332 |
| ## 128 | Associate Professor | m | Graduate      | 2004 | fall | 9  | 19.112 |
| ## 129 | Associate Professor | m | Graduate      | 2003 | fall | 8  | 18.000 |
| ## 130 | Associate Professor | m | Graduate      | 2002 | fall | 7  | 17.144 |
| ## 131 | Associate Professor | m | Graduate      | 2002 | fall | 6  | 19.200 |
| ## 132 | AssistantProfessor  | f | Undergraduate | 2002 | fall | 77 | 17.196 |
| ## 133 | AssistantProfessor  | f | MBA           | 2002 | fall | 74 | 16.540 |
| ## 134 | AssistantProfessor  | f | Graduate      | 2003 | fall | 69 | 18.940 |
| ## 135 | AssistantProfessor  | f | Undergraduate | 2003 | fall | 65 | 17.048 |
| ## 136 | AssistantProfessor  | f | Undergraduate | 2004 | fall | 63 | 16.888 |
| ## 137 | AssistantProfessor  | f | MBA           | 2004 | fall | 55 | 10.224 |
| ## 138 | AssistantProfessor  | f | MBA           | 2003 | fall | 53 | 16.692 |
| ## 139 | AssistantProfessor  | f | Undergraduate | 2002 | fall | 51 | 16.240 |
| ## 140 | AssistantProfessor  | f | Undergraduate | 2002 | fall | 47 | 17.044 |
| ## 141 | AssistantProfessor  | f | Graduate      | 2002 | fall | 45 | 16.976 |
| ## 142 | AssistantProfessor  | f | Graduate      | 2004 | fall | 42 | 18.476 |
| ## 143 | AssistantProfessor  | f | Undergraduate | 2004 | fall | 39 | 15.156 |
| ## 144 | AssistantProfessor  | f | Undergraduate | 2004 | fall | 35 | 16.456 |
| ## 145 | AssistantProfessor  | f | Undergraduate | 2004 | fall | 34 | 17.764 |
| ## 146 | AssistantProfessor  | f | MBA           | 2003 | fall | 33 | 13.576 |
| ## 147 | AssistantProfessor  | f | MBA           | 2002 | fall | 30 | 15.312 |
| ## 148 | AssistantProfessor  | f | Undergraduate | 2003 | fall | 30 | 17.516 |
| ## 149 | AssistantProfessor  | f | Undergraduate | 2004 | fall | 26 | 16.616 |
| ## 150 | AssistantProfessor  | f | Undergraduate | 2003 | fall | 27 | 16.000 |
| ## 151 | AssistantProfessor  | m | MBA           | 2002 | fall | 81 | 17.300 |
| ## 152 | AssistantProfessor  | m | Undergraduate | 2003 | fall | 81 | 14.800 |
| ## 153 | AssistantProfessor  | m | MBA           | 2002 | fall | 79 | 18.076 |
| ## 154 | AssistantProfessor  | m | Undergraduate | 2003 | fall | 74 | 13.040 |
| ## 155 | AssistantProfessor  | m | Undergraduate | 2002 | fall | 67 | 14.608 |
| ## 156 | AssistantProfessor  | m | MBA           | 2003 | fall | 63 | 13.904 |
| ## 157 | AssistantProfessor  | m | MBA           | 2002 | fall | 59 | 17.084 |
| ## 158 | AssistantProfessor  | m | Undergraduate | 2004 | fall | 58 | 13.864 |

|        |                    |   |                    |      |     |        |
|--------|--------------------|---|--------------------|------|-----|--------|
| ## 159 | AssistantProfessor | m | MBA 2002           | fall | 51  | 18.432 |
| ## 160 | AssistantProfessor | m | MBA 2003           | fall | 50  | 14.204 |
| ## 161 | AssistantProfessor | m | MBA 2004           | fall | 46  | 14.784 |
| ## 162 | AssistantProfessor | m | MBA 2003           | fall | 37  | 17.728 |
| ## 163 | AssistantProfessor | m | MBA 2002           | fall | 37  | 12.332 |
| ## 164 | AssistantProfessor | m | Undergraduate 2002 | fall | 35  | 18.236 |
| ## 165 | AssistantProfessor | m | Undergraduate 2004 | fall | 31  | 16.388 |
| ## 166 | AssistantProfessor | m | MBA 2004           | fall | 31  | 14.268 |
| ## 167 | AssistantProfessor | m | Undergraduate 2004 | fall | 29  | 14.208 |
| ## 168 | AssistantProfessor | m | MBA 2004           | fall | 28  | 18.428 |
| ## 169 | AssistantProfessor | m | Undergraduate 2003 | fall | 29  | 18.572 |
| ## 170 | AssistantProfessor | m | Graduate 2004      | fall | 29  | 19.284 |
| ## 171 | AssistantProfessor | m | Graduate 2004      | fall | 29  | 17.716 |
| ## 172 | AssistantProfessor | m | Graduate 2002      | fall | 27  | 18.308 |
| ## 173 | AssistantProfessor | m | MBA 2002           | fall | 22  | 14.856 |
| ## 174 | AssistantProfessor | m | Graduate 2003      | fall | 21  | 17.716 |
| ## 175 | AssistantProfessor | m | Graduate 2002      | fall | 17  | 16.252 |
| ## 176 | AssistantProfessor | m | Graduate 2002      | fall | 17  | 15.500 |
| ## 177 | AssistantProfessor | m | MBA 2003           | fall | 15  | 14.400 |
| ## 178 | AssistantProfessor | m | Graduate 2004      | fall | 15  | 18.856 |
| ## 179 | AssistantProfessor | m | Graduate 2003      | fall | 14  | 17.540 |
| ## 180 | AssistantProfessor | m | Graduate 2002      | fall | 13  | 13.848 |
| ## 181 | AssistantProfessor | m | Graduate 2003      | fall | 14  | 19.692 |
| ## 182 | AssistantProfessor | m | MBA 2002           | fall | 12  | 17.820 |
| ## 183 | AssistantProfessor | m | MBA 2004           | fall | 12  | 19.272 |
| ## 184 | AssistantProfessor | m | Undergraduate 2004 | fall | 11  | 17.820 |
| ## 185 | AssistantProfessor | m | Graduate 2002      | fall | 11  | 16.728 |
| ## 186 | AssistantProfessor | m | Graduate 2003      | fall | 11  | 18.000 |
| ## 187 | AssistantProfessor | m | Undergraduate 2003 | fall | 10  | 19.112 |
| ## 188 | AssistantProfessor | m | Graduate 2003      | fall | 8   | 18.500 |
| ## 189 | AssistantProfessor | m | Graduate 2004      | fall | 7   | 15.428 |
| ## 190 | Professor          | f | MBA 2002           | fall | 102 | 17.568 |
| ## 191 | Professor          | f | Graduate 2002      | fall | 73  | 19.500 |
| ## 192 | Professor          | f | MBA 2002           | fall | 70  | 12.800 |
| ## 193 | Professor          | f | Undergraduate 2002 | fall | 64  | 17.000 |
| ## 194 | Professor          | f | MBA 2003           | fall | 54  | 15.556 |
| ## 195 | Professor          | f | MBA 2004           | fall | 49  | 17.632 |
| ## 196 | Professor          | f | Undergraduate 2003 | fall | 49  | 18.776 |
| ## 197 | Professor          | f | MBA 2002           | fall | 45  | 16.180 |
| ## 198 | Professor          | f | Undergraduate 2002 | fall | 41  | 16.976 |
| ## 199 | Professor          | f | MBA 2002           | fall | 40  | 16.800 |
| ## 200 | Professor          | f | Undergraduate 2004 | fall | 41  | 18.000 |
| ## 201 | Professor          | f | Graduate 2004      | fall | 36  | 18.668 |
| ## 202 | Professor          | f | Graduate 2002      | fall | 35  | 18.284 |
| ## 203 | Professor          | f | Graduate 2002      | fall | 35  | 18.352 |
| ## 204 | Professor          | f | Graduate 2003      | fall | 34  | 19.528 |
| ## 205 | Professor          | f | Undergraduate 2003 | fall | 28  | 17.716 |
| ## 206 | Professor          | f | Graduate 2004      | fall | 13  | 19.668 |
| ## 207 | Professor          | f | Graduate 2002      | fall | 9   | 18.668 |
| ## 208 | Professor          | f | Graduate 2003      | fall | 7   | 18.856 |
| ## 209 | Professor          | m | MBA 2002           | fall | 136 | 17.028 |
| ## 210 | Professor          | m | MBA 2003           | fall | 135 | 18.104 |
| ## 211 | Professor          | m | MBA 2003           | fall | 131 | 18.460 |
| ## 212 | Professor          | m | Graduate 2004      | fall | 125 | 18.208 |

|        |           |   |                    |        |     |        |
|--------|-----------|---|--------------------|--------|-----|--------|
| ## 213 | Professor | m | MBA 2003           | fall   | 106 | 19.696 |
| ## 214 | Professor | m | MBA 2004           | fall   | 101 | 16.440 |
| ## 215 | Professor | m | MBA 2004           | fall   | 86  | 19.304 |
| ## 216 | Professor | m | MBA 2002           | fall   | 84  | 16.904 |
| ## 217 | Professor | m | Graduate 2004      | fall   | 83  | 17.492 |
| ## 218 | Professor | m | MBA 2004           | fall   | 82  | 18.096 |
| ## 219 | Professor | m | Graduate 2002      | fall   | 82  | 10.880 |
| ## 220 | Professor | m | Graduate 2004      | fall   | 82  | 16.636 |
| ## 221 | Professor | m | Undergraduate 2002 | fall   | 81  | 17.400 |
| ## 222 | Professor | m | Graduate 2002      | fall   | 80  | 17.112 |
| ## 223 | Professor | m | MBA 2003           | fall   | 78  | 17.076 |
| ## 224 | Professor | m | MBA 2002           | fall   | 76  | 17.172 |
| ## 225 | Professor | m | Graduate 2003      | fall   | 76  | 16.908 |
| ## 226 | Professor | m | Undergraduate 2002 | fall   | 72  | 15.000 |
| ## 227 | Professor | m | MBA 2004           | fall   | 71  | 15.492 |
| ## 228 | Professor | m | MBA 2002           | fall   | 68  | 19.176 |
| ## 229 | Professor | m | Graduate 2003      | fall   | 68  | 17.296 |
| ## 230 | Professor | m | MBA 2002           | fall   | 67  | 18.028 |
| ## 231 | Professor | m | MBA 2002           | fall   | 67  | 16.120 |
| ## 232 | Professor | m | MBA 2003           | fall   | 67  | 18.788 |
| ## 233 | Professor | m | MBA 2003           | fall   | 65  | 13.624 |
| ## 234 | Professor | m | Undergraduate 2002 | fall   | 60  | 17.732 |
| ## 235 | Professor | m | MBA 2002           | fall   | 58  | 13.864 |
| ## 236 | Professor | m | Undergraduate 2002 | fall   | 58  | 18.276 |
| ## 237 | Professor | m | Undergraduate 2002 | fall   | 56  | 17.572 |
| ## 238 | Professor | m | Undergraduate 2002 | fall   | 56  | 16.944 |
| ## 239 | Professor | m | MBA 2002           | fall   | 54  | 15.184 |
| ## 240 | Professor | m | MBA 2002           | fall   | 53  | 14.568 |
| ## 241 | Professor | m | MBA 2004           | fall   | 53  | 15.924 |
| ## 242 | Professor | m | MBA 2002           | fall   | 50  | 17.920 |
| ## 243 | Professor | m | MBA 2002           | fall   | 51  | 16.880 |
| ## 244 | Professor | m | MBA 2004           | fall   | 51  | 17.680 |
| ## 245 | Professor | m | MBA 2002           | fall   | 47  | 18.980 |
| ## 246 | Professor | m | MBA 2002           | fall   | 48  | 18.128 |
| ## 247 | Professor | m | Undergraduate 2004 | fall   | 48  | 18.892 |
| ## 248 | Professor | m | MBA 2004           | fall   | 46  | 19.044 |
| ## 249 | Professor | m | MBA 2002           | fall   | 45  | 18.756 |
| ## 250 | Professor | m | Graduate 2002      | fall   | 45  | 16.888 |
| ## 251 | Professor | m | Graduate 2002      | fall   | 45  | 18.456 |
| ## 252 | Professor | m | Graduate 2002      | fall   | 44  | 15.440 |
| ## 253 | Professor | m | Graduate 2002      | fall   | 44  | 18.512 |
| ## 254 | Professor | m | Graduate 2002      | fall   | 42  | 15.620 |
| ## 255 | Professor | m | MBA 2002           | fall   | 41  | 16.196 |
| ## 256 | Professor | m | MBA 2002           | spring | 42  | 15.708 |
| ## 257 | Professor | m | Graduate 2002      | spring | 42  | 18.244 |
| ## 258 | Professor | m | MBA 2004           | spring | 40  | 18.300 |
| ## 259 | Professor | m | Undergraduate 2004 | spring | 40  | 15.600 |
| ## 260 | Professor | m | Graduate 2002      | spring | 40  | 18.200 |
| ## 261 | Professor | m | Graduate 2003      | spring | 41  | 16.200 |
| ## 262 | Professor | m | Undergraduate 2004 | spring | 39  | 19.488 |
| ## 263 | Professor | m | MBA 2004           | spring | 38  | 18.596 |
| ## 264 | Professor | m | Undergraduate 2003 | spring | 38  | 17.836 |
| ## 265 | Professor | m | MBA 2002           | spring | 36  | 18.516 |
| ## 266 | Professor | m | Undergraduate 2002 | spring | 36  | 17.600 |

|        |           |   |                    |        |    |        |
|--------|-----------|---|--------------------|--------|----|--------|
| ## 267 | Professor | m | MBA 2002           | spring | 34 | 18.352 |
| ## 268 | Professor | m | Undergraduate 2004 | spring | 35 | 18.588 |
| ## 269 | Professor | m | Graduate 2004      | spring | 35 | 18.116 |
| ## 270 | Professor | m | Graduate 2004      | spring | 34 | 19.032 |
| ## 271 | Professor | m | MBA 2004           | spring | 32 | 18.624 |
| ## 272 | Professor | m | Graduate 2002      | spring | 32 | 17.252 |
| ## 273 | Professor | m | Graduate 2002      | spring | 33 | 19.376 |
| ## 274 | Professor | m | Graduate 2002      | spring | 32 | 18.064 |
| ## 275 | Professor | m | MBA 2002           | spring | 31 | 15.468 |
| ## 276 | Professor | m | Graduate 2004      | spring | 31 | 19.468 |
| ## 277 | Professor | m | MBA 2003           | spring | 28 | 14.572 |
| ## 278 | Professor | m | Graduate 2002      | spring | 28 | 16.144 |
| ## 279 | Professor | m | Graduate 2002      | spring | 29 | 17.144 |
| ## 280 | Professor | m | Graduate 2002      | spring | 29 | 19.572 |
| ## 281 | Professor | m | MBA 2002           | spring | 28 | 15.260 |
| ## 282 | Professor | m | Graduate 2002      | spring | 28 | 15.260 |
| ## 283 | Professor | m | Graduate 2004      | spring | 27 | 16.888 |
| ## 284 | Professor | m | MBA 2002           | spring | 26 | 16.768 |
| ## 285 | Professor | m | Graduate 2002      | spring | 27 | 16.924 |
| ## 286 | Professor | m | Graduate 2004      | spring | 27 | 16.152 |
| ## 287 | Professor | m | Graduate 2002      | spring | 26 | 17.920 |
| ## 288 | Professor | m | MBA 2003           | spring | 25 | 16.332 |
| ## 289 | Professor | m | Graduate 2003      | spring | 24 | 17.332 |
| ## 290 | Professor | m | MBA 2003           | spring | 23 | 18.608 |
| ## 291 | Professor | m | Graduate 2002      | spring | 24 | 19.132 |
| ## 292 | Professor | m | MBA 2003           | spring | 22 | 18.364 |
| ## 293 | Professor | m | MBA 2004           | spring | 23 | 19.272 |
| ## 294 | Professor | m | Graduate 2002      | spring | 22 | 18.908 |
| ## 295 | Professor | m | Graduate 2003      | spring | 23 | 16.544 |
| ## 296 | Professor | m | Graduate 2003      | spring | 21 | 18.668 |
| ## 297 | Professor | m | MBA 2002           | spring | 21 | 18.400 |
| ## 298 | Professor | m | Graduate 2002      | spring | 20 | 16.800 |
| ## 299 | Professor | m | Graduate 2002      | spring | 20 | 15.600 |
| ## 300 | Professor | m | Graduate 2002      | spring | 19 | 18.104 |
| ## 301 | Professor | m | Graduate 2002      | spring | 18 | 12.224 |
| ## 302 | Professor | m | Graduate 2002      | spring | 19 | 18.668 |
| ## 303 | Professor | m | Graduate 2002      | spring | 19 | 19.112 |
| ## 304 | Professor | m | MBA 2004           | spring | 17 | 15.764 |
| ## 305 | Professor | m | MBA 2003           | spring | 18 | 18.824 |
| ## 306 | Professor | m | Graduate 2003      | spring | 17 | 14.352 |
| ## 307 | Professor | m | Graduate 2004      | spring | 17 | 17.176 |
| ## 308 | Professor | m | Graduate 2003      | spring | 16 | 18.752 |
| ## 309 | Professor | m | Graduate 2003      | spring | 17 | 19.500 |
| ## 310 | Professor | m | Graduate 2002      | spring | 16 | 16.532 |
| ## 311 | Professor | m | Graduate 2004      | spring | 15 | 16.800 |
| ## 312 | Professor | m | Graduate 2003      | spring | 16 | 19.200 |
| ## 313 | Professor | m | Graduate 2002      | spring | 15 | 18.932 |
| ## 314 | Professor | m | Graduate 2002      | spring | 16 | 18.668 |
| ## 315 | Professor | m | MBA 2002           | spring | 15 | 15.428 |
| ## 316 | Professor | m | Graduate 2004      | spring | 15 | 19.428 |
| ## 317 | Professor | m | MBA 2002           | spring | 14 | 16.616 |
| ## 318 | Professor | m | Graduate 2004      | spring | 13 | 19.384 |
| ## 319 | Professor | m | Graduate 2002      | spring | 13 | 17.848 |
| ## 320 | Professor | m | Graduate 2002      | spring | 12 | 15.668 |

|        |           |   |          |      |        |    |        |
|--------|-----------|---|----------|------|--------|----|--------|
| ## 321 | Professor | m | Graduate | 2004 | spring | 13 | 18.332 |
| ## 322 | Professor | m | Graduate | 2002 | spring | 13 | 18.000 |
| ## 323 | Professor | m | Graduate | 2002 | spring | 12 | 15.668 |
| ## 324 | Professor | m | Graduate | 2003 | spring | 12 | 18.908 |
| ## 325 | Professor | m | MBA      | 2002 | spring | 10 | 18.000 |
| ## 326 | Professor | m | Graduate | 2004 | spring | 10 | 20.000 |
| ## 327 | Professor | m | Graduate | 2002 | spring | 11 | 16.400 |
| ## 328 | Professor | m | Graduate | 2002 | spring | 11 | 19.200 |
| ## 329 | Professor | m | Graduate | 2002 | spring | 10 | 18.400 |
| ## 330 | Professor | m | Graduate | 2002 | spring | 11 | 20.000 |
| ## 331 | Professor | m | Graduate | 2004 | spring | 11 | 18.800 |
| ## 332 | Professor | m | MBA      | 2004 | spring | 10 | 20.000 |
| ## 333 | Professor | m | Graduate | 2002 | spring | 9  | 16.888 |
| ## 334 | Professor | m | Graduate | 2002 | spring | 9  | 19.556 |
| ## 335 | Professor | m | Graduate | 2003 | spring | 7  | 19.428 |
| ## 336 | Professor | m | Graduate | 2002 | spring | 7  | 19.428 |
| ## 337 | Professor | m | Graduate | 2002 | spring | 7  | 17.332 |
| ## 338 | Professor | m | Graduate | 2004 | spring | 5  | 20.000 |
| ## 339 | Professor | m | Graduate | 2002 | spring | 5  | 19.000 |
| ## 340 | Professor | m | Graduate | 2002 | spring | 4  | 20.000 |
| ##     | smallSize |   |          |      |        |    |        |
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| ## 2   | 0         |   |          |      |        |    |        |
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| ## 10  | 0         |   |          |      |        |    |        |
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| ## 283 | 1 |
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| ## 285 | 1 |
| ## 286 | 1 |
| ## 287 | 1 |
| ## 288 | 1 |
| ## 289 | 1 |
| ## 290 | 1 |
| ## 291 | 1 |
| ## 292 | 1 |
| ## 293 | 1 |
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| ## 297 | 1 |
| ## 298 | 1 |
| ## 299 | 1 |
| ## 300 | 1 |
| ## 301 | 1 |
| ## 302 | 1 |
| ## 303 | 1 |

```
## 304      1
## 305      1
## 306      1
## 307      1
## 308      1
## 309      1
## 310      1
## 311      1
## 312      1
## 313      1
## 314      1
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## 328      1
## 329      1
## 330      1
## 331      1
## 332      1
## 333      1
## 334      1
## 335      1
## 336      1
## 337      1
## 338      1
## 339      1
## 340      1
```

```
summarize(group_by(bschool, student), mean(score))
```

```
## # A tibble: 3 x 2
##   student      'mean(score)'  
##   <chr>          <dbl>  
## 1 Graduate      17.7  
## 2 MBA           16.5  
## 3 Undergraduate 16.8
```

## Using pipe operators

```
bschool |> filter(student == "MBA") # using the filter function
```

| ##    |                     | title | gender | student | year | semester | size   | score |
|-------|---------------------|-------|--------|---------|------|----------|--------|-------|
| ## 1  | Associate Professor | f     | MBA    | 2002    | fall | 114      | 12.700 |       |
| ## 2  | Associate Professor | f     | MBA    | 2002    | fall | 89       | 14.092 |       |
| ## 3  | Associate Professor | f     | MBA    | 2003    | fall | 84       | 17.832 |       |
| ## 4  | Associate Professor | f     | MBA    | 2002    | fall | 67       | 13.880 |       |
| ## 5  | Associate Professor | f     | MBA    | 2003    | fall | 46       | 18.520 |       |
| ## 6  | Associate Professor | f     | MBA    | 2004    | fall | 45       | 18.044 |       |
| ## 7  | Associate Professor | f     | MBA    | 2002    | fall | 38       | 12.736 |       |
| ## 8  | Associate Professor | f     | MBA    | 2002    | fall | 32       | 14.192 |       |
| ## 9  | Associate Professor | f     | MBA    | 2003    | fall | 31       | 17.732 |       |
| ## 10 | Associate Professor | f     | MBA    | 2004    | fall | 30       | 17.200 |       |
| ## 11 | Associate Professor | f     | MBA    | 2003    | fall | 29       | 18.144 |       |
| ## 12 | Associate Professor | f     | MBA    | 2004    | fall | 29       | 18.144 |       |
| ## 13 | Associate Professor | f     | MBA    | 2003    | fall | 23       | 13.740 |       |
| ## 14 | Associate Professor | f     | MBA    | 2004    | fall | 20       | 19.000 |       |
| ## 15 | Associate Professor | f     | MBA    | 2004    | fall | 20       | 18.104 |       |
| ## 16 | Associate Professor | f     | MBA    | 2002    | fall | 6        | 19.332 |       |
| ## 17 | Associate Professor | m     | MBA    | 2003    | fall | 135      | 17.196 |       |
| ## 18 | Associate Professor | m     | MBA    | 2004    | fall | 121      | 18.512 |       |
| ## 19 | Associate Professor | m     | MBA    | 2003    | fall | 110      | 13.176 |       |
| ## 20 | Associate Professor | m     | MBA    | 2003    | fall | 107      | 19.020 |       |
| ## 21 | Associate Professor | m     | MBA    | 2004    | fall | 92       | 13.956 |       |
| ## 22 | Associate Professor | m     | MBA    | 2003    | fall | 92       | 16.912 |       |
| ## 23 | Associate Professor | m     | MBA    | 2002    | fall | 88       | 18.668 |       |
| ## 24 | Associate Professor | m     | MBA    | 2003    | fall | 86       | 16.000 |       |
| ## 25 | Associate Professor | m     | MBA    | 2002    | fall | 76       | 16.852 |       |
| ## 26 | Associate Professor | m     | MBA    | 2003    | fall | 67       | 12.544 |       |
| ## 27 | Associate Professor | m     | MBA    | 2002    | fall | 66       | 11.692 |       |
| ## 28 | Associate Professor | m     | MBA    | 2003    | fall | 66       | 16.308 |       |
| ## 29 | Associate Professor | m     | MBA    | 2002    | fall | 62       | 17.444 |       |
| ## 30 | Associate Professor | m     | MBA    | 2002    | fall | 62       | 16.852 |       |
| ## 31 | Associate Professor | m     | MBA    | 2002    | fall | 61       | 19.068 |       |
| ## 32 | Associate Professor | m     | MBA    | 2004    | fall | 54       | 17.208 |       |
| ## 33 | Associate Professor | m     | MBA    | 2002    | fall | 54       | 13.508 |       |
| ## 34 | Associate Professor | m     | MBA    | 2002    | fall | 52       | 18.152 |       |
| ## 35 | Associate Professor | m     | MBA    | 2002    | fall | 52       | 17.492 |       |
| ## 36 | Associate Professor | m     | MBA    | 2003    | fall | 48       | 17.916 |       |
| ## 37 | Associate Professor | m     | MBA    | 2004    | fall | 45       | 18.488 |       |
| ## 38 | Associate Professor | m     | MBA    | 2002    | fall | 42       | 13.072 |       |
| ## 39 | Associate Professor | m     | MBA    | 2003    | fall | 41       | 16.000 |       |
| ## 40 | Associate Professor | m     | MBA    | 2004    | fall | 40       | 17.300 |       |
| ## 41 | Associate Professor | m     | MBA    | 2003    | fall | 41       | 16.500 |       |
| ## 42 | Associate Professor | m     | MBA    | 2002    | fall | 37       | 10.668 |       |
| ## 43 | Associate Professor | m     | MBA    | 2004    | fall | 35       | 16.916 |       |
| ## 44 | Associate Professor | m     | MBA    | 2002    | fall | 33       | 10.544 |       |
| ## 45 | Associate Professor | m     | MBA    | 2002    | fall | 33       | 14.624 |       |
| ## 46 | Associate Professor | m     | MBA    | 2002    | fall | 33       | 15.500 |       |
| ## 47 | Associate Professor | m     | MBA    | 2004    | fall | 32       | 18.708 |       |
| ## 48 | Associate Professor | m     | MBA    | 2004    | fall | 27       | 16.296 |       |
| ## 49 | Associate Professor | m     | MBA    | 2003    | fall | 27       | 15.076 |       |
| ## 50 | Associate Professor | m     | MBA    | 2004    | fall | 24       | 18.668 |       |
| ## 51 | Associate Professor | m     | MBA    | 2004    | fall | 22       | 16.000 |       |
| ## 52 | Associate Professor | m     | MBA    | 2003    | fall | 20       | 15.400 |       |
| ## 53 | Associate Professor | m     | MBA    | 2004    | fall | 15       | 14.132 |       |

|        |                     |   |          |      |     |        |
|--------|---------------------|---|----------|------|-----|--------|
| ## 54  | Associate Professor | m | MBA 2002 | fall | 16  | 14.132 |
| ## 55  | Associate Professor | m | MBA 2002 | fall | 15  | 14.000 |
| ## 56  | Associate Professor | m | MBA 2002 | fall | 15  | 18.572 |
| ## 57  | Associate Professor | m | MBA 2002 | fall | 15  | 17.144 |
| ## 58  | AssistantProfessor  | f | MBA 2002 | fall | 74  | 16.540 |
| ## 59  | AssistantProfessor  | f | MBA 2004 | fall | 55  | 10.224 |
| ## 60  | AssistantProfessor  | f | MBA 2003 | fall | 53  | 16.692 |
| ## 61  | AssistantProfessor  | f | MBA 2003 | fall | 33  | 13.576 |
| ## 62  | AssistantProfessor  | f | MBA 2002 | fall | 30  | 15.312 |
| ## 63  | AssistantProfessor  | m | MBA 2002 | fall | 81  | 17.300 |
| ## 64  | AssistantProfessor  | m | MBA 2002 | fall | 79  | 18.076 |
| ## 65  | AssistantProfessor  | m | MBA 2003 | fall | 63  | 13.904 |
| ## 66  | AssistantProfessor  | m | MBA 2002 | fall | 59  | 17.084 |
| ## 67  | AssistantProfessor  | m | MBA 2002 | fall | 51  | 18.432 |
| ## 68  | AssistantProfessor  | m | MBA 2003 | fall | 50  | 14.204 |
| ## 69  | AssistantProfessor  | m | MBA 2004 | fall | 46  | 14.784 |
| ## 70  | AssistantProfessor  | m | MBA 2003 | fall | 37  | 17.728 |
| ## 71  | AssistantProfessor  | m | MBA 2002 | fall | 37  | 12.332 |
| ## 72  | AssistantProfessor  | m | MBA 2004 | fall | 31  | 14.268 |
| ## 73  | AssistantProfessor  | m | MBA 2004 | fall | 28  | 18.428 |
| ## 74  | AssistantProfessor  | m | MBA 2002 | fall | 22  | 14.856 |
| ## 75  | AssistantProfessor  | m | MBA 2003 | fall | 15  | 14.400 |
| ## 76  | AssistantProfessor  | m | MBA 2002 | fall | 12  | 17.820 |
| ## 77  | AssistantProfessor  | m | MBA 2004 | fall | 12  | 19.272 |
| ## 78  | Professor           | f | MBA 2002 | fall | 102 | 17.568 |
| ## 79  | Professor           | f | MBA 2002 | fall | 70  | 12.800 |
| ## 80  | Professor           | f | MBA 2003 | fall | 54  | 15.556 |
| ## 81  | Professor           | f | MBA 2004 | fall | 49  | 17.632 |
| ## 82  | Professor           | f | MBA 2002 | fall | 45  | 16.180 |
| ## 83  | Professor           | f | MBA 2002 | fall | 40  | 16.800 |
| ## 84  | Professor           | m | MBA 2002 | fall | 136 | 17.028 |
| ## 85  | Professor           | m | MBA 2003 | fall | 135 | 18.104 |
| ## 86  | Professor           | m | MBA 2003 | fall | 131 | 18.460 |
| ## 87  | Professor           | m | MBA 2003 | fall | 106 | 19.696 |
| ## 88  | Professor           | m | MBA 2004 | fall | 101 | 16.440 |
| ## 89  | Professor           | m | MBA 2004 | fall | 86  | 19.304 |
| ## 90  | Professor           | m | MBA 2002 | fall | 84  | 16.904 |
| ## 91  | Professor           | m | MBA 2004 | fall | 82  | 18.096 |
| ## 92  | Professor           | m | MBA 2003 | fall | 78  | 17.076 |
| ## 93  | Professor           | m | MBA 2002 | fall | 76  | 17.172 |
| ## 94  | Professor           | m | MBA 2004 | fall | 71  | 15.492 |
| ## 95  | Professor           | m | MBA 2002 | fall | 68  | 19.176 |
| ## 96  | Professor           | m | MBA 2002 | fall | 67  | 18.028 |
| ## 97  | Professor           | m | MBA 2002 | fall | 67  | 16.120 |
| ## 98  | Professor           | m | MBA 2003 | fall | 67  | 18.788 |
| ## 99  | Professor           | m | MBA 2003 | fall | 65  | 13.624 |
| ## 100 | Professor           | m | MBA 2002 | fall | 58  | 13.864 |
| ## 101 | Professor           | m | MBA 2002 | fall | 54  | 15.184 |
| ## 102 | Professor           | m | MBA 2002 | fall | 53  | 14.568 |
| ## 103 | Professor           | m | MBA 2004 | fall | 53  | 15.924 |
| ## 104 | Professor           | m | MBA 2002 | fall | 50  | 17.920 |
| ## 105 | Professor           | m | MBA 2002 | fall | 51  | 16.880 |
| ## 106 | Professor           | m | MBA 2004 | fall | 51  | 17.680 |
| ## 107 | Professor           | m | MBA 2002 | fall | 47  | 18.980 |

|        |           |   |          |        |    |        |
|--------|-----------|---|----------|--------|----|--------|
| ## 108 | Professor | m | MBA 2002 | fall   | 48 | 18.128 |
| ## 109 | Professor | m | MBA 2004 | fall   | 46 | 19.044 |
| ## 110 | Professor | m | MBA 2002 | fall   | 45 | 18.756 |
| ## 111 | Professor | m | MBA 2002 | fall   | 41 | 16.196 |
| ## 112 | Professor | m | MBA 2002 | spring | 42 | 15.708 |
| ## 113 | Professor | m | MBA 2004 | spring | 40 | 18.300 |
| ## 114 | Professor | m | MBA 2004 | spring | 38 | 18.596 |
| ## 115 | Professor | m | MBA 2002 | spring | 36 | 18.516 |
| ## 116 | Professor | m | MBA 2002 | spring | 34 | 18.352 |
| ## 117 | Professor | m | MBA 2004 | spring | 32 | 18.624 |
| ## 118 | Professor | m | MBA 2002 | spring | 31 | 15.468 |
| ## 119 | Professor | m | MBA 2003 | spring | 28 | 14.572 |
| ## 120 | Professor | m | MBA 2002 | spring | 28 | 15.260 |
| ## 121 | Professor | m | MBA 2002 | spring | 26 | 16.768 |
| ## 122 | Professor | m | MBA 2003 | spring | 25 | 16.332 |
| ## 123 | Professor | m | MBA 2003 | spring | 23 | 18.608 |
| ## 124 | Professor | m | MBA 2003 | spring | 22 | 18.364 |
| ## 125 | Professor | m | MBA 2004 | spring | 23 | 19.272 |
| ## 126 | Professor | m | MBA 2002 | spring | 21 | 18.400 |
| ## 127 | Professor | m | MBA 2004 | spring | 17 | 15.764 |
| ## 128 | Professor | m | MBA 2003 | spring | 18 | 18.824 |
| ## 129 | Professor | m | MBA 2002 | spring | 15 | 15.428 |
| ## 130 | Professor | m | MBA 2002 | spring | 14 | 16.616 |
| ## 131 | Professor | m | MBA 2002 | spring | 10 | 18.000 |
| ## 132 | Professor | m | MBA 2004 | spring | 10 | 20.000 |

```
bschool |> mutate(
  smallsize = 1*(size<10),
  largesize = 1*(size>100)# using the mutate function
)
```

| ##    |           | title     | gender | student       | year | semester | size | score  |
|-------|-----------|-----------|--------|---------------|------|----------|------|--------|
| ## 1  | Associate | Professor | f      | MBA           | 2002 | fall     | 114  | 12.700 |
| ## 2  | Associate | Professor | f      | MBA           | 2002 | fall     | 89   | 14.092 |
| ## 3  | Associate | Professor | f      | MBA           | 2003 | fall     | 84   | 17.832 |
| ## 4  | Associate | Professor | f      | MBA           | 2002 | fall     | 67   | 13.880 |
| ## 5  | Associate | Professor | f      | MBA           | 2003 | fall     | 46   | 18.520 |
| ## 6  | Associate | Professor | f      | MBA           | 2004 | fall     | 45   | 18.044 |
| ## 7  | Associate | Professor | f      | MBA           | 2002 | fall     | 38   | 12.736 |
| ## 8  | Associate | Professor | f      | MBA           | 2002 | fall     | 32   | 14.192 |
| ## 9  | Associate | Professor | f      | MBA           | 2003 | fall     | 31   | 17.732 |
| ## 10 | Associate | Professor | f      | MBA           | 2004 | fall     | 30   | 17.200 |
| ## 11 | Associate | Professor | f      | Graduate      | 2002 | fall     | 29   | 16.412 |
| ## 12 | Associate | Professor | f      | MBA           | 2003 | fall     | 29   | 18.144 |
| ## 13 | Associate | Professor | f      | MBA           | 2004 | fall     | 29   | 18.144 |
| ## 14 | Associate | Professor | f      | Undergraduate | 2004 | fall     | 28   | 15.852 |
| ## 15 | Associate | Professor | f      | MBA           | 2003 | fall     | 23   | 13.740 |
| ## 16 | Associate | Professor | f      | Undergraduate | 2003 | fall     | 22   | 16.544 |
| ## 17 | Associate | Professor | f      | MBA           | 2004 | fall     | 20   | 19.000 |
| ## 18 | Associate | Professor | f      | MBA           | 2004 | fall     | 20   | 18.104 |
| ## 19 | Associate | Professor | f      | Graduate      | 2003 | fall     | 19   | 17.556 |
| ## 20 | Associate | Professor | f      | Graduate      | 2004 | fall     | 15   | 18.856 |
| ## 21 | Associate | Professor | f      | MBA           | 2002 | fall     | 6    | 19.332 |
| ## 22 | Associate | Professor | m      | MBA           | 2003 | fall     | 135  | 17.196 |

|       |                     |                 |               |      |     |        |
|-------|---------------------|-----------------|---------------|------|-----|--------|
| ## 23 | Associate Professor | m Undergraduate | 2002          | fall | 130 | 17.924 |
| ## 24 | Associate Professor | m               | MBA 2004      | fall | 121 | 18.512 |
| ## 25 | Associate Professor | m Undergraduate | 2002          | fall | 119 | 16.780 |
| ## 26 | Associate Professor | m Undergraduate | 2003          | fall | 113 | 18.964 |
| ## 27 | Associate Professor | m               | MBA 2003      | fall | 110 | 13.176 |
| ## 28 | Associate Professor | m               | MBA 2003      | fall | 107 | 19.020 |
| ## 29 | Associate Professor | m Undergraduate | 2002          | fall | 98  | 16.452 |
| ## 30 | Associate Professor | m               | MBA 2004      | fall | 92  | 13.956 |
| ## 31 | Associate Professor | m               | MBA 2003      | fall | 92  | 16.912 |
| ## 32 | Associate Professor | m Undergraduate | 2003          | fall | 91  | 16.312 |
| ## 33 | Associate Professor | m               | MBA 2002      | fall | 88  | 18.668 |
| ## 34 | Associate Professor | m               | Graduate 2003 | fall | 88  | 18.436 |
| ## 35 | Associate Professor | m               | MBA 2003      | fall | 86  | 16.000 |
| ## 36 | Associate Professor | m               | Graduate 2002 | fall | 82  | 18.780 |
| ## 37 | Associate Professor | m               | MBA 2002      | fall | 76  | 16.852 |
| ## 38 | Associate Professor | m               | Graduate 2003 | fall | 75  | 18.488 |
| ## 39 | Associate Professor | m               | Graduate 2002 | fall | 72  | 18.424 |
| ## 40 | Associate Professor | m               | Graduate 2004 | fall | 70  | 14.628 |
| ## 41 | Associate Professor | m Undergraduate | 2004          | fall | 68  | 17.972 |
| ## 42 | Associate Professor | m               | MBA 2003      | fall | 67  | 12.544 |
| ## 43 | Associate Professor | m               | MBA 2002      | fall | 66  | 11.692 |
| ## 44 | Associate Professor | m               | MBA 2003      | fall | 66  | 16.308 |
| ## 45 | Associate Professor | m               | Graduate 2004 | fall | 63  | 18.452 |
| ## 46 | Associate Professor | m               | Graduate 2004 | fall | 62  | 19.096 |
| ## 47 | Associate Professor | m               | MBA 2002      | fall | 62  | 17.444 |
| ## 48 | Associate Professor | m               | MBA 2002      | fall | 62  | 16.852 |
| ## 49 | Associate Professor | m               | MBA 2002      | fall | 61  | 19.068 |
| ## 50 | Associate Professor | m               | Graduate 2002 | fall | 59  | 18.068 |
| ## 51 | Associate Professor | m               | Graduate 2002 | fall | 58  | 16.280 |
| ## 52 | Associate Professor | m               | Graduate 2004 | fall | 57  | 18.456 |
| ## 53 | Associate Professor | m Undergraduate | 2003          | fall | 54  | 16.964 |
| ## 54 | Associate Professor | m               | MBA 2004      | fall | 54  | 17.208 |
| ## 55 | Associate Professor | m               | MBA 2002      | fall | 54  | 13.508 |
| ## 56 | Associate Professor | m               | MBA 2002      | fall | 52  | 18.152 |
| ## 57 | Associate Professor | m               | MBA 2002      | fall | 52  | 17.492 |
| ## 58 | Associate Professor | m               | MBA 2003      | fall | 48  | 17.916 |
| ## 59 | Associate Professor | m               | Graduate 2002 | fall | 48  | 17.360 |
| ## 60 | Associate Professor | m               | Graduate 2003 | fall | 48  | 17.532 |
| ## 61 | Associate Professor | m Undergraduate | 2002          | fall | 46  | 15.392 |
| ## 62 | Associate Professor | m               | MBA 2004      | fall | 45  | 18.488 |
| ## 63 | Associate Professor | m Undergraduate | 2003          | fall | 46  | 17.156 |
| ## 64 | Associate Professor | m               | MBA 2002      | fall | 42  | 13.072 |
| ## 65 | Associate Professor | m               | MBA 2003      | fall | 41  | 16.000 |
| ## 66 | Associate Professor | m               | MBA 2004      | fall | 40  | 17.300 |
| ## 67 | Associate Professor | m               | MBA 2003      | fall | 41  | 16.500 |
| ## 68 | Associate Professor | m Undergraduate | 2004          | fall | 41  | 12.900 |
| ## 69 | Associate Professor | m Undergraduate | 2002          | fall | 40  | 16.000 |
| ## 70 | Associate Professor | m               | Graduate 2003 | fall | 39  | 14.564 |
| ## 71 | Associate Professor | m               | Graduate 2002 | fall | 40  | 17.948 |
| ## 72 | Associate Professor | m               | Graduate 2003 | fall | 39  | 16.924 |
| ## 73 | Associate Professor | m Undergraduate | 2004          | fall | 39  | 16.844 |
| ## 74 | Associate Professor | m Undergraduate | 2002          | fall | 38  | 16.864 |
| ## 75 | Associate Professor | m               | MBA 2002      | fall | 37  | 10.668 |
| ## 76 | Associate Professor | m               | MBA 2004      | fall | 35  | 16.916 |

|        |                     |   |               |      |      |    |        |
|--------|---------------------|---|---------------|------|------|----|--------|
| ## 77  | Associate Professor | m | Undergraduate | 2002 | fall | 36 | 18.056 |
| ## 78  | Associate Professor | m | Graduate      | 2003 | fall | 36 | 18.856 |
| ## 79  | Associate Professor | m | Graduate      | 2004 | fall | 35 | 17.764 |
| ## 80  | Associate Professor | m | Graduate      | 2003 | fall | 34 | 18.824 |
| ## 81  | Associate Professor | m | MBA           | 2002 | fall | 33 | 10.544 |
| ## 82  | Associate Professor | m | Undergraduate | 2003 | fall | 33 | 16.364 |
| ## 83  | Associate Professor | m | MBA           | 2002 | fall | 33 | 14.624 |
| ## 84  | Associate Professor | m | MBA           | 2002 | fall | 33 | 15.500 |
| ## 85  | Associate Professor | m | Undergraduate | 2002 | fall | 32 | 17.500 |
| ## 86  | Associate Professor | m | Graduate      | 2002 | fall | 33 | 17.752 |
| ## 87  | Associate Professor | m | MBA           | 2004 | fall | 32 | 18.708 |
| ## 88  | Associate Professor | m | Undergraduate | 2004 | fall | 28 | 18.144 |
| ## 89  | Associate Professor | m | Graduate      | 2002 | fall | 28 | 16.856 |
| ## 90  | Associate Professor | m | MBA           | 2004 | fall | 27 | 16.296 |
| ## 91  | Associate Professor | m | MBA           | 2003 | fall | 27 | 15.076 |
| ## 92  | Associate Professor | m | Undergraduate | 2002 | fall | 26 | 16.152 |
| ## 93  | Associate Professor | m | Undergraduate | 2002 | fall | 26 | 13.280 |
| ## 94  | Associate Professor | m | MBA           | 2004 | fall | 24 | 18.668 |
| ## 95  | Associate Professor | m | Undergraduate | 2004 | fall | 25 | 16.332 |
| ## 96  | Associate Professor | m | Graduate      | 2004 | fall | 25 | 18.500 |
| ## 97  | Associate Professor | m | Graduate      | 2002 | fall | 24 | 17.740 |
| ## 98  | Associate Professor | m | MBA           | 2004 | fall | 22 | 16.000 |
| ## 99  | Associate Professor | m | Undergraduate | 2003 | fall | 22 | 13.456 |
| ## 100 | Associate Professor | m | Undergraduate | 2003 | fall | 21 | 17.144 |
| ## 101 | Associate Professor | m | MBA           | 2003 | fall | 20 | 15.400 |
| ## 102 | Associate Professor | m | Graduate      | 2004 | fall | 20 | 17.800 |
| ## 103 | Associate Professor | m | Undergraduate | 2004 | fall | 20 | 19.580 |
| ## 104 | Associate Professor | m | Graduate      | 2004 | fall | 20 | 18.736 |
| ## 105 | Associate Professor | m | Graduate      | 2002 | fall | 19 | 13.896 |
| ## 106 | Associate Professor | m | Undergraduate | 2002 | fall | 19 | 13.332 |
| ## 107 | Associate Professor | m | Graduate      | 2004 | fall | 18 | 19.296 |
| ## 108 | Associate Professor | m | Graduate      | 2002 | fall | 17 | 17.752 |
| ## 109 | Associate Professor | m | Graduate      | 2002 | fall | 16 | 16.500 |
| ## 110 | Associate Professor | m | Graduate      | 2002 | fall | 17 | 17.500 |
| ## 111 | Associate Professor | m | Graduate      | 2002 | fall | 17 | 16.752 |
| ## 112 | Associate Professor | m | MBA           | 2004 | fall | 15 | 14.132 |
| ## 113 | Associate Professor | m | MBA           | 2002 | fall | 16 | 14.132 |
| ## 114 | Associate Professor | m | Graduate      | 2002 | fall | 15 | 18.932 |
| ## 115 | Associate Professor | m | Graduate      | 2002 | fall | 15 | 13.600 |
| ## 116 | Associate Professor | m | Graduate      | 2004 | fall | 15 | 19.468 |
| ## 117 | Associate Professor | m | MBA           | 2002 | fall | 15 | 14.000 |
| ## 118 | Associate Professor | m | MBA           | 2002 | fall | 15 | 18.572 |
| ## 119 | Associate Professor | m | MBA           | 2002 | fall | 15 | 17.144 |
| ## 120 | Associate Professor | m | Graduate      | 2002 | fall | 15 | 16.000 |
| ## 121 | Associate Professor | m | Graduate      | 2002 | fall | 13 | 16.308 |
| ## 122 | Associate Professor | m | Graduate      | 2004 | fall | 13 | 16.668 |
| ## 123 | Associate Professor | m | Graduate      | 2002 | fall | 12 | 20.000 |
| ## 124 | Associate Professor | m | Graduate      | 2003 | fall | 12 | 18.908 |
| ## 125 | Associate Professor | m | Graduate      | 2002 | fall | 12 | 16.364 |
| ## 126 | Associate Professor | m | Graduate      | 2002 | fall | 11 | 19.200 |
| ## 127 | Associate Professor | m | Graduate      | 2002 | fall | 9  | 17.332 |
| ## 128 | Associate Professor | m | Graduate      | 2004 | fall | 9  | 19.112 |
| ## 129 | Associate Professor | m | Graduate      | 2003 | fall | 8  | 18.000 |
| ## 130 | Associate Professor | m | Graduate      | 2002 | fall | 7  | 17.144 |

|        |                     |   |               |      |      |    |        |
|--------|---------------------|---|---------------|------|------|----|--------|
| ## 131 | Associate Professor | m | Graduate      | 2002 | fall | 6  | 19.200 |
| ## 132 | AssistantProfessor  | f | Undergraduate | 2002 | fall | 77 | 17.196 |
| ## 133 | AssistantProfessor  | f | MBA           | 2002 | fall | 74 | 16.540 |
| ## 134 | AssistantProfessor  | f | Graduate      | 2003 | fall | 69 | 18.940 |
| ## 135 | AssistantProfessor  | f | Undergraduate | 2003 | fall | 65 | 17.048 |
| ## 136 | AssistantProfessor  | f | Undergraduate | 2004 | fall | 63 | 16.888 |
| ## 137 | AssistantProfessor  | f | MBA           | 2004 | fall | 55 | 10.224 |
| ## 138 | AssistantProfessor  | f | MBA           | 2003 | fall | 53 | 16.692 |
| ## 139 | AssistantProfessor  | f | Undergraduate | 2002 | fall | 51 | 16.240 |
| ## 140 | AssistantProfessor  | f | Undergraduate | 2002 | fall | 47 | 17.044 |
| ## 141 | AssistantProfessor  | f | Graduate      | 2002 | fall | 45 | 16.976 |
| ## 142 | AssistantProfessor  | f | Graduate      | 2004 | fall | 42 | 18.476 |
| ## 143 | AssistantProfessor  | f | Undergraduate | 2004 | fall | 39 | 15.156 |
| ## 144 | AssistantProfessor  | f | Undergraduate | 2004 | fall | 35 | 16.456 |
| ## 145 | AssistantProfessor  | f | Undergraduate | 2004 | fall | 34 | 17.764 |
| ## 146 | AssistantProfessor  | f | MBA           | 2003 | fall | 33 | 13.576 |
| ## 147 | AssistantProfessor  | f | MBA           | 2002 | fall | 30 | 15.312 |
| ## 148 | AssistantProfessor  | f | Undergraduate | 2003 | fall | 30 | 17.516 |
| ## 149 | AssistantProfessor  | f | Undergraduate | 2004 | fall | 26 | 16.616 |
| ## 150 | AssistantProfessor  | f | Undergraduate | 2003 | fall | 27 | 16.000 |
| ## 151 | AssistantProfessor  | m | MBA           | 2002 | fall | 81 | 17.300 |
| ## 152 | AssistantProfessor  | m | Undergraduate | 2003 | fall | 81 | 14.800 |
| ## 153 | AssistantProfessor  | m | MBA           | 2002 | fall | 79 | 18.076 |
| ## 154 | AssistantProfessor  | m | Undergraduate | 2003 | fall | 74 | 13.040 |
| ## 155 | AssistantProfessor  | m | Undergraduate | 2002 | fall | 67 | 14.608 |
| ## 156 | AssistantProfessor  | m | MBA           | 2003 | fall | 63 | 13.904 |
| ## 157 | AssistantProfessor  | m | MBA           | 2002 | fall | 59 | 17.084 |
| ## 158 | AssistantProfessor  | m | Undergraduate | 2004 | fall | 58 | 13.864 |
| ## 159 | AssistantProfessor  | m | MBA           | 2002 | fall | 51 | 18.432 |
| ## 160 | AssistantProfessor  | m | MBA           | 2003 | fall | 50 | 14.204 |
| ## 161 | AssistantProfessor  | m | MBA           | 2004 | fall | 46 | 14.784 |
| ## 162 | AssistantProfessor  | m | MBA           | 2003 | fall | 37 | 17.728 |
| ## 163 | AssistantProfessor  | m | MBA           | 2002 | fall | 37 | 12.332 |
| ## 164 | AssistantProfessor  | m | Undergraduate | 2002 | fall | 35 | 18.236 |
| ## 165 | AssistantProfessor  | m | Undergraduate | 2004 | fall | 31 | 16.388 |
| ## 166 | AssistantProfessor  | m | MBA           | 2004 | fall | 31 | 14.268 |
| ## 167 | AssistantProfessor  | m | Undergraduate | 2004 | fall | 29 | 14.208 |
| ## 168 | AssistantProfessor  | m | MBA           | 2004 | fall | 28 | 18.428 |
| ## 169 | AssistantProfessor  | m | Undergraduate | 2003 | fall | 29 | 18.572 |
| ## 170 | AssistantProfessor  | m | Graduate      | 2004 | fall | 29 | 19.284 |
| ## 171 | AssistantProfessor  | m | Graduate      | 2004 | fall | 29 | 17.716 |
| ## 172 | AssistantProfessor  | m | Graduate      | 2002 | fall | 27 | 18.308 |
| ## 173 | AssistantProfessor  | m | MBA           | 2002 | fall | 22 | 14.856 |
| ## 174 | AssistantProfessor  | m | Graduate      | 2003 | fall | 21 | 17.716 |
| ## 175 | AssistantProfessor  | m | Graduate      | 2002 | fall | 17 | 16.252 |
| ## 176 | AssistantProfessor  | m | Graduate      | 2002 | fall | 17 | 15.500 |
| ## 177 | AssistantProfessor  | m | MBA           | 2003 | fall | 15 | 14.400 |
| ## 178 | AssistantProfessor  | m | Graduate      | 2004 | fall | 15 | 18.856 |
| ## 179 | AssistantProfessor  | m | Graduate      | 2003 | fall | 14 | 17.540 |
| ## 180 | AssistantProfessor  | m | Graduate      | 2002 | fall | 13 | 13.848 |
| ## 181 | AssistantProfessor  | m | Graduate      | 2003 | fall | 14 | 19.692 |
| ## 182 | AssistantProfessor  | m | MBA           | 2002 | fall | 12 | 17.820 |
| ## 183 | AssistantProfessor  | m | MBA           | 2004 | fall | 12 | 19.272 |
| ## 184 | AssistantProfessor  | m | Undergraduate | 2004 | fall | 11 | 17.820 |

|        |                    |   |               |      |      |     |        |
|--------|--------------------|---|---------------|------|------|-----|--------|
| ## 185 | AssistantProfessor | m | Graduate      | 2002 | fall | 11  | 16.728 |
| ## 186 | AssistantProfessor | m | Graduate      | 2003 | fall | 11  | 18.000 |
| ## 187 | AssistantProfessor | m | Undergraduate | 2003 | fall | 10  | 19.112 |
| ## 188 | AssistantProfessor | m | Graduate      | 2003 | fall | 8   | 18.500 |
| ## 189 | AssistantProfessor | m | Graduate      | 2004 | fall | 7   | 15.428 |
| ## 190 | Professor          | f | MBA           | 2002 | fall | 102 | 17.568 |
| ## 191 | Professor          | f | Graduate      | 2002 | fall | 73  | 19.500 |
| ## 192 | Professor          | f | MBA           | 2002 | fall | 70  | 12.800 |
| ## 193 | Professor          | f | Undergraduate | 2002 | fall | 64  | 17.000 |
| ## 194 | Professor          | f | MBA           | 2003 | fall | 54  | 15.556 |
| ## 195 | Professor          | f | MBA           | 2004 | fall | 49  | 17.632 |
| ## 196 | Professor          | f | Undergraduate | 2003 | fall | 49  | 18.776 |
| ## 197 | Professor          | f | MBA           | 2002 | fall | 45  | 16.180 |
| ## 198 | Professor          | f | Undergraduate | 2002 | fall | 41  | 16.976 |
| ## 199 | Professor          | f | MBA           | 2002 | fall | 40  | 16.800 |
| ## 200 | Professor          | f | Undergraduate | 2004 | fall | 41  | 18.000 |
| ## 201 | Professor          | f | Graduate      | 2004 | fall | 36  | 18.668 |
| ## 202 | Professor          | f | Graduate      | 2002 | fall | 35  | 18.284 |
| ## 203 | Professor          | f | Graduate      | 2002 | fall | 35  | 18.352 |
| ## 204 | Professor          | f | Graduate      | 2003 | fall | 34  | 19.528 |
| ## 205 | Professor          | f | Undergraduate | 2003 | fall | 28  | 17.716 |
| ## 206 | Professor          | f | Graduate      | 2004 | fall | 13  | 19.668 |
| ## 207 | Professor          | f | Graduate      | 2002 | fall | 9   | 18.668 |
| ## 208 | Professor          | f | Graduate      | 2003 | fall | 7   | 18.856 |
| ## 209 | Professor          | m | MBA           | 2002 | fall | 136 | 17.028 |
| ## 210 | Professor          | m | MBA           | 2003 | fall | 135 | 18.104 |
| ## 211 | Professor          | m | MBA           | 2003 | fall | 131 | 18.460 |
| ## 212 | Professor          | m | Graduate      | 2004 | fall | 125 | 18.208 |
| ## 213 | Professor          | m | MBA           | 2003 | fall | 106 | 19.696 |
| ## 214 | Professor          | m | MBA           | 2004 | fall | 101 | 16.440 |
| ## 215 | Professor          | m | MBA           | 2004 | fall | 86  | 19.304 |
| ## 216 | Professor          | m | MBA           | 2002 | fall | 84  | 16.904 |
| ## 217 | Professor          | m | Graduate      | 2004 | fall | 83  | 17.492 |
| ## 218 | Professor          | m | MBA           | 2004 | fall | 82  | 18.096 |
| ## 219 | Professor          | m | Graduate      | 2002 | fall | 82  | 10.880 |
| ## 220 | Professor          | m | Graduate      | 2004 | fall | 82  | 16.636 |
| ## 221 | Professor          | m | Undergraduate | 2002 | fall | 81  | 17.400 |
| ## 222 | Professor          | m | Graduate      | 2002 | fall | 80  | 17.112 |
| ## 223 | Professor          | m | MBA           | 2003 | fall | 78  | 17.076 |
| ## 224 | Professor          | m | MBA           | 2002 | fall | 76  | 17.172 |
| ## 225 | Professor          | m | Graduate      | 2003 | fall | 76  | 16.908 |
| ## 226 | Professor          | m | Undergraduate | 2002 | fall | 72  | 15.000 |
| ## 227 | Professor          | m | MBA           | 2004 | fall | 71  | 15.492 |
| ## 228 | Professor          | m | MBA           | 2002 | fall | 68  | 19.176 |
| ## 229 | Professor          | m | Graduate      | 2003 | fall | 68  | 17.296 |
| ## 230 | Professor          | m | MBA           | 2002 | fall | 67  | 18.028 |
| ## 231 | Professor          | m | MBA           | 2002 | fall | 67  | 16.120 |
| ## 232 | Professor          | m | MBA           | 2003 | fall | 67  | 18.788 |
| ## 233 | Professor          | m | MBA           | 2003 | fall | 65  | 13.624 |
| ## 234 | Professor          | m | Undergraduate | 2002 | fall | 60  | 17.732 |
| ## 235 | Professor          | m | MBA           | 2002 | fall | 58  | 13.864 |
| ## 236 | Professor          | m | Undergraduate | 2002 | fall | 58  | 18.276 |
| ## 237 | Professor          | m | Undergraduate | 2002 | fall | 56  | 17.572 |
| ## 238 | Professor          | m | Undergraduate | 2002 | fall | 56  | 16.944 |

|        |           |   |                    |        |    |        |
|--------|-----------|---|--------------------|--------|----|--------|
| ## 239 | Professor | m | MBA 2002           | fall   | 54 | 15.184 |
| ## 240 | Professor | m | MBA 2002           | fall   | 53 | 14.568 |
| ## 241 | Professor | m | MBA 2004           | fall   | 53 | 15.924 |
| ## 242 | Professor | m | MBA 2002           | fall   | 50 | 17.920 |
| ## 243 | Professor | m | MBA 2002           | fall   | 51 | 16.880 |
| ## 244 | Professor | m | MBA 2004           | fall   | 51 | 17.680 |
| ## 245 | Professor | m | MBA 2002           | fall   | 47 | 18.980 |
| ## 246 | Professor | m | MBA 2002           | fall   | 48 | 18.128 |
| ## 247 | Professor | m | Undergraduate 2004 | fall   | 48 | 18.892 |
| ## 248 | Professor | m | MBA 2004           | fall   | 46 | 19.044 |
| ## 249 | Professor | m | MBA 2002           | fall   | 45 | 18.756 |
| ## 250 | Professor | m | Graduate 2002      | fall   | 45 | 16.888 |
| ## 251 | Professor | m | Graduate 2002      | fall   | 45 | 18.456 |
| ## 252 | Professor | m | Graduate 2002      | fall   | 44 | 15.440 |
| ## 253 | Professor | m | Graduate 2002      | fall   | 44 | 18.512 |
| ## 254 | Professor | m | Graduate 2002      | fall   | 42 | 15.620 |
| ## 255 | Professor | m | MBA 2002           | fall   | 41 | 16.196 |
| ## 256 | Professor | m | MBA 2002           | spring | 42 | 15.708 |
| ## 257 | Professor | m | Graduate 2002      | spring | 42 | 18.244 |
| ## 258 | Professor | m | MBA 2004           | spring | 40 | 18.300 |
| ## 259 | Professor | m | Undergraduate 2004 | spring | 40 | 15.600 |
| ## 260 | Professor | m | Graduate 2002      | spring | 40 | 18.200 |
| ## 261 | Professor | m | Graduate 2003      | spring | 41 | 16.200 |
| ## 262 | Professor | m | Undergraduate 2004 | spring | 39 | 19.488 |
| ## 263 | Professor | m | MBA 2004           | spring | 38 | 18.596 |
| ## 264 | Professor | m | Undergraduate 2003 | spring | 38 | 17.836 |
| ## 265 | Professor | m | MBA 2002           | spring | 36 | 18.516 |
| ## 266 | Professor | m | Undergraduate 2002 | spring | 36 | 17.600 |
| ## 267 | Professor | m | MBA 2002           | spring | 34 | 18.352 |
| ## 268 | Professor | m | Undergraduate 2004 | spring | 35 | 18.588 |
| ## 269 | Professor | m | Graduate 2004      | spring | 35 | 18.116 |
| ## 270 | Professor | m | Graduate 2004      | spring | 34 | 19.032 |
| ## 271 | Professor | m | MBA 2004           | spring | 32 | 18.624 |
| ## 272 | Professor | m | Graduate 2002      | spring | 32 | 17.252 |
| ## 273 | Professor | m | Graduate 2002      | spring | 33 | 19.376 |
| ## 274 | Professor | m | Graduate 2002      | spring | 32 | 18.064 |
| ## 275 | Professor | m | MBA 2002           | spring | 31 | 15.468 |
| ## 276 | Professor | m | Graduate 2004      | spring | 31 | 19.468 |
| ## 277 | Professor | m | MBA 2003           | spring | 28 | 14.572 |
| ## 278 | Professor | m | Graduate 2002      | spring | 28 | 16.144 |
| ## 279 | Professor | m | Graduate 2002      | spring | 29 | 17.144 |
| ## 280 | Professor | m | Graduate 2002      | spring | 29 | 19.572 |
| ## 281 | Professor | m | MBA 2002           | spring | 28 | 15.260 |
| ## 282 | Professor | m | Graduate 2002      | spring | 28 | 15.260 |
| ## 283 | Professor | m | Graduate 2004      | spring | 27 | 16.888 |
| ## 284 | Professor | m | MBA 2002           | spring | 26 | 16.768 |
| ## 285 | Professor | m | Graduate 2002      | spring | 27 | 16.924 |
| ## 286 | Professor | m | Graduate 2004      | spring | 27 | 16.152 |
| ## 287 | Professor | m | Graduate 2002      | spring | 26 | 17.920 |
| ## 288 | Professor | m | MBA 2003           | spring | 25 | 16.332 |
| ## 289 | Professor | m | Graduate 2003      | spring | 24 | 17.332 |
| ## 290 | Professor | m | MBA 2003           | spring | 23 | 18.608 |
| ## 291 | Professor | m | Graduate 2002      | spring | 24 | 19.132 |
| ## 292 | Professor | m | MBA 2003           | spring | 22 | 18.364 |

|        |           |           |               |        |    |        |
|--------|-----------|-----------|---------------|--------|----|--------|
| ## 293 | Professor | m         | MBA 2004      | spring | 23 | 19.272 |
| ## 294 | Professor | m         | Graduate 2002 | spring | 22 | 18.908 |
| ## 295 | Professor | m         | Graduate 2003 | spring | 23 | 16.544 |
| ## 296 | Professor | m         | Graduate 2003 | spring | 21 | 18.668 |
| ## 297 | Professor | m         | MBA 2002      | spring | 21 | 18.400 |
| ## 298 | Professor | m         | Graduate 2002 | spring | 20 | 16.800 |
| ## 299 | Professor | m         | Graduate 2002 | spring | 20 | 15.600 |
| ## 300 | Professor | m         | Graduate 2002 | spring | 19 | 18.104 |
| ## 301 | Professor | m         | Graduate 2002 | spring | 18 | 12.224 |
| ## 302 | Professor | m         | Graduate 2002 | spring | 19 | 18.668 |
| ## 303 | Professor | m         | Graduate 2002 | spring | 19 | 19.112 |
| ## 304 | Professor | m         | MBA 2004      | spring | 17 | 15.764 |
| ## 305 | Professor | m         | MBA 2003      | spring | 18 | 18.824 |
| ## 306 | Professor | m         | Graduate 2003 | spring | 17 | 14.352 |
| ## 307 | Professor | m         | Graduate 2004 | spring | 17 | 17.176 |
| ## 308 | Professor | m         | Graduate 2003 | spring | 16 | 18.752 |
| ## 309 | Professor | m         | Graduate 2003 | spring | 17 | 19.500 |
| ## 310 | Professor | m         | Graduate 2002 | spring | 16 | 16.532 |
| ## 311 | Professor | m         | Graduate 2004 | spring | 15 | 16.800 |
| ## 312 | Professor | m         | Graduate 2003 | spring | 16 | 19.200 |
| ## 313 | Professor | m         | Graduate 2002 | spring | 15 | 18.932 |
| ## 314 | Professor | m         | Graduate 2002 | spring | 16 | 18.668 |
| ## 315 | Professor | m         | MBA 2002      | spring | 15 | 15.428 |
| ## 316 | Professor | m         | Graduate 2004 | spring | 15 | 19.428 |
| ## 317 | Professor | m         | MBA 2002      | spring | 14 | 16.616 |
| ## 318 | Professor | m         | Graduate 2004 | spring | 13 | 19.384 |
| ## 319 | Professor | m         | Graduate 2002 | spring | 13 | 17.848 |
| ## 320 | Professor | m         | Graduate 2002 | spring | 12 | 15.668 |
| ## 321 | Professor | m         | Graduate 2004 | spring | 13 | 18.332 |
| ## 322 | Professor | m         | Graduate 2002 | spring | 13 | 18.000 |
| ## 323 | Professor | m         | Graduate 2002 | spring | 12 | 15.668 |
| ## 324 | Professor | m         | Graduate 2003 | spring | 12 | 18.908 |
| ## 325 | Professor | m         | MBA 2002      | spring | 10 | 18.000 |
| ## 326 | Professor | m         | Graduate 2004 | spring | 10 | 20.000 |
| ## 327 | Professor | m         | Graduate 2002 | spring | 11 | 16.400 |
| ## 328 | Professor | m         | Graduate 2002 | spring | 11 | 19.200 |
| ## 329 | Professor | m         | Graduate 2002 | spring | 10 | 18.400 |
| ## 330 | Professor | m         | Graduate 2002 | spring | 11 | 20.000 |
| ## 331 | Professor | m         | Graduate 2004 | spring | 11 | 18.800 |
| ## 332 | Professor | m         | MBA 2004      | spring | 10 | 20.000 |
| ## 333 | Professor | m         | Graduate 2002 | spring | 9  | 16.888 |
| ## 334 | Professor | m         | Graduate 2002 | spring | 9  | 19.556 |
| ## 335 | Professor | m         | Graduate 2003 | spring | 7  | 19.428 |
| ## 336 | Professor | m         | Graduate 2002 | spring | 7  | 19.428 |
| ## 337 | Professor | m         | Graduate 2002 | spring | 7  | 17.332 |
| ## 338 | Professor | m         | Graduate 2004 | spring | 5  | 20.000 |
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| ## 340 | Professor | m         | Graduate 2002 | spring | 4  | 20.000 |
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| ## 323 | 0 | 0 |
| ## 324 | 0 | 0 |
| ## 325 | 0 | 0 |
| ## 326 | 0 | 0 |
| ## 327 | 0 | 0 |
| ## 328 | 0 | 0 |
| ## 329 | 0 | 0 |

```
## 330      0      0
## 331      0      0
## 332      0      0
## 333      1      0
## 334      1      0
## 335      1      0
## 336      1      0
## 337      1      0
## 338      1      0
## 339      1      0
## 340      1      0
```

```
# summarize
bschool |>
  group_by(gender, student)|>
  summarize(
    mean_score = mean(score),
    sd_score = sd(score),
    n = length(score)
  )
```

```
## 'summarise()' has regrouped the output.
## i Summaries were computed grouped by gender and student.
## i Output is grouped by gender.
## i Use 'summarise(groups = "drop_last")' to silence this message.
## i Use 'summarise(.by = c(gender, student))' for per-operation grouping
## ('?dplyr::dplyr_by') instead.
```

```
## # A tibble: 6 x 5
## # Groups:   gender [2]
##   gender student      mean_score sd_score      n
##   <chr>   <chr>          <dbl>    <dbl> <int>
## 1 f      Graduate        18.5     0.943    14
## 2 f      MBA             16.0     2.38     27
## 3 f      Undergraduate    16.9     0.855    18
## 4 m      Graduate        17.7     1.63    130
## 5 m      MBA             16.6     2.06   105
## 6 m      Undergraduate    16.7     1.82    46
```

```
# group_by function: how many MBA courses were taught by female associate professor
# average score for graduate courses taught by male assistant professor
bschool |> group_by(gender, student, title)|> count()
```

```
## # A tibble: 18 x 4
## # Groups:   gender, student, title [18]
##   gender student      title      n
##   <chr>   <chr>          <chr>    <int>
## 1 f      Graduate    " Professor"      8
## 2 f      Graduate    "AssistantProfessor"  3
## 3 f      Graduate    "Associate Professor"  3
## 4 f      MBA         " Professor"      6
## 5 f      MBA         "AssistantProfessor"  5
## 6 f      MBA         "Associate Professor" 16
```

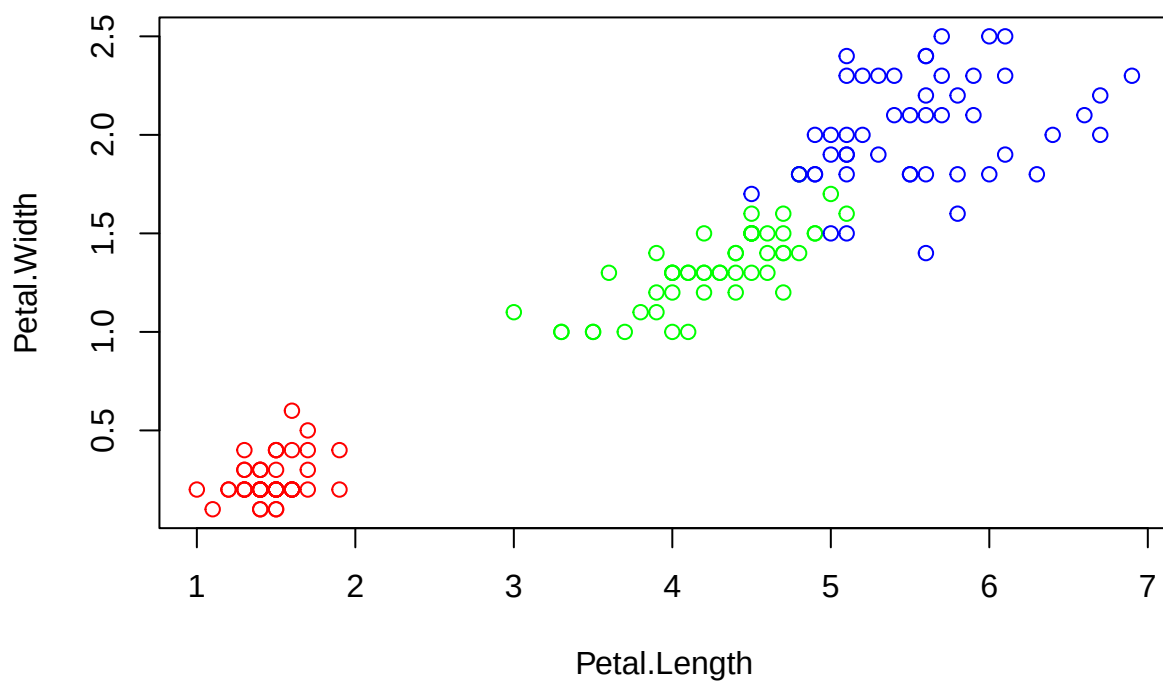
```
## 7 f Undergraduate " Professor" 5
## 8 f Undergraduate "AssistantProfessor" 11
## 9 f Undergraduate "Associate Professor" 2
## 10 m Graduate " Professor" 71
## 11 m Graduate "AssistantProfessor" 14
## 12 m Graduate "Associate Professor" 45
## 13 m MBA " Professor" 49
## 14 m MBA "AssistantProfessor" 15
## 15 m MBA "Associate Professor" 41
## 16 m Undergraduate " Professor" 12
## 17 m Undergraduate "AssistantProfessor" 10
## 18 m Undergraduate "Associate Professor" 24
```

```
# rank has the average highest score
bschool |> filter(student == "MBA") |>
  group_by(title)|>
  summarize(
    mean_score = mean(score)
  )
```

```
## # A tibble: 3 x 2
##   title          mean_score
##   <chr>          <dbl>
## 1 " Professor"      17.2
## 2 "AssistantProfessor" 15.8
## 3 "Associate Professor" 16.1
```

```
## Graph with basic R functions
```

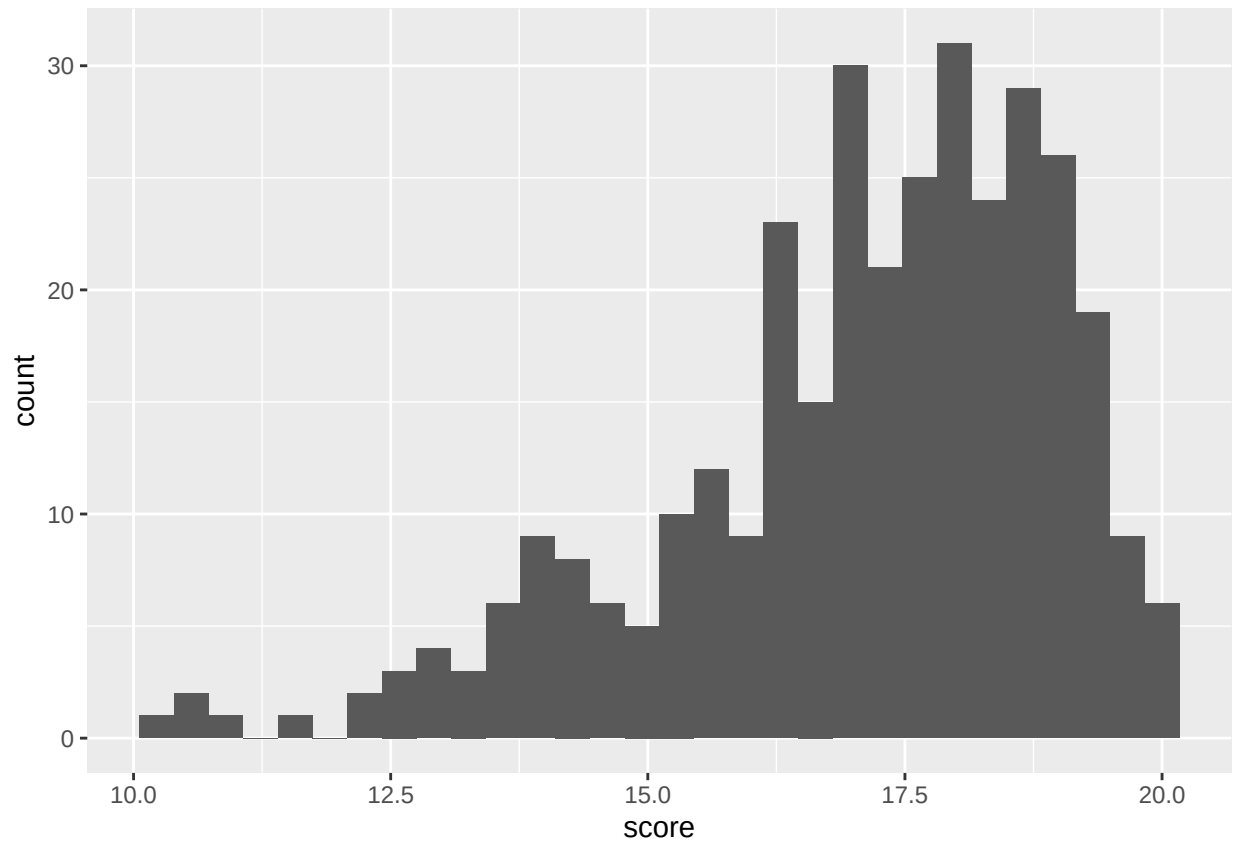
```
cols <- c("red", "green", "blue")
plot(Petal.Width~Petal.Length, data = iris,
     col = cols[as.numeric(Species)])
```



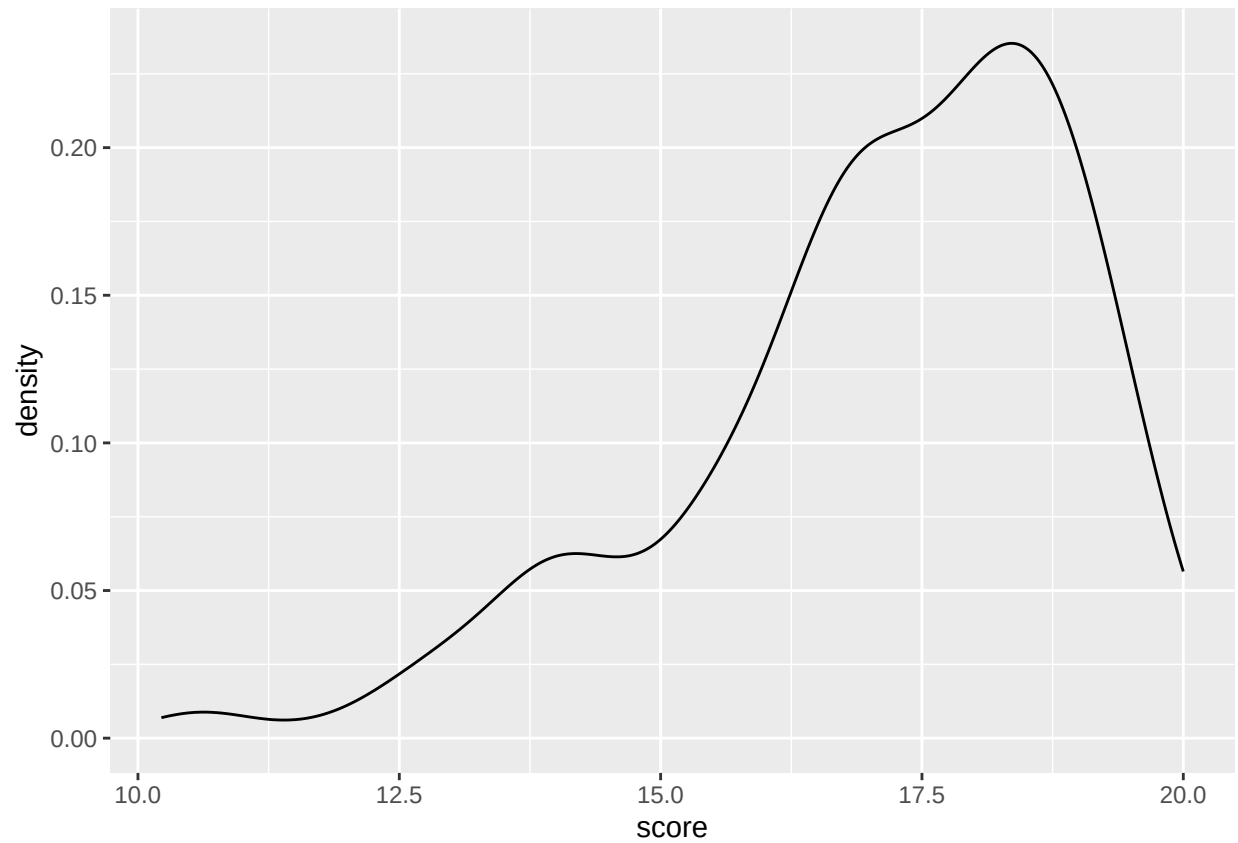
```
#legend("topleft", levels(iris$Species), col=cols, pch=1)
```

```
#GGPLOT  
p1 <- bschool |>  
  ggplot(aes(x = score))  
p1 + geom_histogram()
```

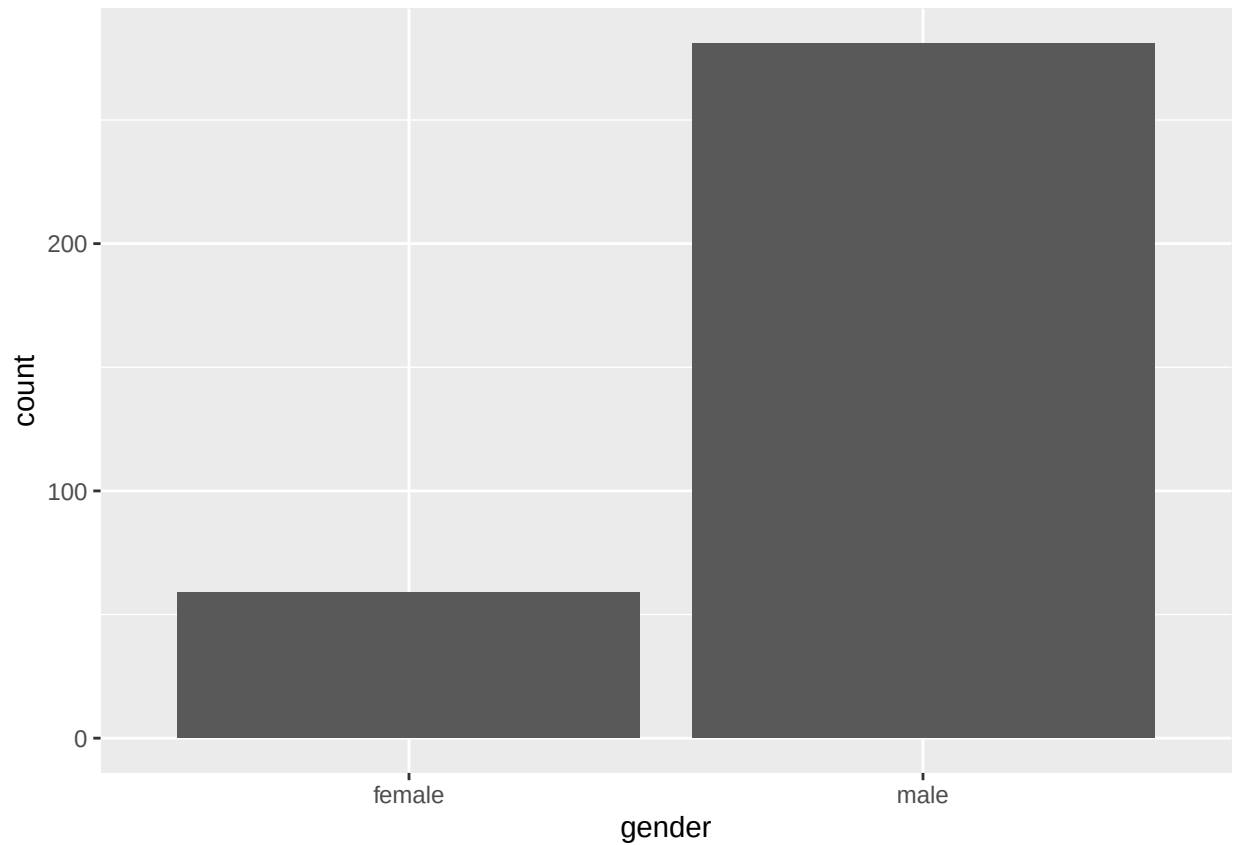
```
## 'stat_bin()' using 'bins = 30'. Pick better value 'binwidth'.
```



```
p1 + geom_density()
```

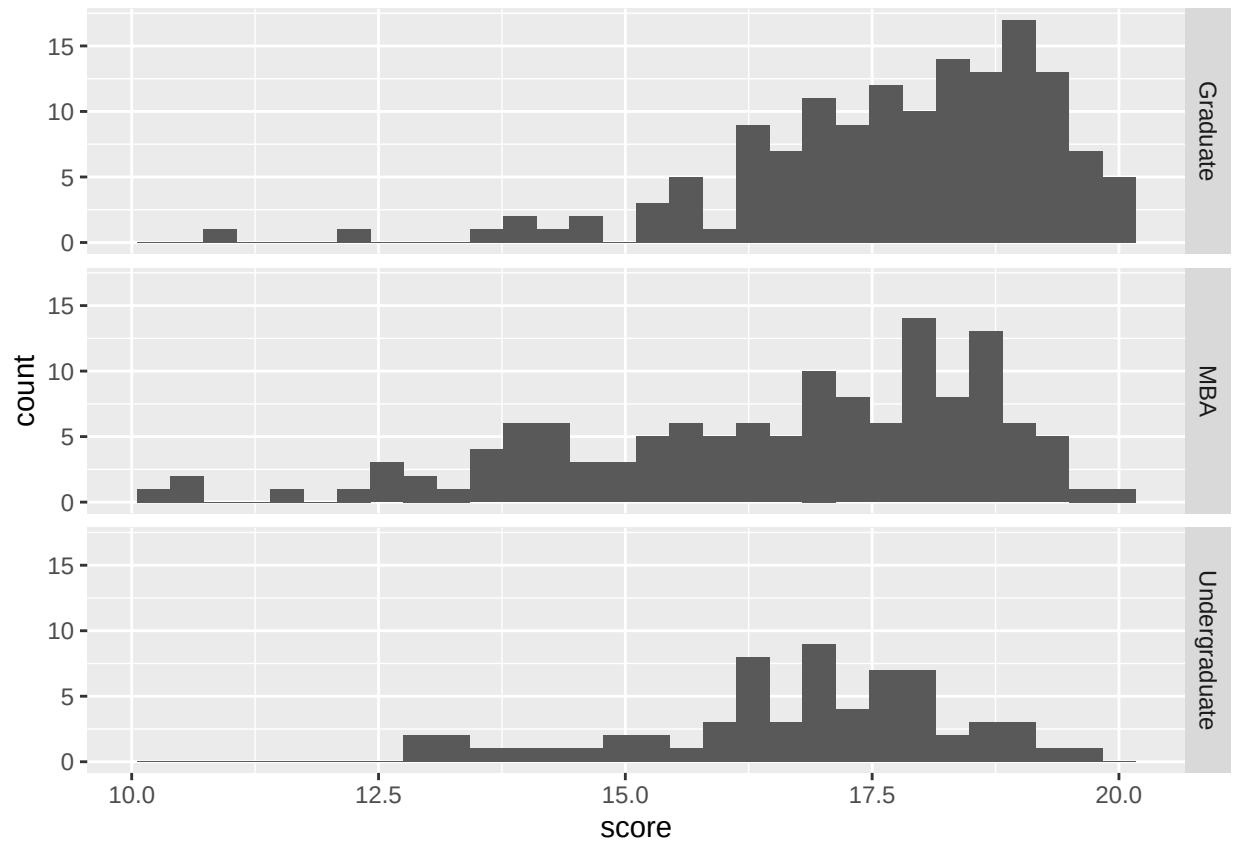


```
# qualitative
p2 <- bschool |>
  mutate(
    gender = ifelse(gender == "f",
                    "female", "male")
  ) |>
  ggplot(aes(x = gender))
p2 + geom_bar()
```

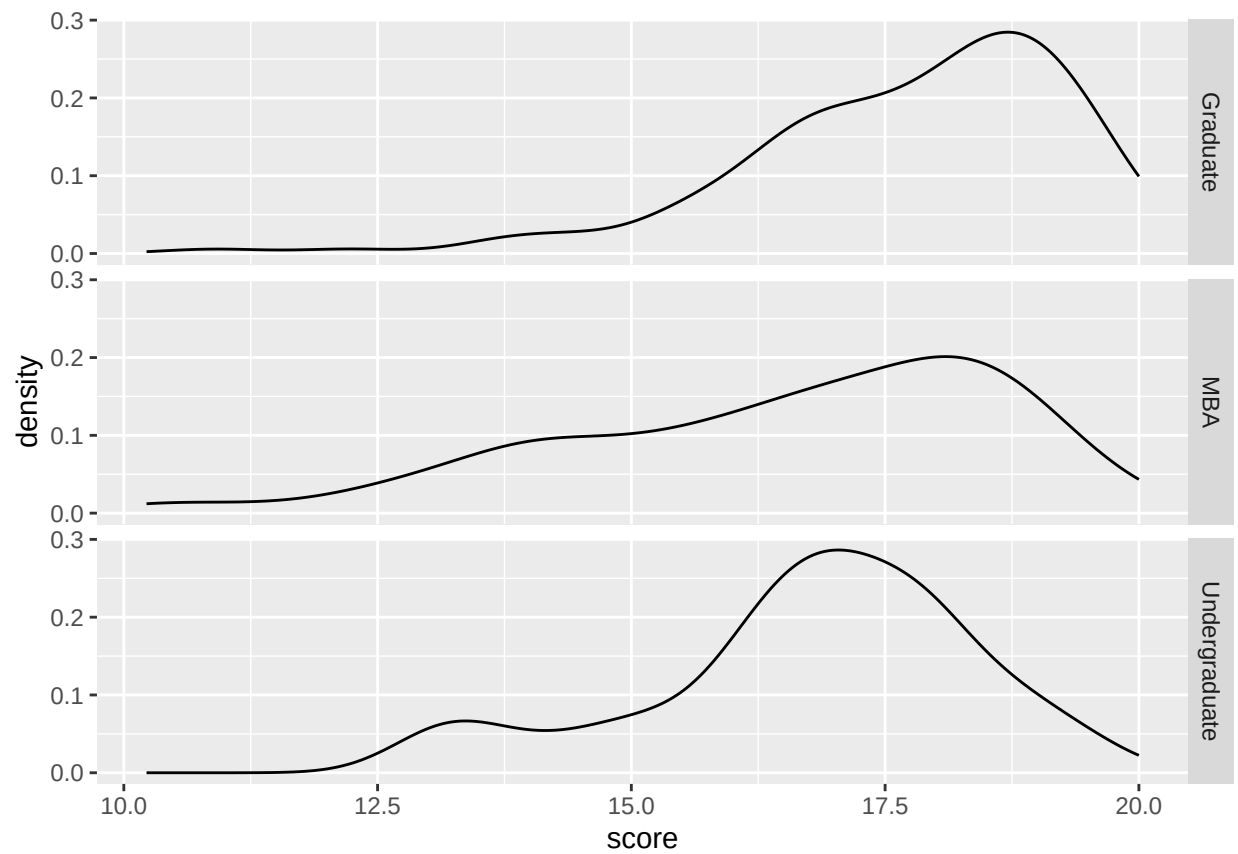


```
# Quantitative vs Qualitative  
p12 <- ggplot(bschool, aes(x=score))  
p12 + geom_histogram() +  
  facet_grid(student~.)
```

```
## 'stat_bin()' using 'bins = 30'. Pick better value 'binwidth'.
```

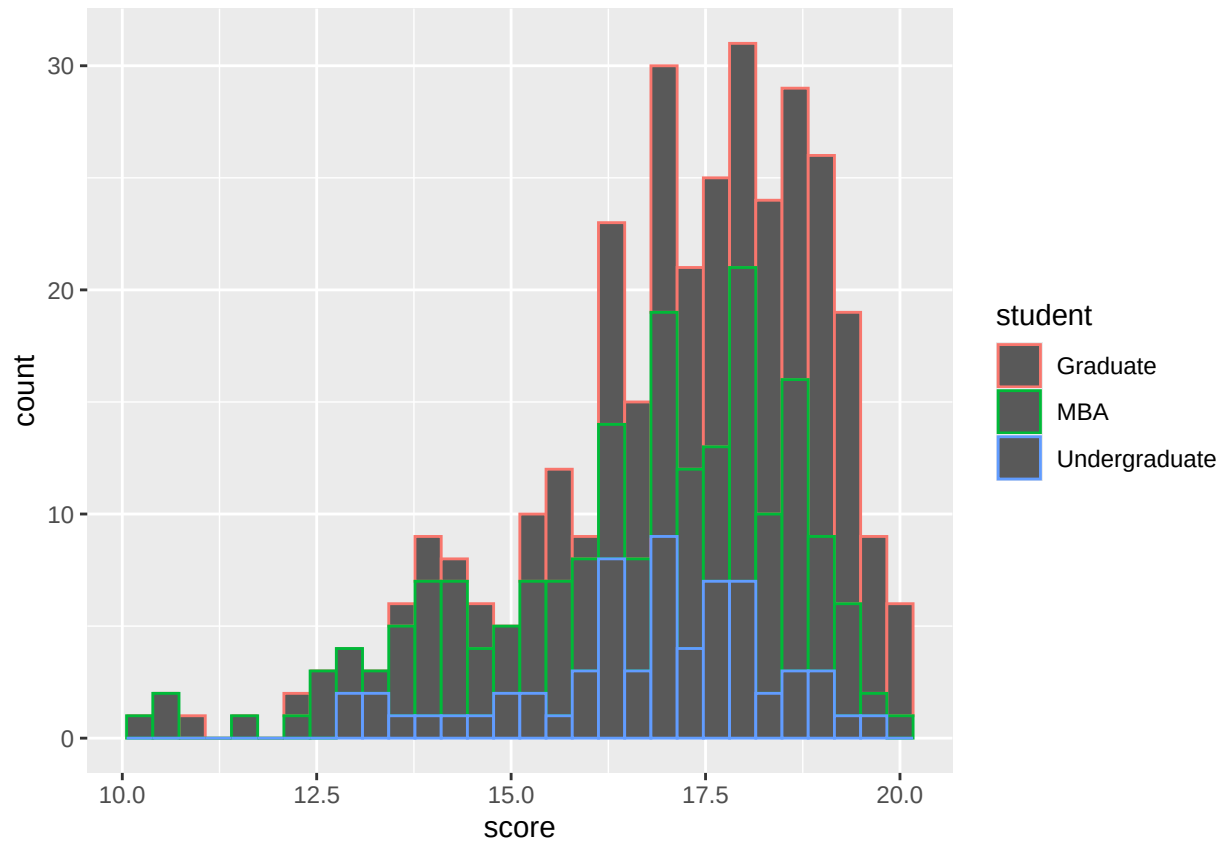


```
p12 + geom_density() +  
      facet_grid(student~.)
```

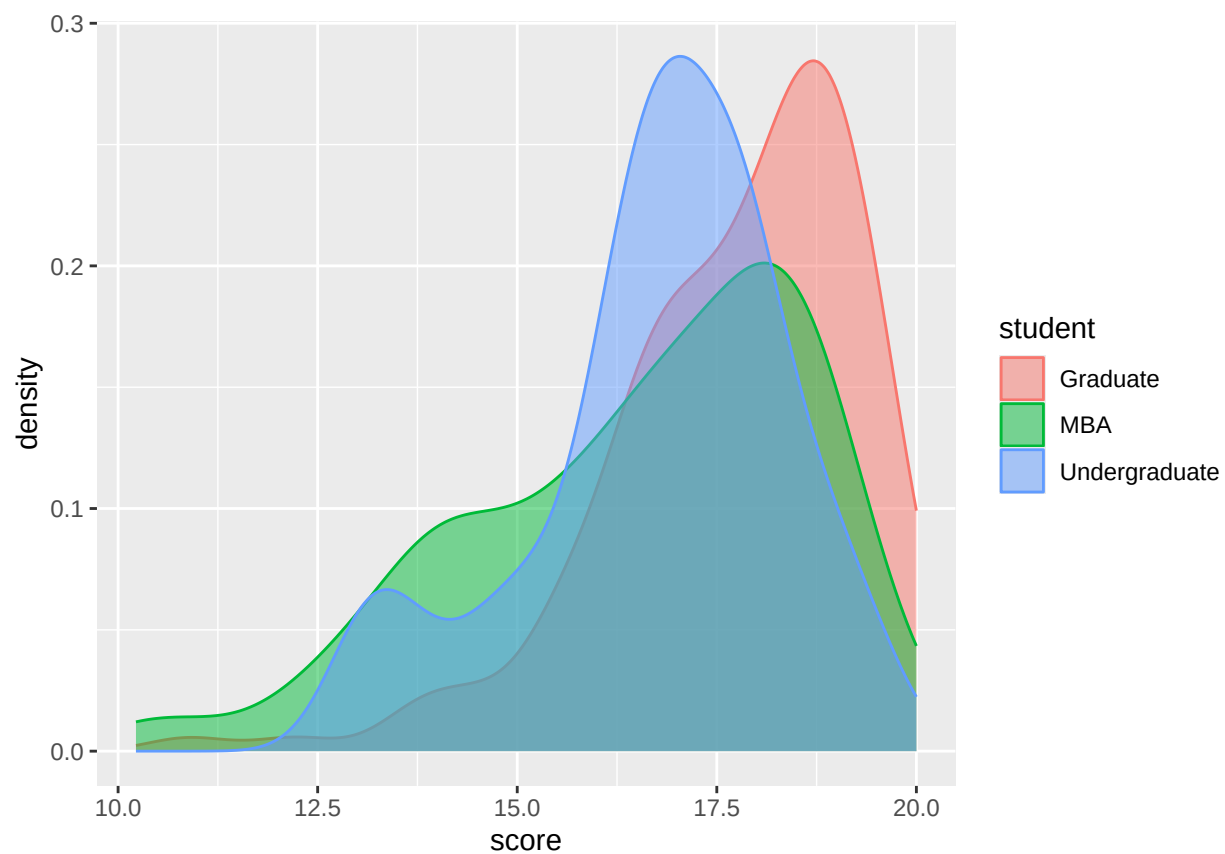


```
p12 +  
  geom_histogram(aes(color=student))
```

```
## 'stat_bin()' using 'bins = 30'. Pick better value 'binwidth'.
```

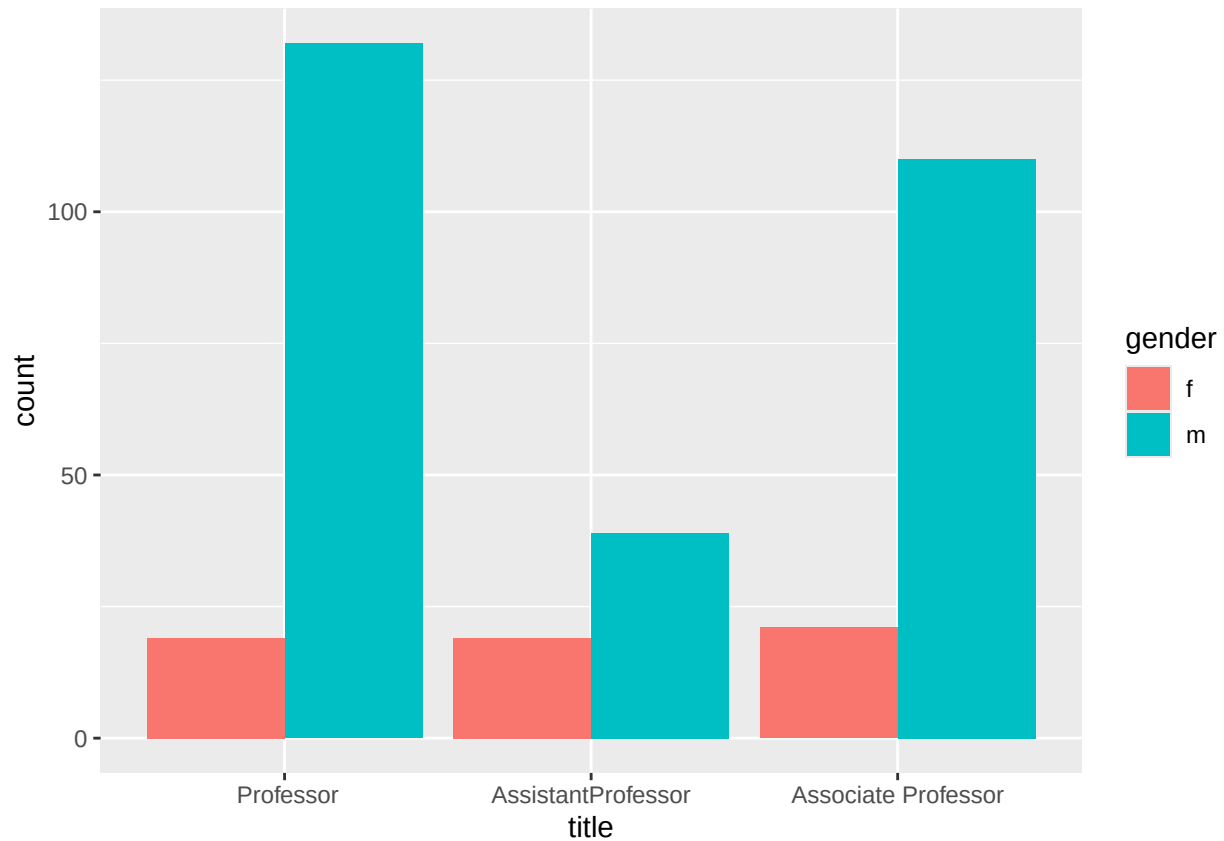


```
p12 + geom_density(aes(color=student, fill=student), alpha=0.5)
```



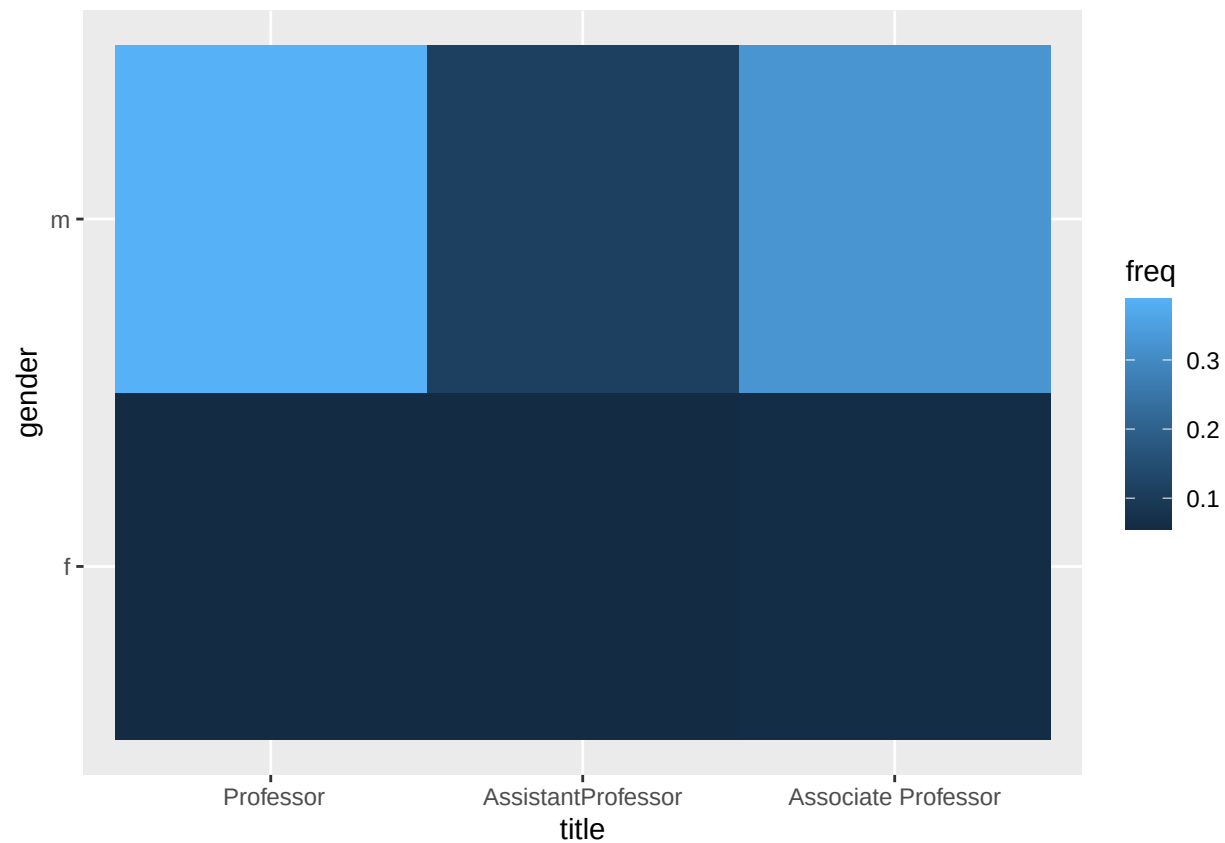
```
p12 <- p12 + geom_boxplot(aes(y=student))
```

```
# qualitative vs qualitative  
bschool |>  
  ggplot(aes(x=title, fill=gender)) +  
  geom_bar(position="dodge")
```



```
bschool |> group_by(title, gender) |>
  summarize(
    freq = n()/nrow(bschool)
  )|>
  ggplot(aes(x=title, y=gender))+
  geom_tile(aes(fill=freq))
```

```
## 'summarise()' has regrouped the output.
## i Summaries were computed grouped by title and gender.
## i Output is grouped by title.
## i Use 'summarise(groups = "drop_last")' to silence this message.
## i Use 'summarise(by = c(title, gender))' for per-operation grouping
## ('?dplyr::dplyr_by') instead.
```



```
p22 <- bschool |>
  ggplot(aes(x=size, y=score)) +
  geom_point()+
  geom_smooth(method = "lm")
p22
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```

